Options research: from order events to decisions

Purpose

Explain the data, reconstructed books, correctly mapped volatility surface, skew/liquidity diagnostics, time-of-day evidence and research-to-paper strategy workflow.

Dataset, not report date

35,045,335 MBO records; 22 sessions from 2026-03-09 to 2026-04-08. Surface snapshot: 2026-04-08T14:00:00 UTC.

How to use this document

Present the opening narrative, the EDA chapter and the surface/strategy slides first. PDF bookmarks navigate sections. The appendices reproduce every dashboard data table (dashboard 2 as differences from dashboard 1), all metric definitions and formulas. The companion plot collection contains every unique saved PNG once, with readable tiles for tall figures.

Executive conclusions: measurement is not alpha

Sources: phase01 facts; surface summary; both opportunities manifests; time_of_day summary.

Surface and skew

106 usable IV observations across 6 expiries. Calls/puts and ITM/OTM are covered, but some 25-delta wings remain unobserved.

Liquidity and execution

Strict analytical gate: 68/111 quotes; wider research gate: 106/111. Neither gate proves that a multi-leg package can fill.

Time-of-day evidence

RV slot rank consistency is 0.85; liquidity slot consistency is -0.34. Timing merits hedge-overlay research, not a standalone profitability claim.

Do not overclaim

No live orders, verified dealer positions, open interest or SHAP prediction model. Original momentum hypothesis was rejected; exploratory surface work is not a new clean holdout.

Data and instrument identity

Source: outputs/phase01/facts_step01.json; surface instrument mapping.

MBO is not a ready-made quote table

Add/cancel/modify/reset events must be replayed into bid and ask books. Trade messages and displayed resting liquidity are different objects.

Identity is session-aware

1,056 distinct raw symbols: 985 options, 21 futures and 50 spread/strategy or unparsed symbols. Instrument IDs can be reused: join on session_date and instrument_id.

Products and quote units

EUCO options are analyzed against their same-month FCEU futures. Prices are decimal quote units; one EUR/USD pip is 0.0001. Do not convert to currency profit without a verified contract multiplier.

Method: causal data-to-decision pipeline

1. Reconstruct and align

Replay feed events within session/instrument, respect resets and feed sequence, reconstruct two-sided books, and align underlying information without looking forward.

2. Price and measure

Map each option to its own monthly future; solve Black-76 IV from bid/mid/ask; apply analytical quote gates; derive Greeks, ATM level, skew, curvature and term diagnostics.

3. Research before execution

Use strictly prior same-expiry histories for z-scores; form cost-aware packages; stress model marks. Paper entry additionally checks quote age, leg skew, depth, timing and future identity.

4. Separate evidence tracks

The earlier frozen momentum workflow and the newer exploratory surface/time-of-day work are distinct. April has been viewed by exploratory analysis and is not a fresh global holdout.

EDA chapter: what was explored and in which order

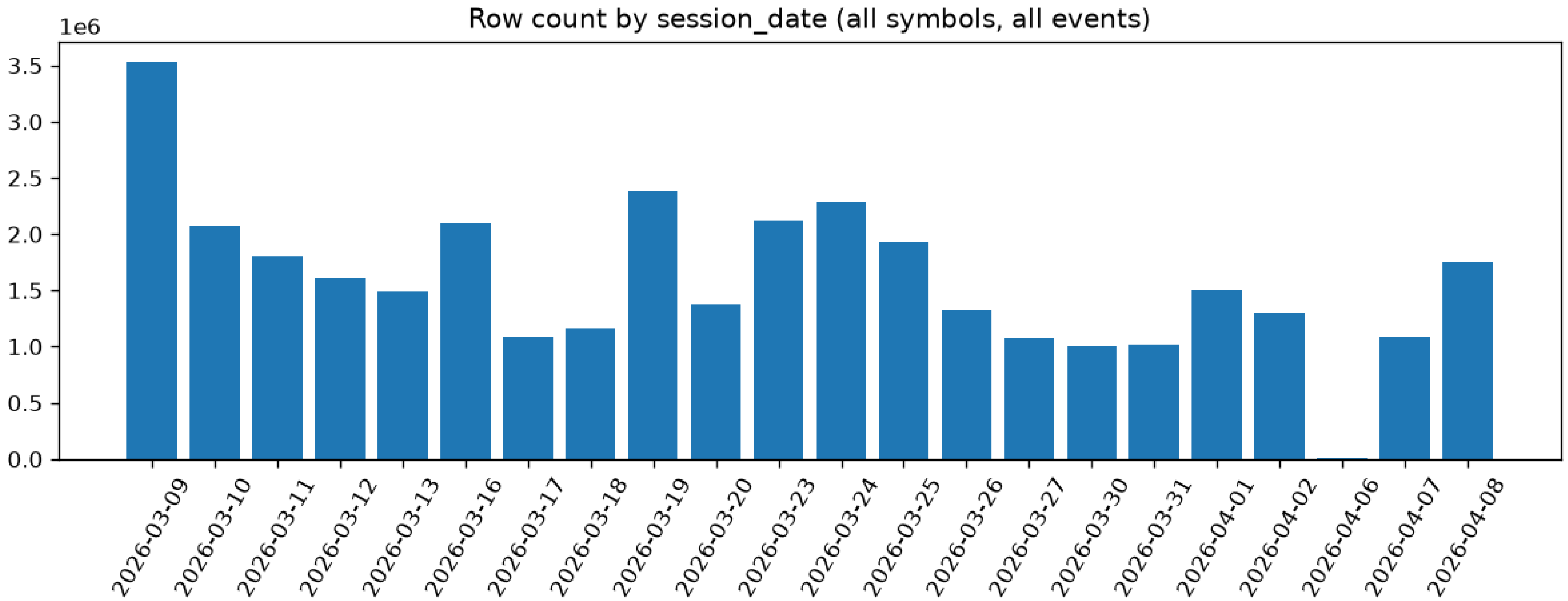
Sources: phase01, eda_returns, phase04 and phase06 facts. EDA is descriptive; nothing here is a trading signal.

Step	Question	Evidence used	Outcome
1. Inventory	What is in the feed?	Action, symbol and session counts	35,045,335 events; 22 sessions; 1,056 symbols
2. Coverage and timing	Are sessions complete and timestamps usable?	Rows per session; ts_recv - ts_event	58,925 ts_recv reversals; latency heavy-tailed
3. Price and returns	Which transform is stationary?	Front-future prices, ADF, ACF, fractional differencing	Levels non-stationary; returns stationary
4. Distributions	Are returns normal?	Histograms, boxplots, Q-Q by instrument class	Fat tails and quote-driven jumps
5. Instrument paths	Do options behave per expiry/strike?	Future, option and matched call/put paths	Sparse, expiry-specific; many strikes unmatched
6. Execution realism	Can signals be traded?	Spreads, entry gate, trade counts	Median spread 163 bps; 4 option trades
7. Hypothesis test	Did the EDA-led momentum idea work?	Phase 7-10 backtest and causality checks	REJECT: net -0.194

MBO action counts: add 17,515,881; cancel 17,516,289; reset 11,249; trade 996; fill 867; modify 53. Symbols: 985 options, 21 futures, 50 spreads/unparsed. Adds and cancels dominate: almost all information is quote revision, not trading.

EDA 1: session coverage before any statistic

Source: outputs\eda_returns\06_session_coverage.png

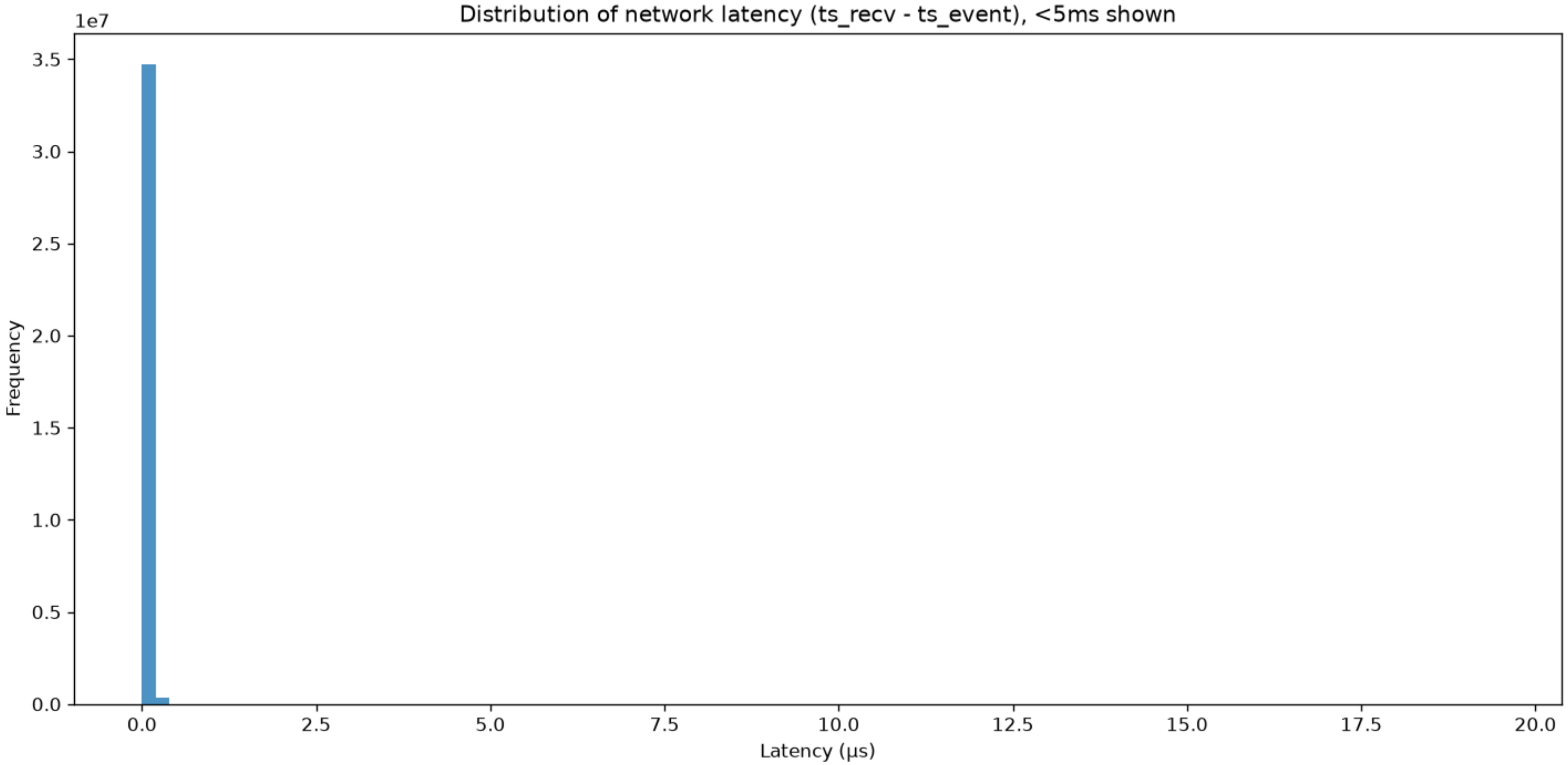


- Method: count all MBO rows per session_date across every symbol and event type.
- Result: 22 sessions from 2026-03-09 to 2026-04-08; activity varies by day, so pooled statistics are weighted toward busy sessions.
- Why it matters: missing or thin sessions bias averages and time-of-day profiles; per-session medians are used later for robustness.

Instrument/session-level exploratory exhibit. Price levels are descriptive; return distributions and paths require spread, stale-quote and sampling checks before strategy interpretation.

EDA 2: feed timing and latency

Source: outputs\eda_returns\01_network_latency_distribution.png

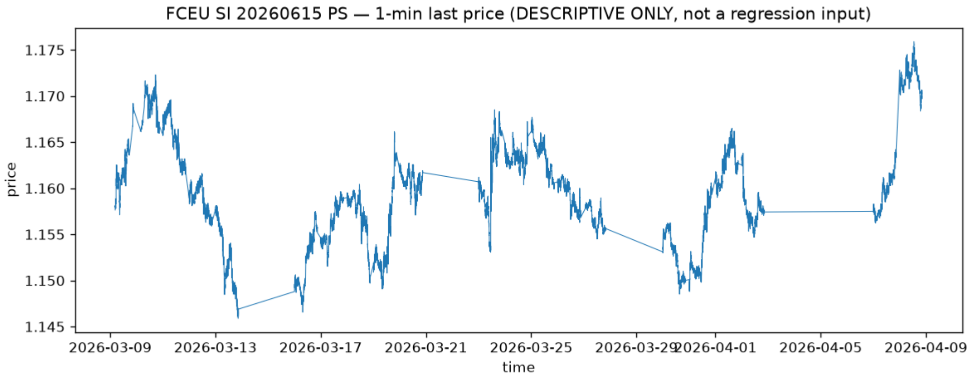


- Method: network latency = ts_recv - ts_event per record (shown below 5 ms); ts_in_delta is the venue-side processing delta.
- Result: median 0.021 us, p95 0.123 us; the raw mean 11,152 us and max 2.3e+10 us are driven by outliers/resets. ts_in_delta median 1.248 us, p99 4.308 us.
- Why it matters: use medians/quantiles, not means; feed latency is not our end-to-end order latency or fill probability.

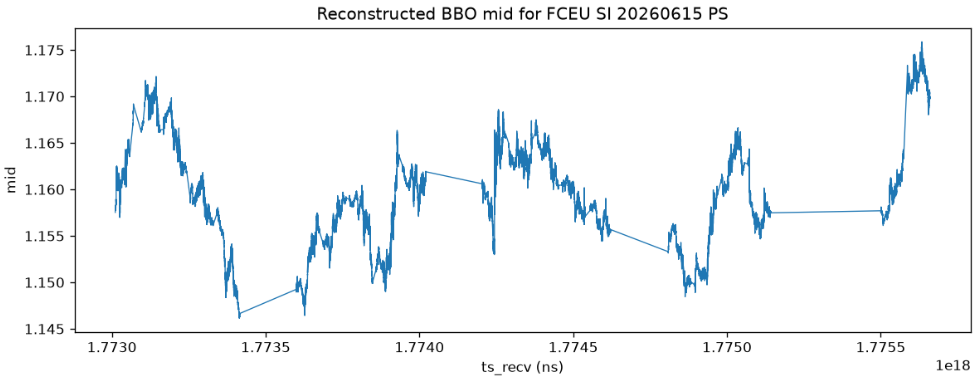
Receive-minus-event latency describes feed timing. It is not end-to-end strategy latency or a fill probability.

EDA 3: price level - raw events versus reconstructed BBO

Sources: outputs\eda_returns\01_price_level_descriptive.png; outputs\phase04\01_bbo_mid.png



FCEU SI 20260615 PS — 1-min last price (DESCRIPTIVE ONLY, not a regression input)



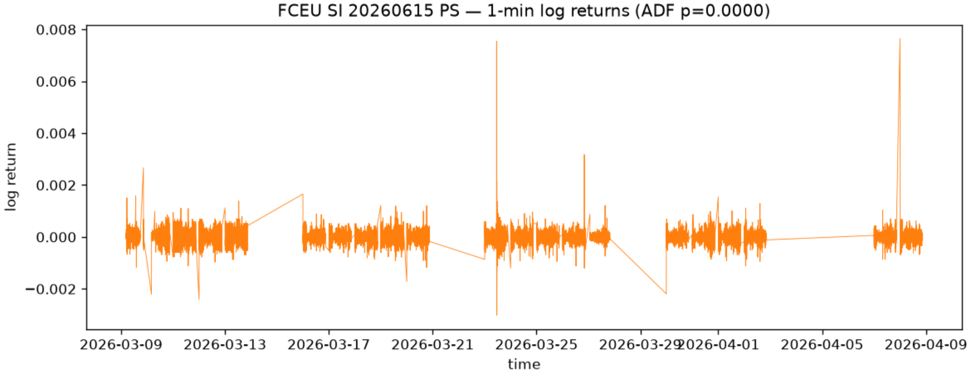
Reconstructed BBO mid for FCEU SI 20260615 PS

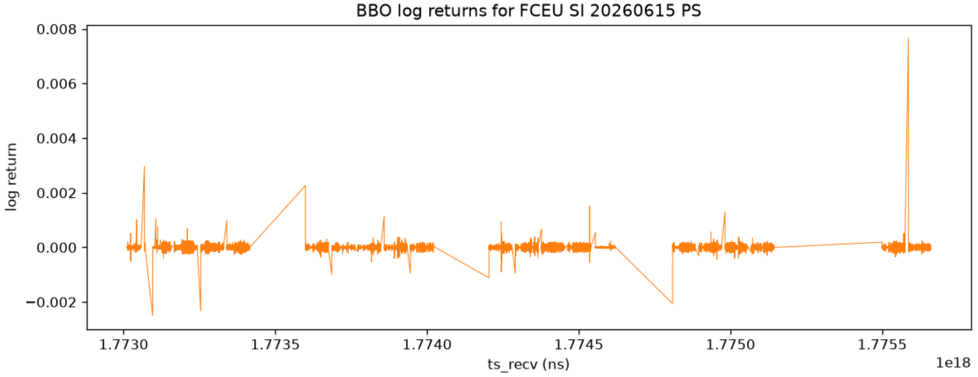
- Left (preliminary): 1-minute last price from raw add/trade events of FCEU SI 20260615 PS (12,486,196 rows, 24,964 bars). ADF -1.93, p=0.319.
- Right (phase04, supersedes left): reconstructed best-bid/offer mid, 4,104,473 BBO rows, 4,022,597 valid mids. ADF -2.00, p=0.285.
- Both fail to reject a unit root: price levels are non-stationary, so levels are descriptive only and never used as regression inputs.

Phase04 clock labels are 19,800 s early (naive UTC read as IST); shapes and statistics are unaffected, clock-time reading is not.

EDA 4: log returns - stationary but spiky

Sources: outputs\eda_returns\02_log_returns.png; outputs\phase04\02_bbo_mid_log_returns.png



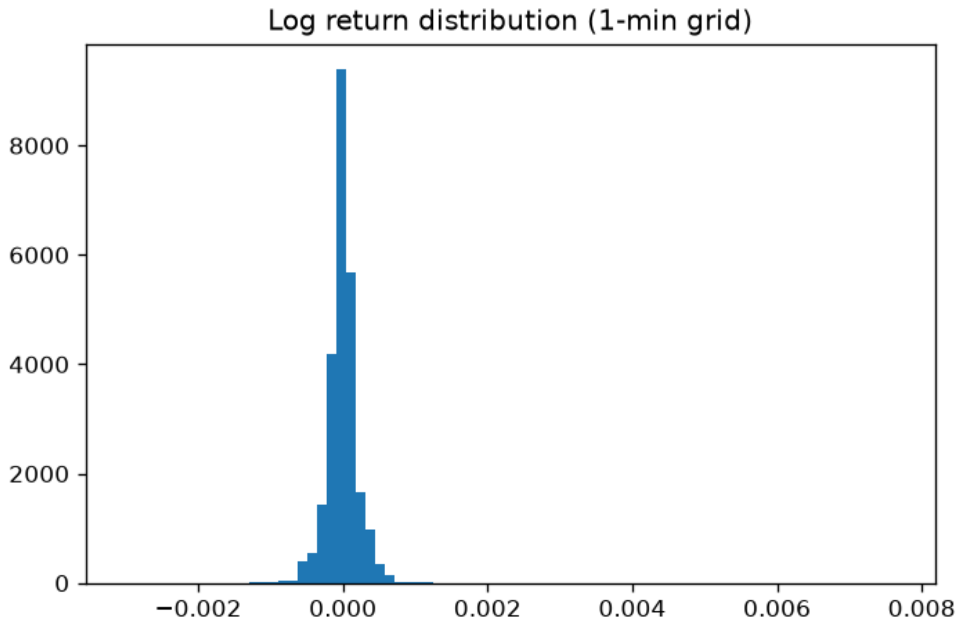


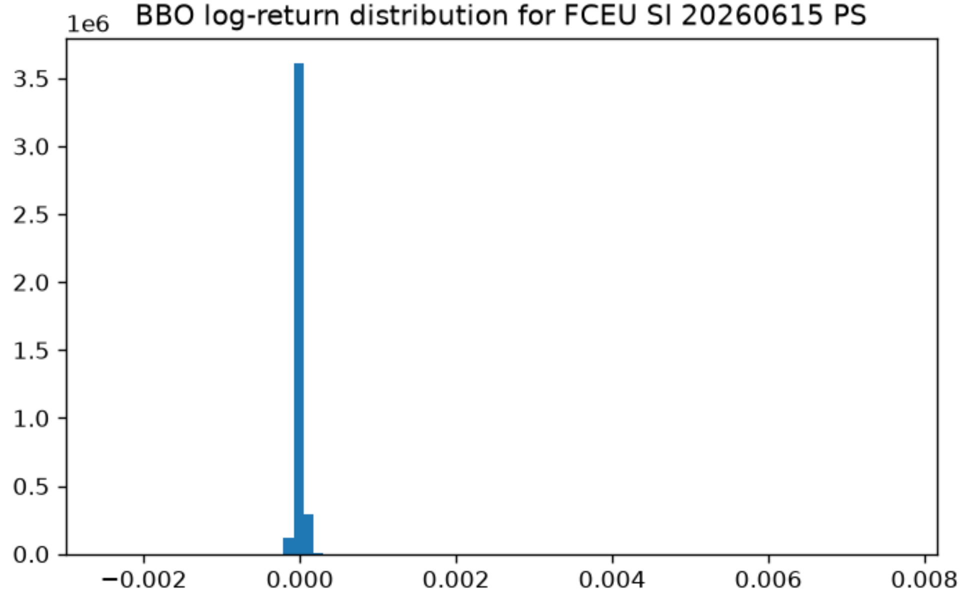
- Method: r_t = ln(P_t / P_t-1) within one symbol and session; never across rolls or different contracts.
- Result: ADF on log returns -26.29 (1-minute grid) and -27.29 (event-time BBO), both p~0: returns are stationary.
- Isolated spikes coincide with book resets, session opens and one-sided books; they are measurement artefacts until proven executable.

Left: preliminary event-price returns. Right: reconstructed BBO-mid returns (phase04 clock offset applies).

EDA 5: return histograms - mass at zero, fat tails

Sources: outputs\eda_returns\03_log_return_histogram.png; outputs\phase04\03_bbo_mid_return_hist.png



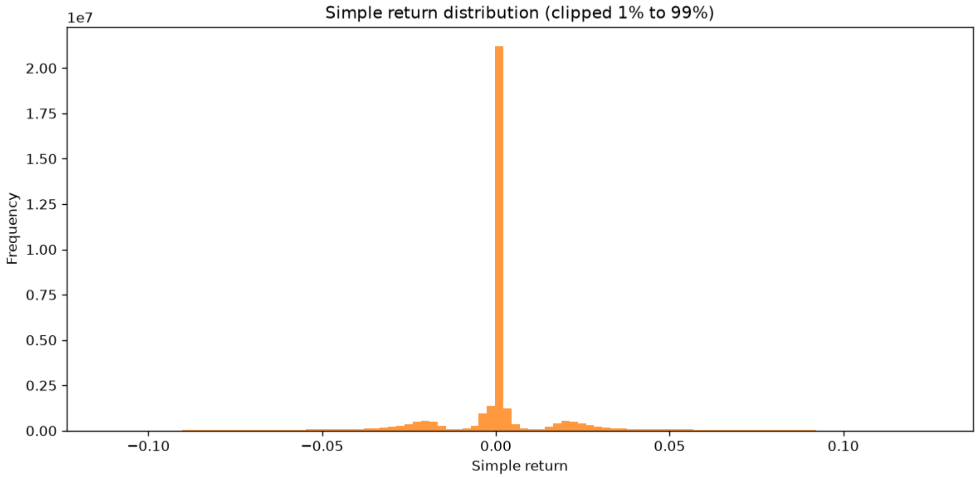


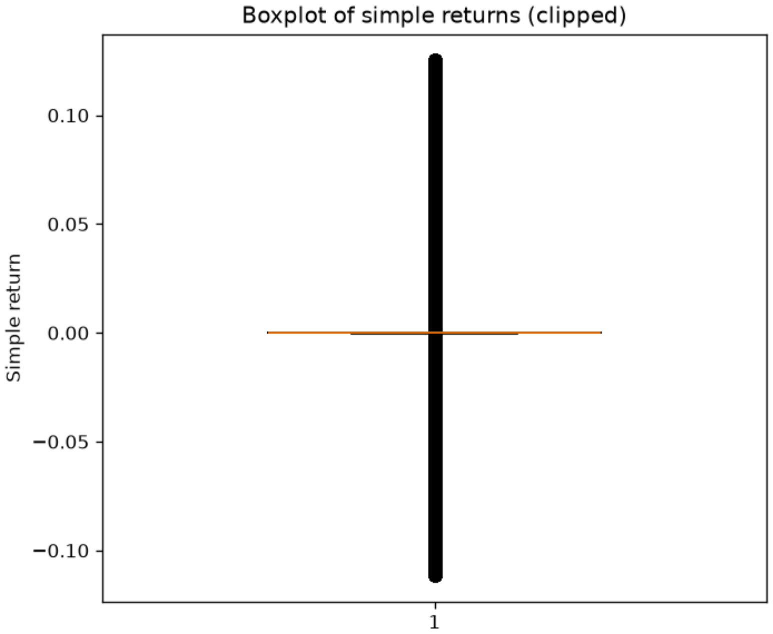
- Method: histogram of 1-minute log returns (left) and event-time BBO-mid log returns (right).
- Result: a tall spike at exactly zero (no mid change between observations) plus long tails; event-time sampling makes the zero spike larger.
- Why it matters: a normal model understates tail risk, and sampling frequency changes realized variance (see the time-of-day 1m vs 5m RV comparison).

Instrument/session-level exploratory exhibit. Price levels are descriptive; return distributions and paths require spread, stale-quote and sampling checks before strategy interpretation.

EDA 6: simple returns across all instruments

Sources: outputs\eda_returns\02_simple_returns_distribution.png; outputs\eda_returns\03_simple_returns_boxplot.png



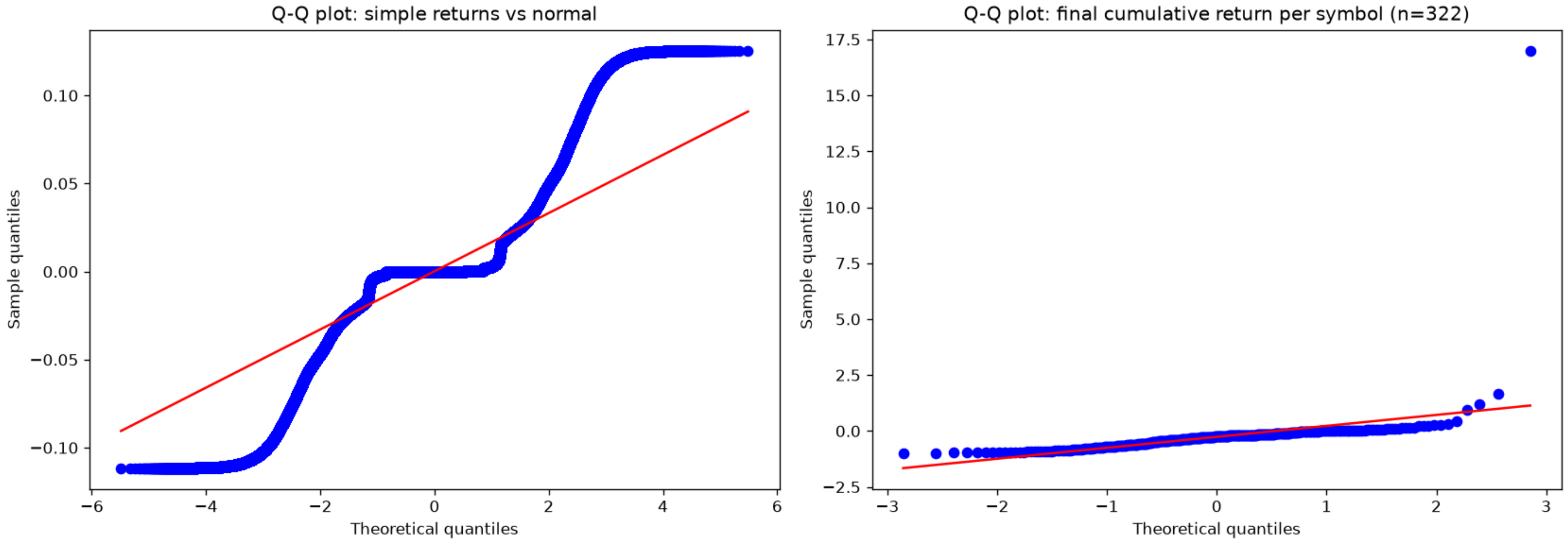


- Method: simple return = P_t / P_t-1 - 1 per symbol, pooled over 320 symbols and 35,033,764 return rows; plots are clipped to the 1%-99% range [-0.112, 0.125].
- Result: mean 0.00127, std 0.0838, median 0.0; unclipped min -44.4 and max 49.0 come from near-zero option prices (a 0.0001 tick on a 0.0002 premium is +50%).
- Why it matters: option percentage returns explode at small premiums; compare options in vol points or price units, not percentages.

Instrument/session-level exploratory exhibit. Price levels are descriptive; return distributions and paths require spread, stale-quote and sampling checks before strategy interpretation.

EDA 7: Q-Q against the normal distribution

Source: outputs\eda_returns\04_return_qq_plots.png

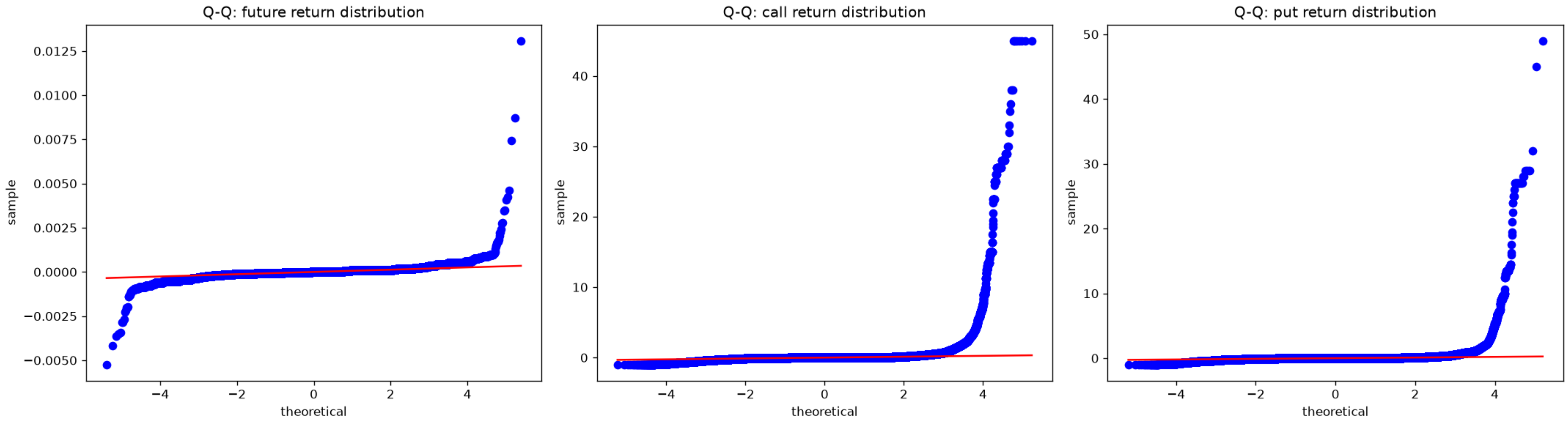


- Method: empirical quantiles of clipped simple returns (left) and final cumulative return per symbol (right) against normal quantiles.
- Result: S-shaped departures in both tails; final cumulative return per symbol has mean -0.262, min -0.98, max 17.0 - skewed by options repricing from tiny premiums.
- Why it matters: z-score thresholds based on normality will fire too often in the tails; robust (quantile) thresholds are preferable.

Q-Q diagnostics compare empirical return quantiles with a normal reference. Tail departures indicate distribution mismatch, not a directional forecast or an executable edge.

EDA 8: Q-Q by instrument class - futures, calls, puts

Source: outputs\eda_returns\12_qq_pooled_by_class.png

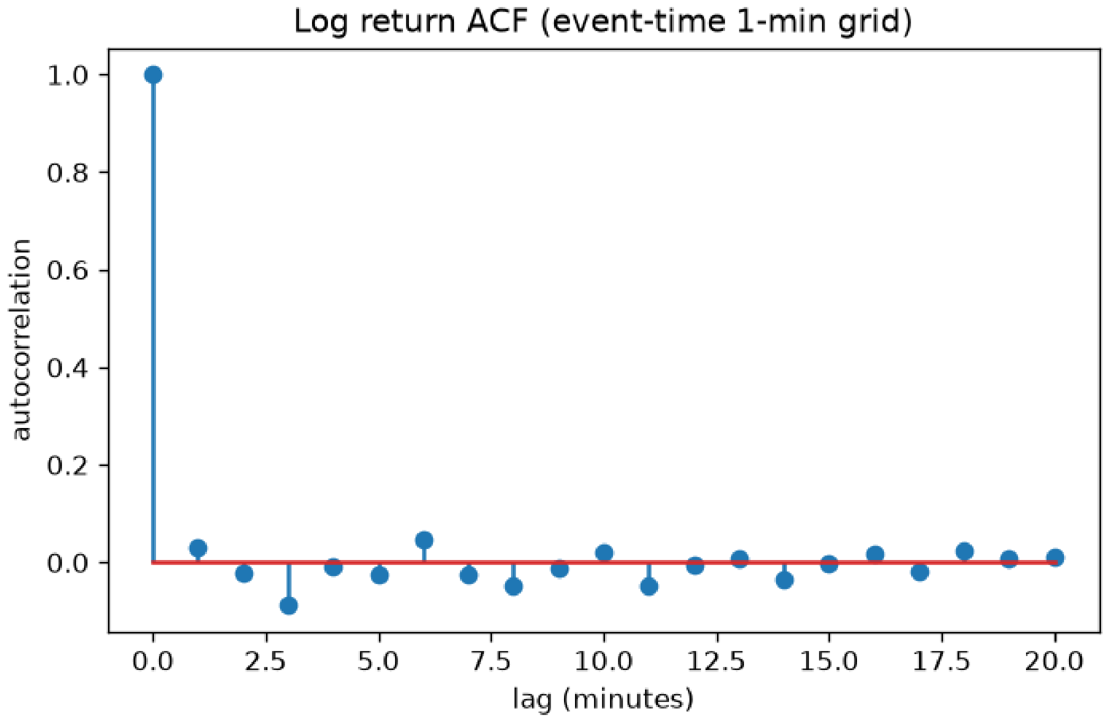


- Method: separate Q-Q plots per class so futures are not mixed with options.
- Result: futures are closest to normal; calls and puts show much heavier tails because tick-size moves on small premiums are large percentages.
- Why it matters: pooled statistics are dominated by options; any future-based signal must be studied on futures only.

Q-Q diagnostics compare empirical return quantiles with a normal reference. Tail departures indicate distribution mismatch, not a directional forecast or an executable edge.

EDA 9: autocorrelation of returns

Source: outputs\eda_returns\04_acf.png; ACF values from eda_returns and phase04 facts.



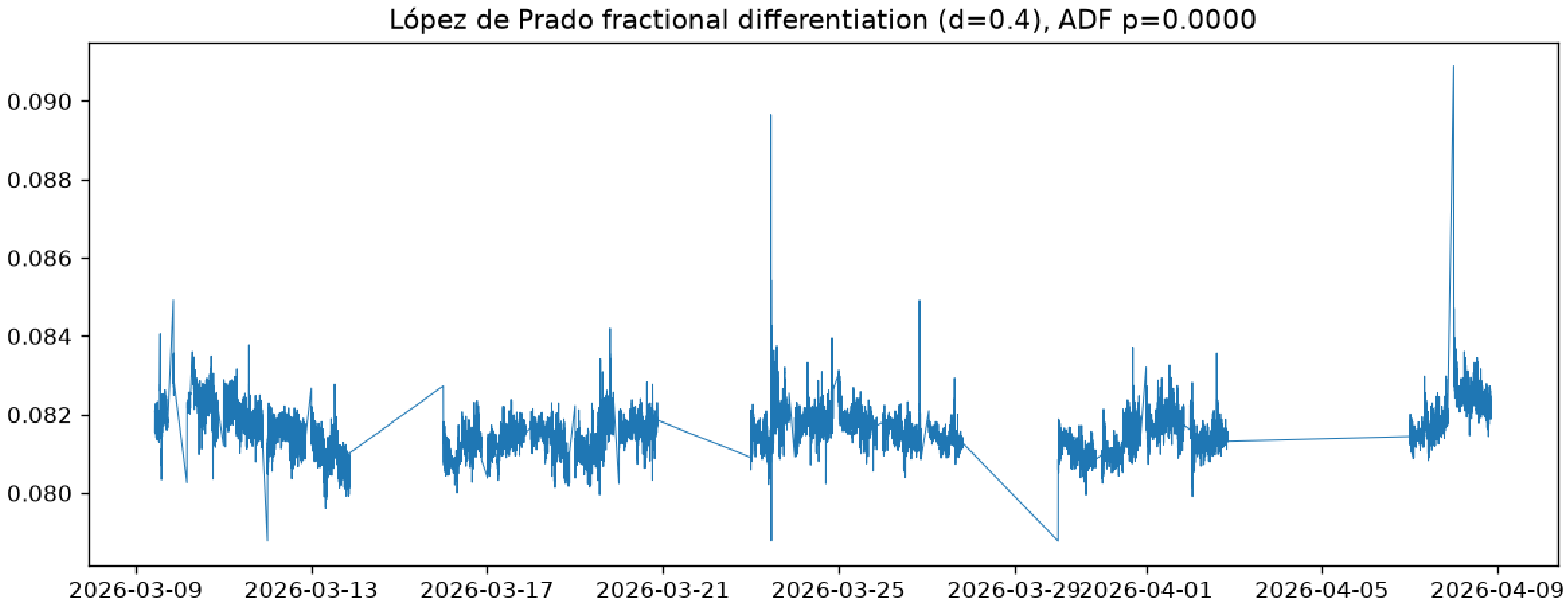
Lag	1-min grid	Event-time BBO
Lag 1	+0.0295	+0.0422
Lag 2	-0.0209	+0.0160
Lag 3	-0.0875	-0.0382
Lag 4	-0.0097	-0.0006
Lag 5	-0.0256	+0.0035

- Method: ACF(k) = corr(r_t, r_t-k); the 95% band for white noise is about +/-2/sqrt(n).
- Result: all |ACF| <= 0.087; largest on the 1-minute grid is lag 3 (-0.087), in event time lag 1 (+0.042), consistent with bid/ask bounce and quote clustering.
- Why it matters: statistically detectable dependence is far too small to pay a 1-2 tick spread; this was confirmed by the rejected momentum backtest.

Autocorrelation measures linear dependence between returns and lagged returns; it is not proof of out-of-sample predictability after spread and costs.

EDA 10: fractional differencing - keep memory, gain stationarity

Source: outputs\eda_returns\05_frac_diff.png

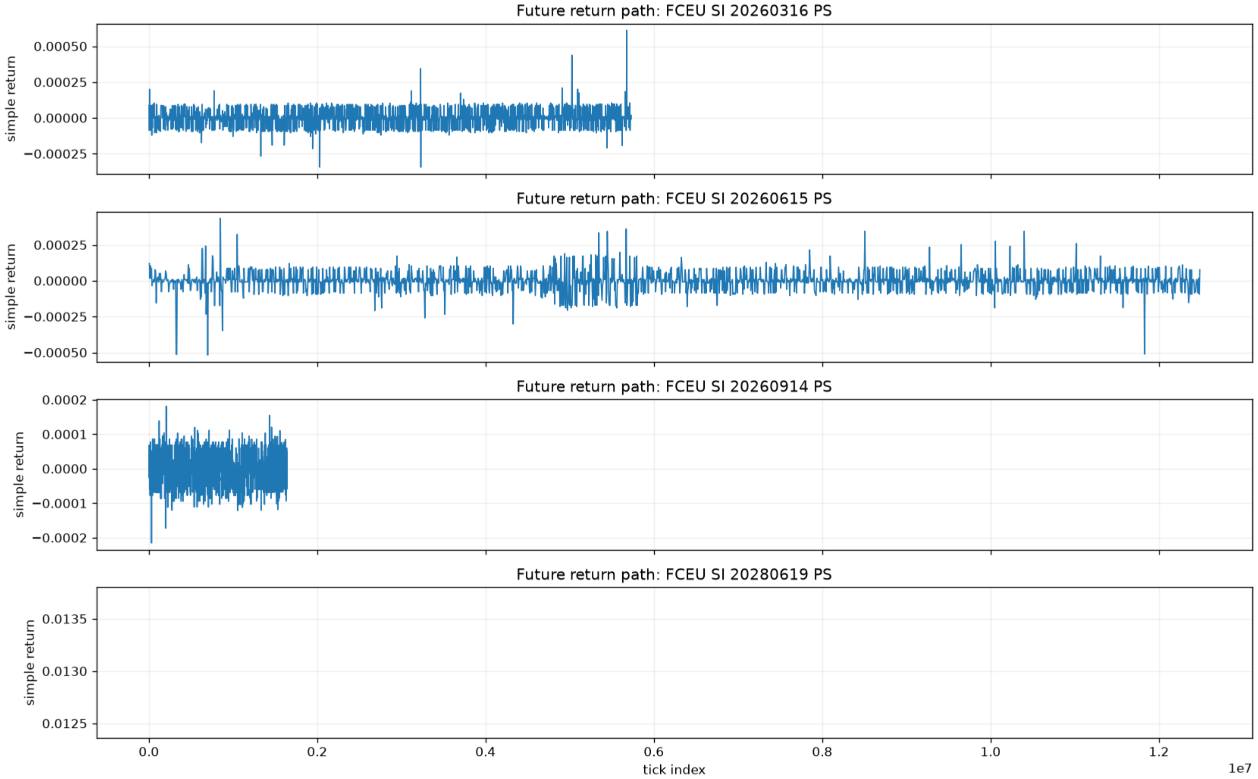


- Method: Lopez de Prado fixed-width fractional differencing of log price with d=0.4; d=0 is the price, d=1 the return.
- Result: ADF -6.52, p=1.1e-08: stationary while retaining more level memory than returns. Exploratory CUSUM filter (threshold 0.00066) marks 1,702 events.
- Why it matters: d and the CUSUM threshold were chosen on the full sample here; for prediction they must be fit on training data only.

Fractional differencing is a stationarity-versus-memory research transform. Any differencing order must be selected on training data before predictive use.

EDA 11: futures cumulative return paths by contract

Source: outputs\eda_returns\07_future_return_paths.png

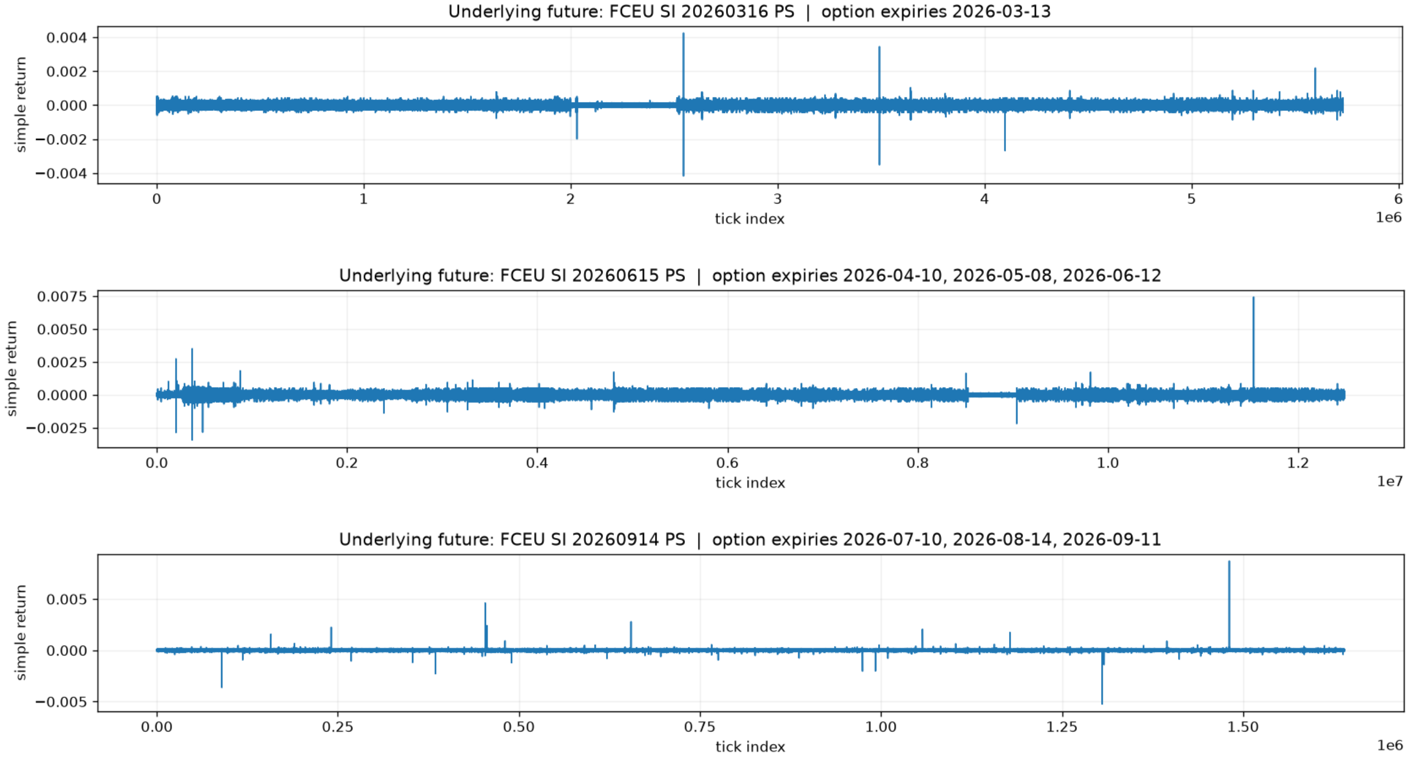


- Method: cumulative simple return per FCEU contract, computed within contract (no roll splicing).
- Result: the June contract is the liquid front with the most continuous path; March expires mid-sample and September/far months are sparse.
- Why it matters: each option expiry must be mapped to its own monthly future (see contract mapping); one generic future is wrong.

Instrument/session-level exploratory exhibit. Price levels are descriptive; return distributions and paths require spread, stale-quote and sampling checks before strategy interpretation.

EDA 12: underlying futures by option-expiry group

Sources: outputs\eda_returns\11_future_underlying_2026-03-16.png; outputs\eda_returns\11_future_underlying_2026-06-15.png; outputs\eda_returns\11_future_underlying_2026-09-14.png

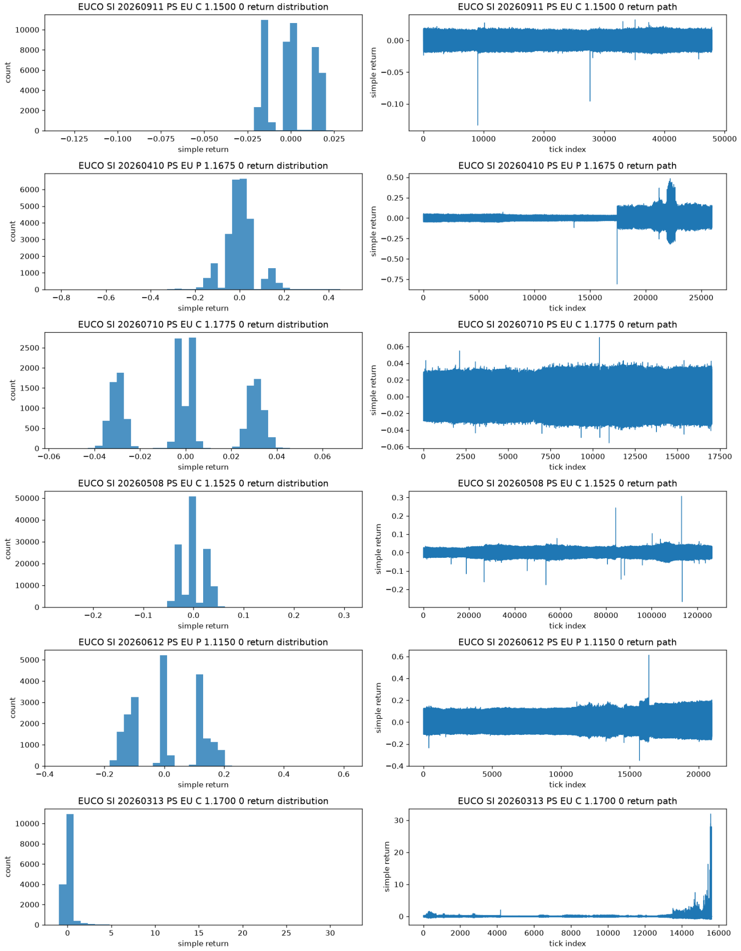


- Method: one strip per quarterly future, labelled with the option expiries that reference it.
- Result: serial-month options (Apr/May, Jul/Aug) have no own future and borrow information from the next quarterly contract (calendar spreads/interpolation).
- Why it matters: this is why forward construction (parity, spreads, interpolation) is part of the IV methodology.

Instrument/session-level exploratory exhibit. Price levels are descriptive; return distributions and paths require spread, stale-quote and sampling checks before strategy interpretation.

EDA 13: per-symbol return distribution and path

Source: outputs\eda_returns\05_symbol_returns_by_symbol.png

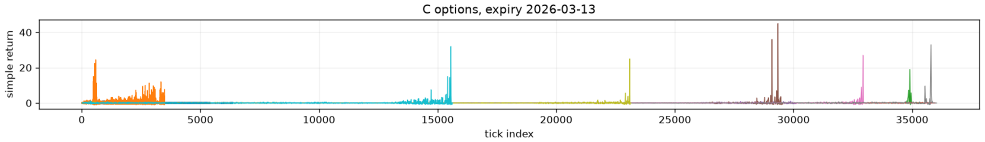


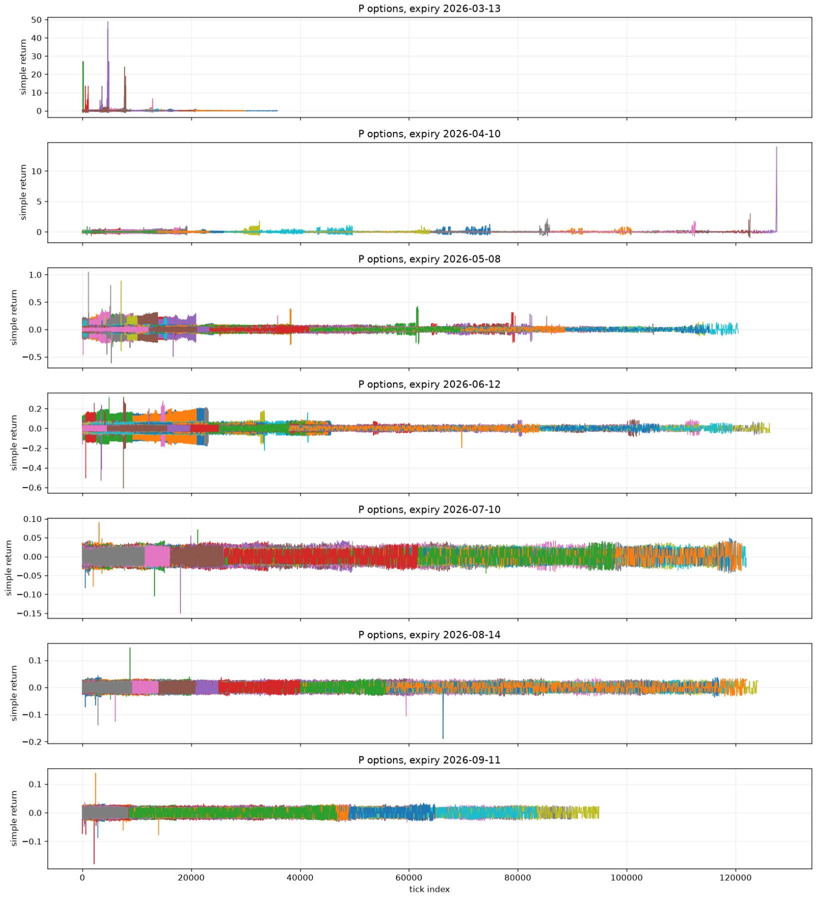
- Method: for the most active symbols, distribution (left) and cumulative path (right) are shown symbol by symbol.
- Result: options show discrete, tick-sized jumps and long flat periods; futures show continuous paths. Full-resolution tiles are in the plot collection.

Instrument/session-level exploratory exhibit. Price levels are descriptive; return distributions and paths require spread, stale-quote and sampling checks before strategy interpretation.

EDA 14: option return paths by expiry - calls and puts

Sources: outputs\eda_returns_subset\08_c_options_by_expiry.png; outputs\eda_returns_subset\08_p_options_by_expiry.png



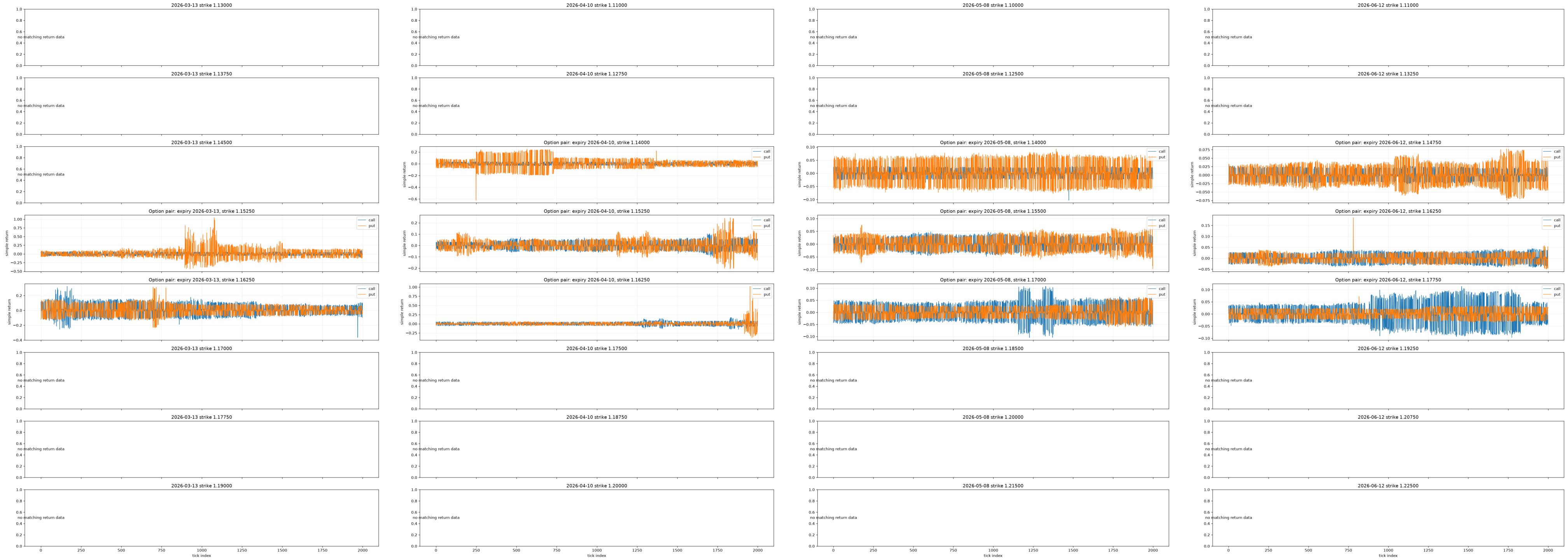


- Method: one panel per expiry, all strikes of that right overlaid in tick-index order (calls figure as generated contains only the 2026-03-13 expiry).
- Result: near expiries (Mar/Apr) show simple-return spikes up to +50 and +14 as cheap options reprice; longer expiries stay within roughly +/-0.1-0.2. Percentage returns scale inversely with premium.
- Why it matters: compare expiries in IV/vol points, and never pool short- and long-dated option returns.

Saved subset figures. Quote-driven moves are not executed option-trade returns.

EDA 15: call/put pairs matched by expiry and strike (1/2)

Sources: outputs\eda_returns\10_call_put_matched_expiry_2026-03-13.png; outputs\eda_returns\10_call_put_matched_expiry_2026-04-10.png; outputs\eda_returns\10_call_put_matched_expiry_2026-05-08.png; outputs\eda_returns\10_call_put_matched_expiry_2026-06-12.png

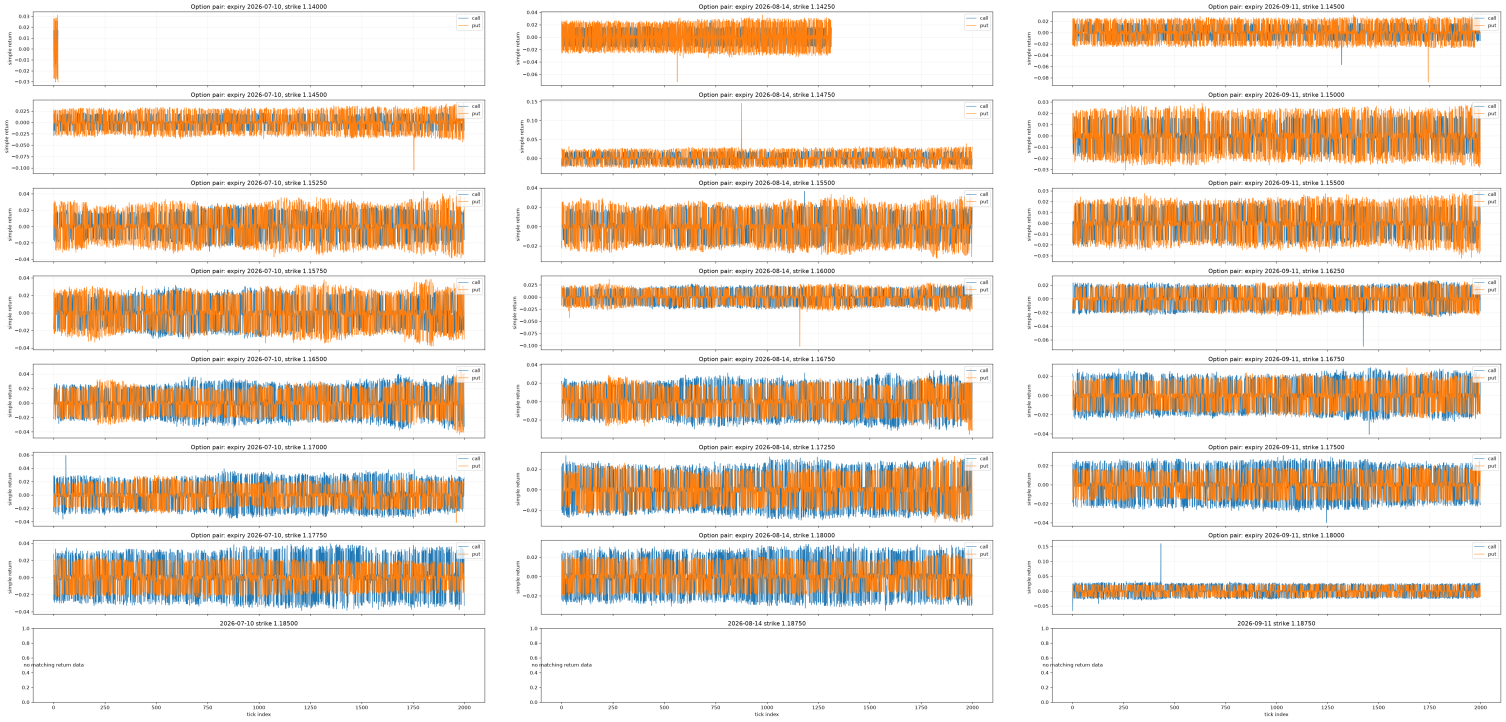


- Method: for each expiry, the call and put at the same strike are overlaid; panels with no matched data are labelled, not dropped.
- Result: only a few near-the-money strikes per expiry have both legs quoted; empty panels show where parity-based checks are impossible. Full-size, readable parts are in all_plots_collection.pdf.

Call/put return paths paired by expiry and strike, not pooled across instruments. Empty panels explicitly mark missing matching data. Quote-driven moves are not executed option-trade returns.

EDA 15: call/put pairs matched by expiry and strike (2/2)

Sources: outputs\eda_returns\10_call_put_matched_expiry_2026-07-10.png; outputs\eda_returns\10_call_put_matched_expiry_2026-08-14.png; outputs\eda_returns\10_call_put_matched_expiry_2026-09-11.png

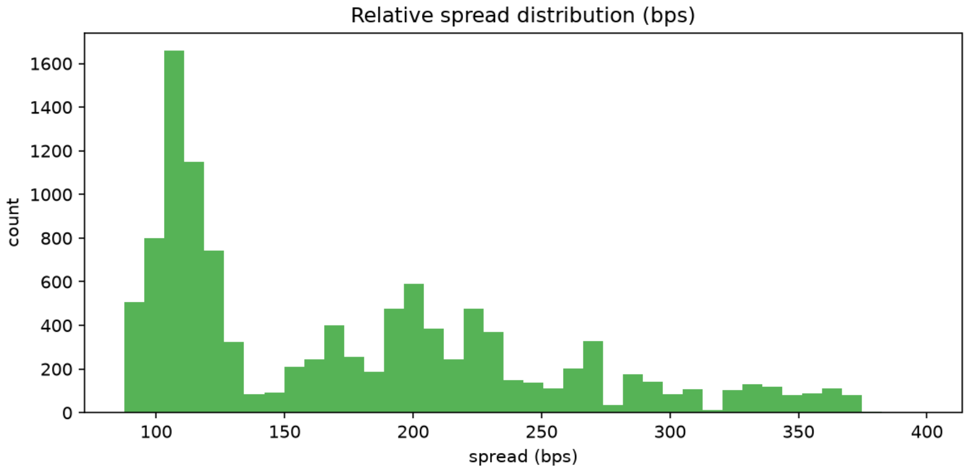


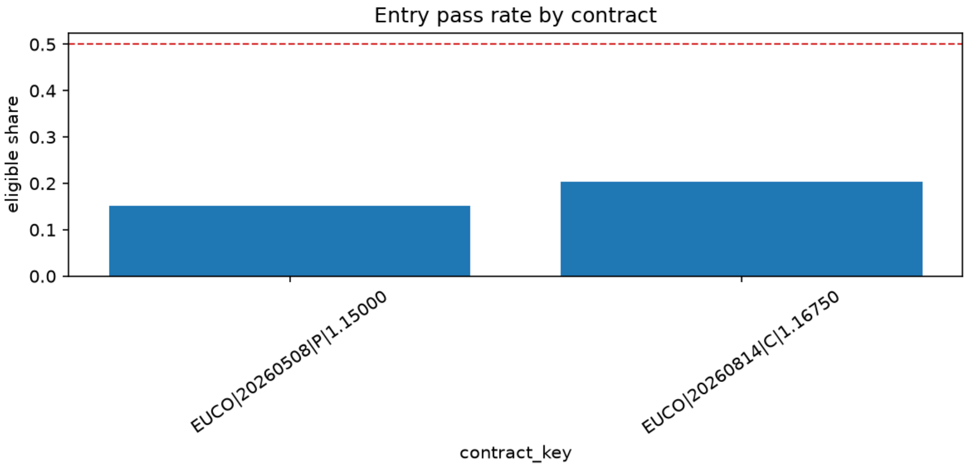
- Method: for each expiry, the call and put at the same strike are overlaid; panels with no matched data are labelled, not dropped.
- Result: only a few near-the-money strikes per expiry have both legs quoted; empty panels show where parity-based checks are impossible. Full-size, readable parts are in all_plots_collection.pdf.

Call/put return paths paired by expiry and strike, not pooled across instruments. Empty panels explicitly mark missing matching data. Quote-driven moves are not executed option-trade returns.

EDA 16: spreads and entry-gate pass rates (development subset)

Sources: outputs\phase06\spread_distribution_bps.png; outputs\phase06\entry_pass_rate_by_contract.png





- Method: relative spread in bps on 12,506 one-minute decision rows of 2 contracts; entry gate = fresh quote + small spread + near-the-money.
- Result: spread p25/p50/p75/p90/p99 = 110/163/224/283/365 bps; 2,220 rows eligible (17.75%): EUCO|20260814|C|1.16750 20.3%; EUCO|20260508|P|1.15000 15.2%.
- Why it matters: a 1.6% median spread is a large hurdle for any short-horizon option signal.

Instrument/session-level exploratory exhibit. Price levels are descriptive; return distributions and paths require spread, stale-quote and sampling checks before strategy interpretation.

EDA conclusions and the hypothesis they produced

Sources: phase06, phase09 and phase10 facts.

Trades are sparse

821 valid trade rows: 817 futures and 4 options; the June future alone has 672. Option analysis must rely on reconstructed quotes.

Hypothesis from EDA

A near-the-money option contract with a fresh quote and a small relative spread should offer a stronger ex ante signal than a stale or wide-spread quote when the futures return exceeds the development-calibrated threshold.

Outcome: rejected

278 trades on development data; net -0.194 price units, win rate 48.9%, bootstrap total-P&L 95% CI [-1.00, 0.57]. Decision: reject - development-only signal failed the Phase 7 hypothesis and the Phase 8 backtest remains negative on dev+validation data.

Pipeline integrity

Phase 10 checks passed: causality, future quote guard, contract fixed while held, round trip cost, missing exit unresolved, real trade examples. The failure is the signal, not look-ahead or accounting errors.

What carried forward

Use returns not levels; robust, class-specific statistics; vol-point or price-unit costs; per-month future mapping; spread-aware gates. These shape the surface, liquidity and time-of-day work that follows.

EDA methods: what the diagnostics do and do not say

Returns

Simple return = P_t/P_previous - 1; log return = ln(P_t/P_previous). Compute within symbol and session, never across instrument rolls or unrelated options.

Distribution and dependence

Mean/median describe location; sample standard deviation describes dispersion. Q-Q plots assess normal-reference tails; ACF at lag k compares r_t with r_(t-k). Neither establishes net trading edge.

Stationarity and transformation

ADF tests a unit-root null under its specified lag/trend settings; a small p-value is not profitability. Fractional differencing trades off persistence and stationarity and needs training-only calibration.

Quote-based limitations

Thin option trading means many observed changes are quotes, not fills. Spreads, cancellations, stale marks and asynchronous observations can distort returns and RV.

Contract mapping: six expiries, six monthly futures

Source: volsurface snapshot summary. Future IDs are historical definitions where the snapshot lacks a current-session definition.

Option expiry	Mapped future / ID	Route	Futures mid	Parity difference (pips)
2026-04-10	FCEU SI 20260413 PS / 434	spread 2026-04/2026-06 vs 2026-06 (quote age 20m)	1.169710	0.602
2026-05-08	FCEU SI 20260518 PS / 435	spread 2026-05/2026-06 vs 2026-06 (quote age 21m)	1.171630	0.755
2026-06-12	FCEU SI 20260615 PS / 436	direct	1.173150	0.393
2026-07-10	FCEU SI 20260713 PS / 437	interp 2026-06..2026-09	1.174452	1.045
2026-08-14	FCEU SI 20260817 PS / 438	interp 2026-06..2026-09	1.176078	1.009
2026-09-11	FCEU SI 20260914 PS / 439	direct	1.177380	0.417

Direct June/September quotes anchor the curve. April/May use calendar spreads; July/August use interpolation. A small put-call parity difference is a consistency check, not proof of fresh executable prices.

Method: why Black-76 and how IV is solved

Exact formulas, reasons and data-availability qualifications are reproduced in the dashboard appendix.

Futures option model

Black-76 uses the mapped forward/future F, strike K, ACT/365 time T and discount factor DF = exp(-rT). This avoids substituting the same spot or one generic future for all expiry months.

Pricing and inversion

d1 = [ln(F/K) + 0.5*sigma^2*T]/(sigma*sqrt(T)); d2 = d1 - sigma*sqrt(T). C = DF*[F*N(d1)-K*N(d2)], with the corresponding put formula. Bisection finds sigma separately for bid, mid and ask.

Explicit assumptions

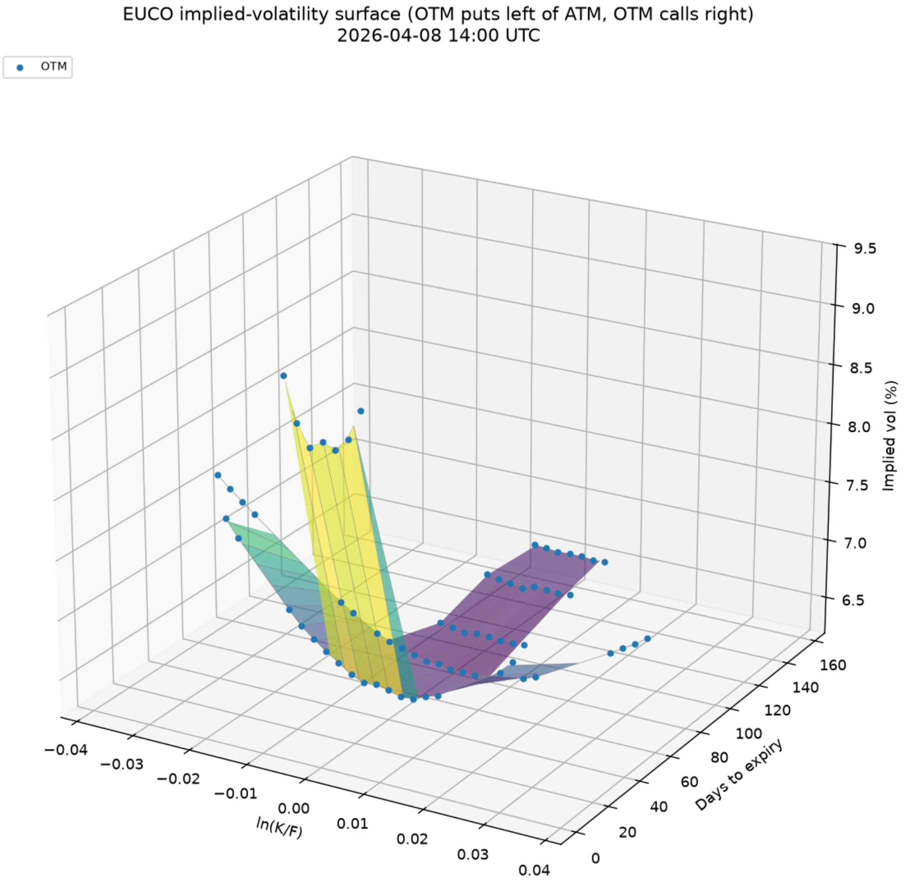
Continuous rate r = 3.5%; expiry time 13:00 UTC. Quotes outside model bounds, invalid books and insufficient strike coverage stay missing rather than receiving invented IVs.

Observed points versus visual surface

Both ITM and OTM options are retained. OTM puts below F and OTM calls above F give the canonical smile; interpolation between observations is visualization, not an observed executable quote.

Volatility surface: moneyness, maturity and IV

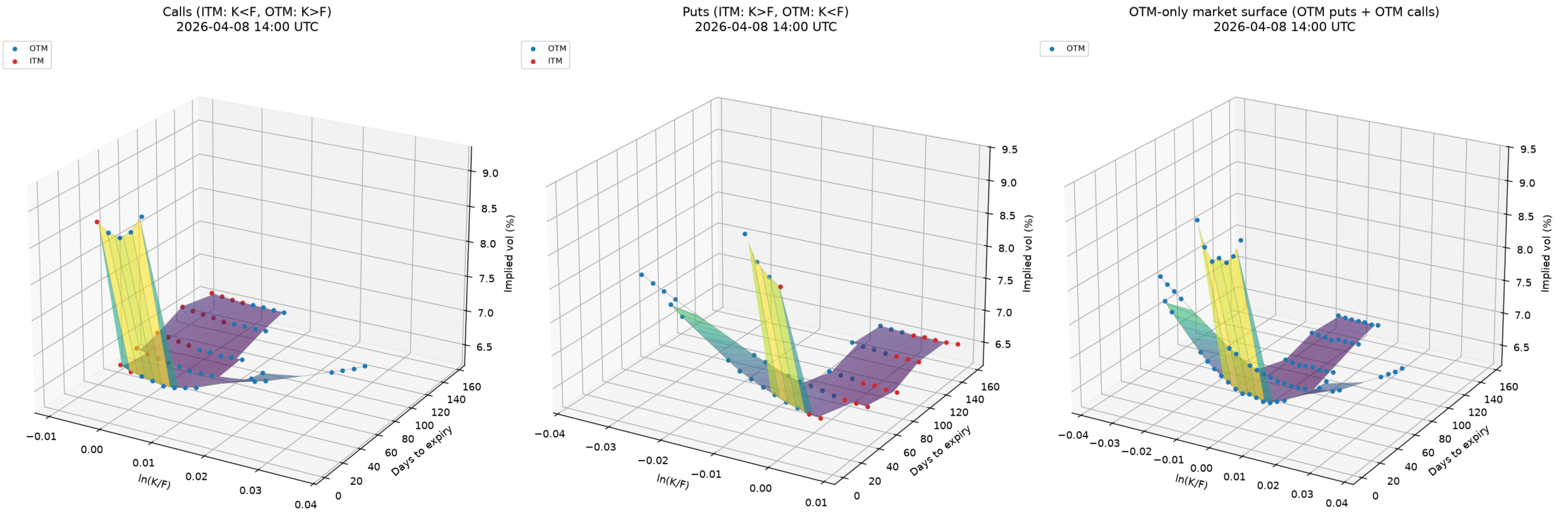
Source: outputs\volsurface\2026-04-08_1400\00_iv_surface_main.png



2026-04-08 14:00 UTC snapshot. ITM and OTM observations are retained in the dataset; the main fitted surface uses OTM wings to avoid duplicate/noisy parity information. Interpolation is not a new market quote.

Calls, puts and coverage in three dimensions

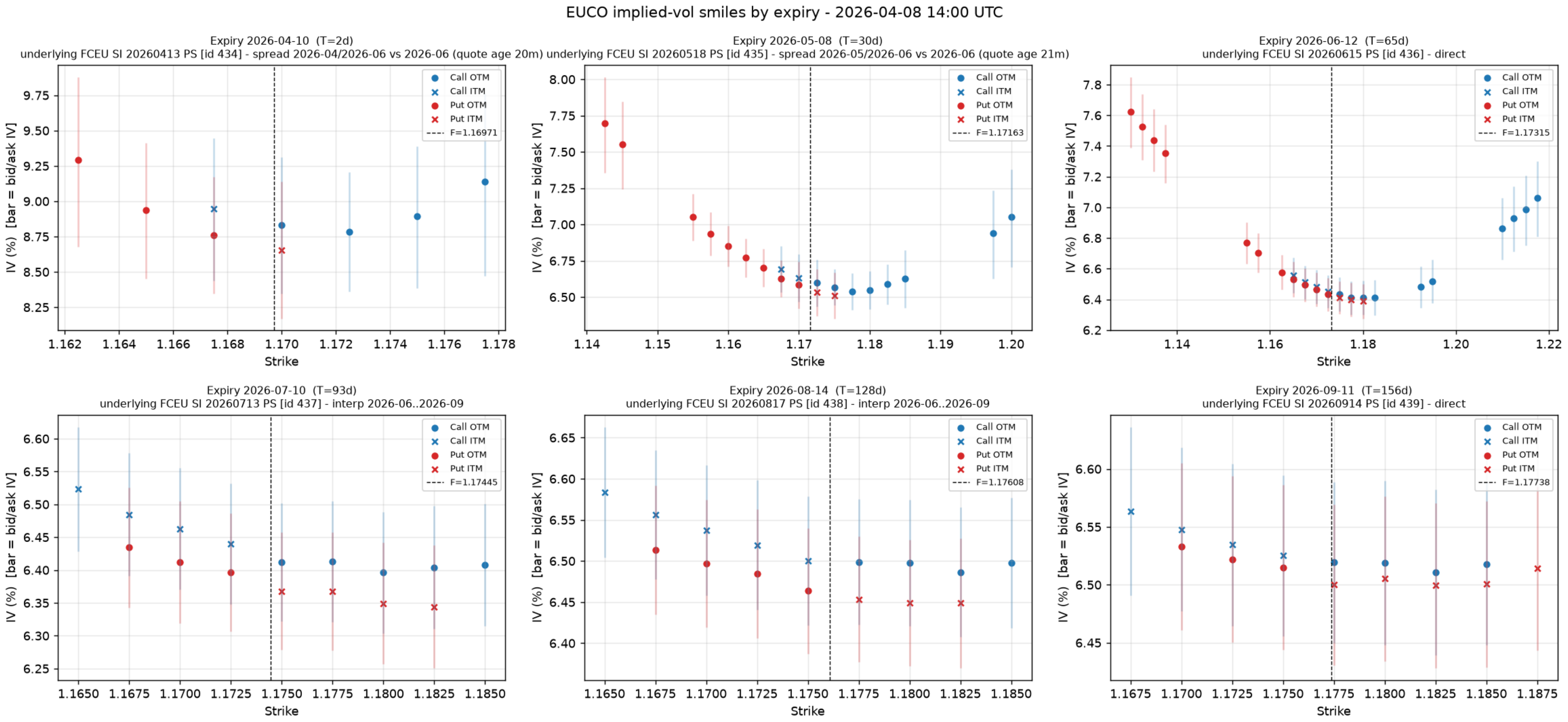
Source: outputs\volsurface\2026-04-08_1400\01_iv_surface_3d.png



2026-04-08 14:00 UTC snapshot. ITM and OTM observations are retained in the dataset; the main fitted surface uses OTM wings to avoid duplicate/noisy parity information. Interpolation is not a new market quote.

Smile by expiry: compare like with like

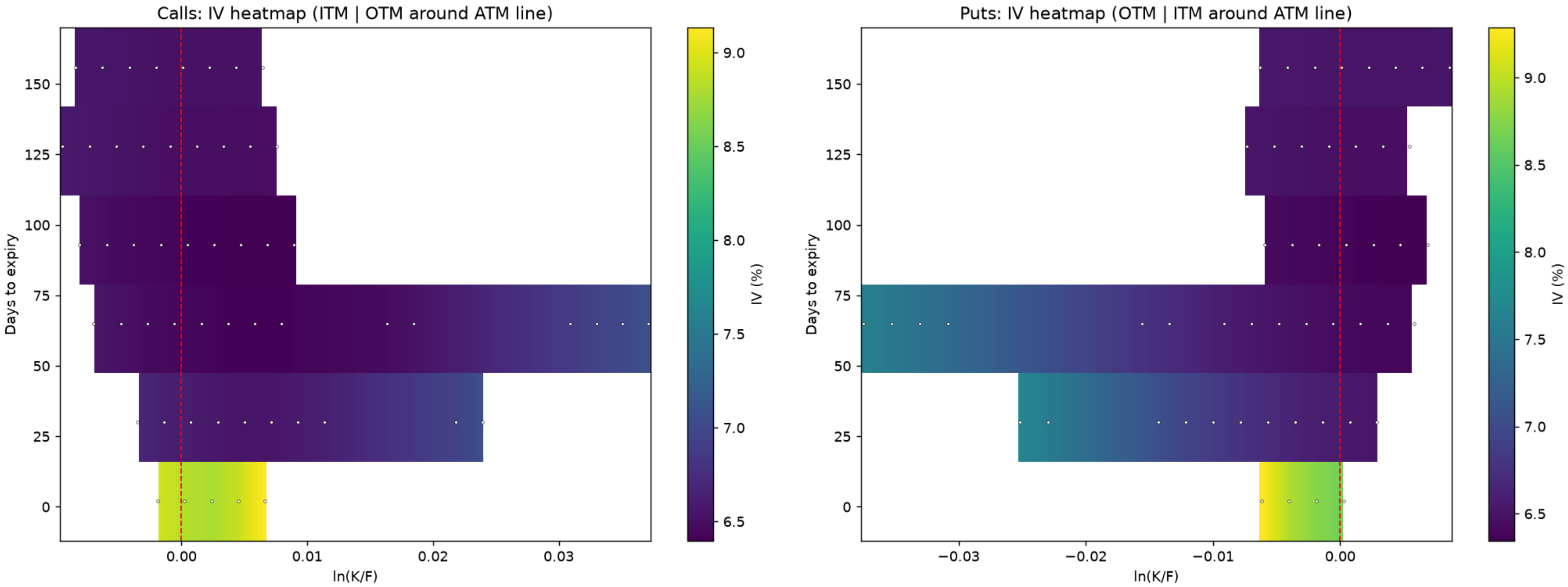
Source: outputs\volsurface\2026-04-08_1400\02_iv_smiles_by_expiry.png



2026-04-08 14:00 UTC snapshot. ITM and OTM observations are retained in the dataset; the main fitted surface uses OTM wings to avoid duplicate/noisy parity information. Interpolation is not a new market quote.

Heatmap: locate observed and interpolated regions

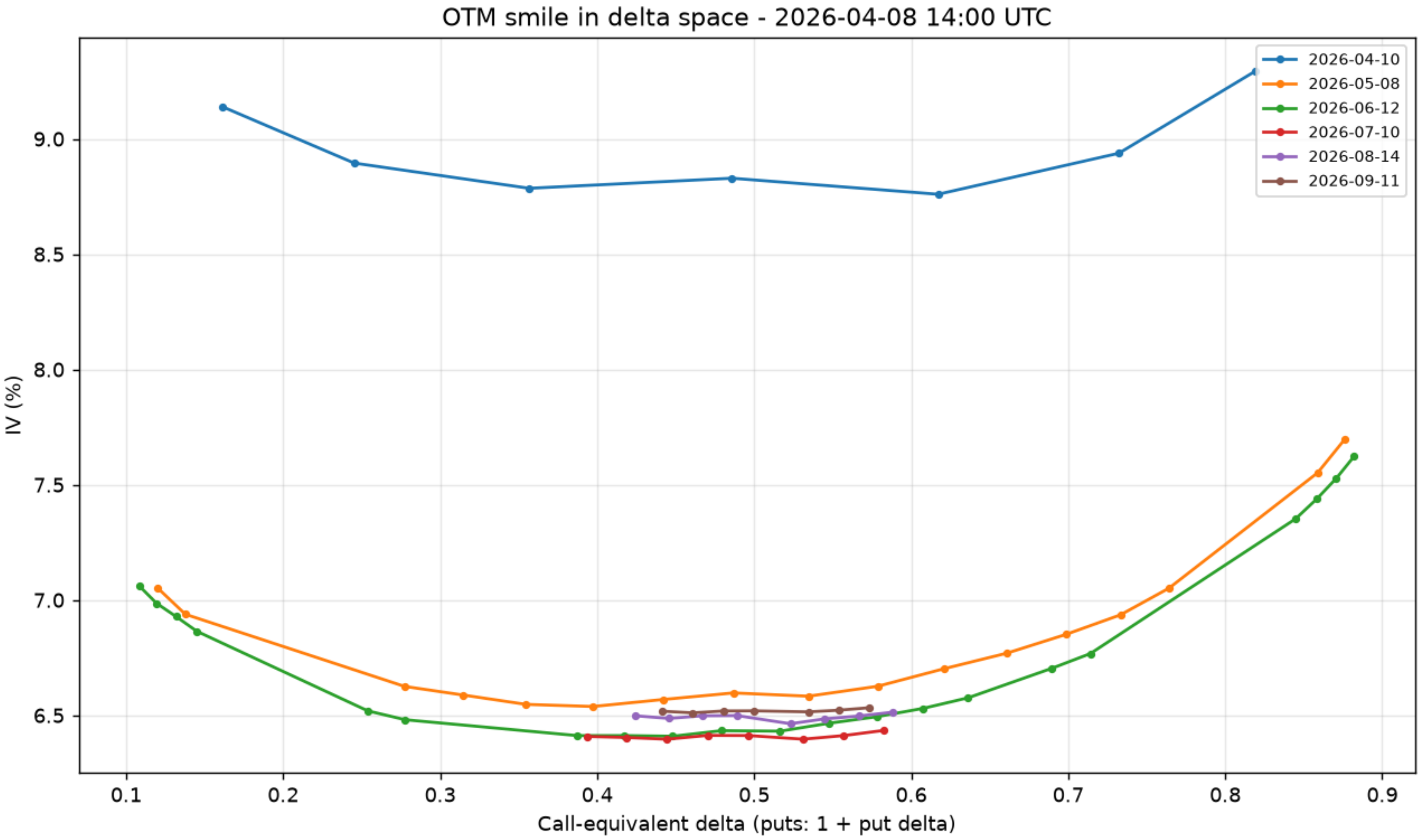
Source: outputs\volsurface\2026-04-08_1400\03_iv_heatmap.png



2026-04-08 14:00 UTC snapshot. ITM and OTM observations are retained in the dataset; the main fitted surface uses OTM wings to avoid duplicate/noisy parity information. Interpolation is not a new market quote.

Delta-space smile: comparable wing exposure

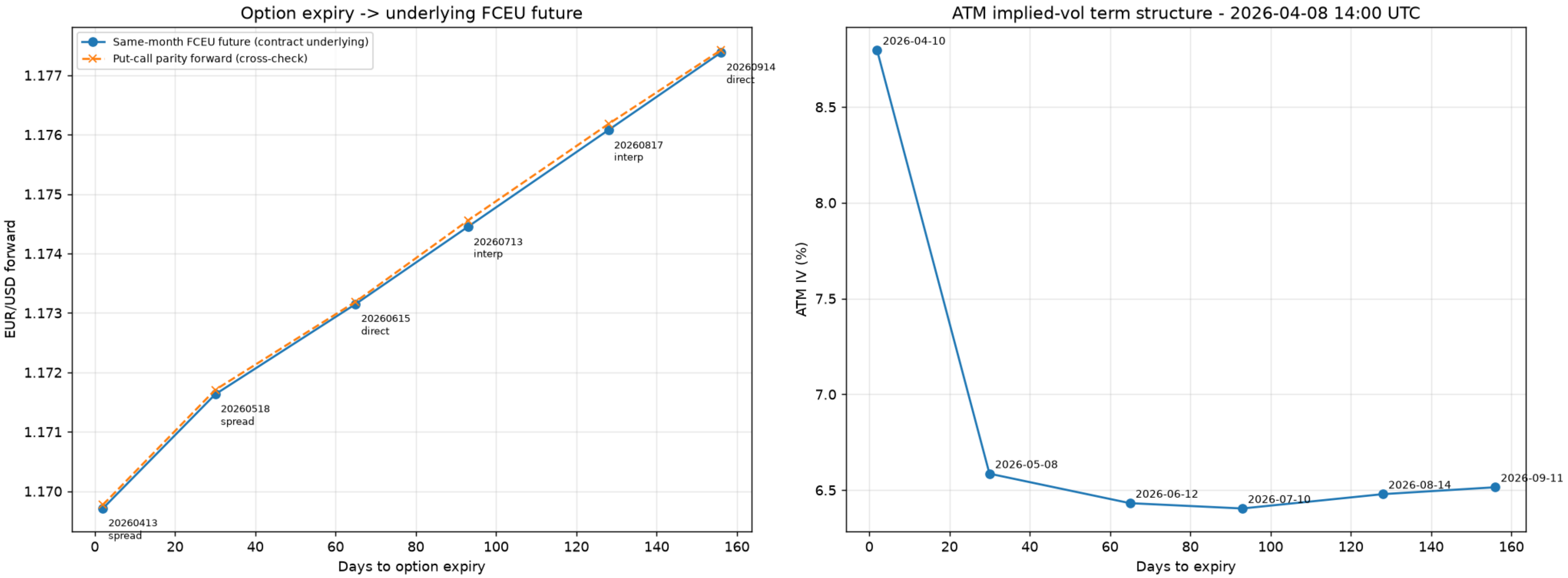
Source: outputs\volsurface\2026-04-08_1400\04_otm_smile_delta_space.png



2026-04-08 14:00 UTC snapshot. ITM and OTM observations are retained in the dataset; the main fitted surface uses OTM wings to avoid duplicate/noisy parity information. Interpolation is not a new market quote.

Underlying mapping and ATM term profile

Source: outputs\volsurface\2026-04-08_1400\05_underlying_map_and_atm_term.png



2026-04-08 14:00 UTC snapshot. ITM and OTM observations are retained in the dataset; the main fitted surface uses OTM wings to avoid duplicate/noisy parity information. Interpolation is not a new market quote.

Reading ATM, risk reversal and butterfly

ATM level

atm_iv interpolates OTM mid IV at K=F. A decimal 0.0731 means 7.31% annualized. A change of 0.001 is 0.10 volatility points, not 0.001%.

Risk reversal: directional skew

RR25 = 25-delta call IV - 25-delta put IV. Negative means puts are richer. Wing selection uses absolute undiscounted forward delta abs(delta)/DF, not days-to-expiry.

Butterfly: curvature

BF25 = (25-delta call IV + 25-delta put IV)/2 - ATM IV. Positive means the wings are richer than ATM. This is an IV shape statistic, not the cash price of an executable butterfly.

Coverage discipline

No extrapolation to unavailable 10D/25D wings. NaN means unavailable, not zero skew. A rich put alone is neither a mean-reversion forecast nor a sell recommendation.

Snapshot results: ATM level, skew and curvature

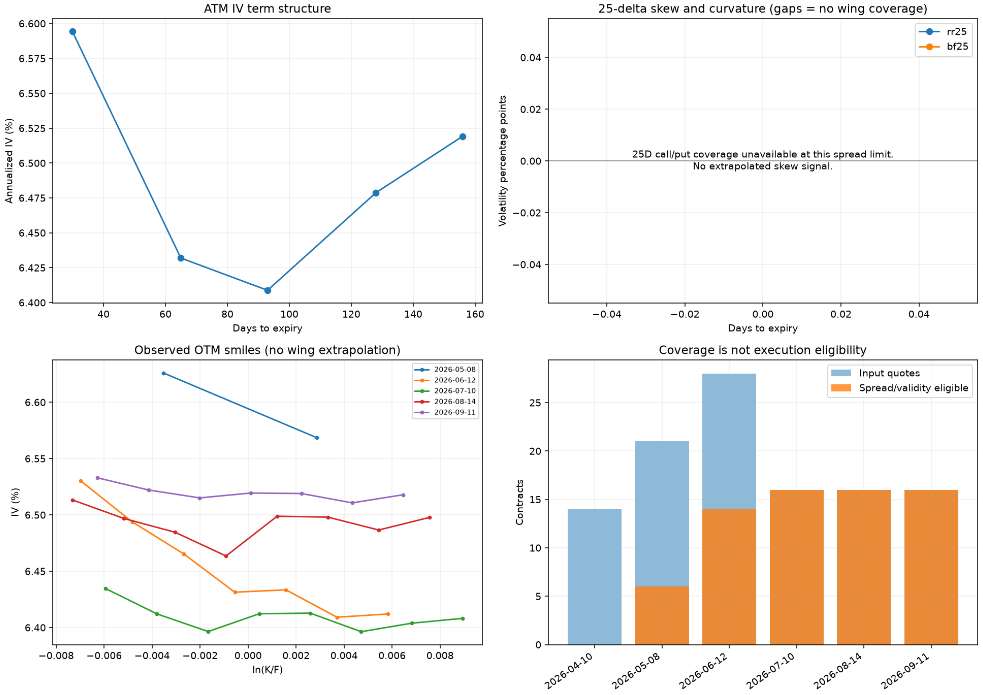
Source: surface_framework_research/surface_metrics.csv. ATM is percent; RR/BF are volatility points, explicitly converted from stored decimals.

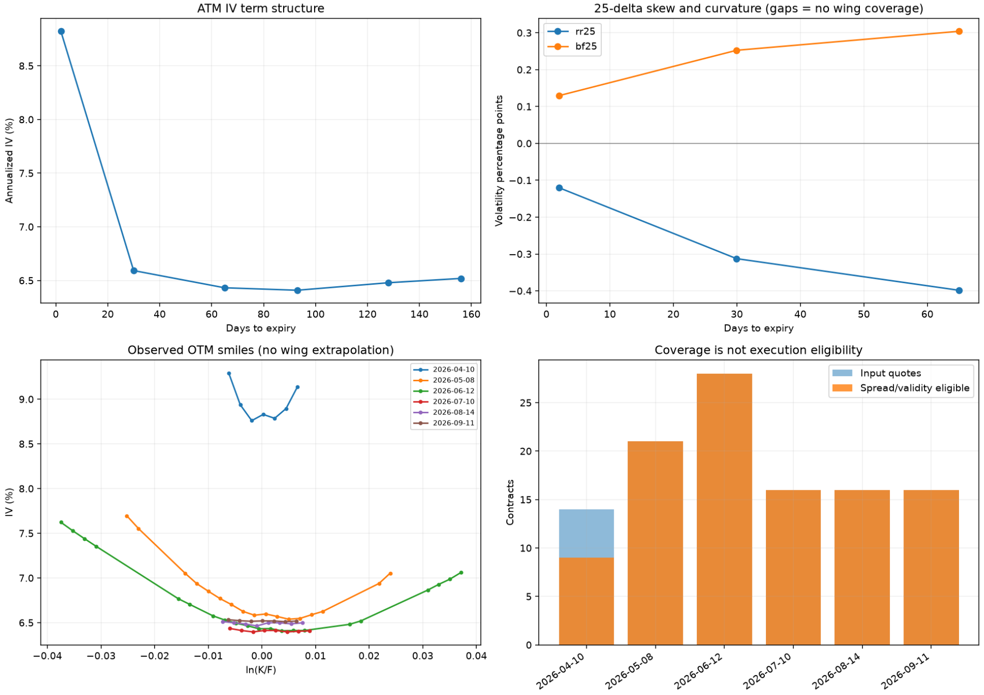
Expiry	ATM IV (%)	RR25 (vp)	BF25 (vp)	Eligible / inputs
2026-04-10	8.821	-0.120	0.129	9 / 14
2026-05-08	6.592	-0.313	0.252	21 / 21
2026-06-12	6.432	-0.398	0.304	28 / 28
2026-07-10	6.409	Missing	Missing	16 / 16
2026-08-14	6.479	Missing	Missing	16 / 16
2026-09-11	6.519	Missing	Missing	16 / 16

The near April expiry has the highest observed ATM IV. April-June skew is put-rich; later 25-delta coverage is absent. Across different monthly futures, inverted IV or negative forward variance is a research flag, not proof of calendar arbitrage.

Dashboards 1 and 2: same snapshot, two quote-quality gates

Sources: outputs\surface_framework\dashboard.png; outputs\surface_framework_research\dashboard.png





- Left - dashboard 1 (conservative): 5% gate: 68/111 eligible, 0 price alerts, 9 candidate records.
- Right - dashboard 2 (research): 50% gate: 106/111 eligible, 0 price alerts, 17 candidate records.

A wider analytical universe improves coverage but does not relax paper-entry safeguards. Full tables of both dashboards are in the appendix; dashboard 2 is shown there as differences from dashboard 1.

Liquidity: cost is part of the hypothesis

Source: surface_framework_research/liquidity_by_expiry.csv. Relative spread is percent of premium; volatility spread is in vol points.

Expiry	Relative spread (%)	IV spread (vp)	Tight / valid	Bid / ask size
2026-04-10	29.41	1.104	0/14	280/280
2026-05-08	6.32	0.320	6/21	420/420
2026-06-12	4.88	0.227	14/28	560/560
2026-07-10	2.80	0.185	16/16	320/320
2026-08-14	2.39	0.157	16/16	320/320
2026-09-11	2.09	0.142	16/16	320/320

Displayed size is symmetric in this snapshot: median imbalance is zero. April's wide volatility spread can consume apparent edge. Depth is cancellable resting liquidity, not inventory or open interest.

Liquidity methods and implementation choices

Compare costs in the right units

mid=(bid+ask)/2; relative spread=(ask-bid)/mid. IV spread=100*(IV_ask-IV_bid), with price-spread/vega fallback. This makes costs comparable across strikes and maturities.

Top-of-book execution overlays

Imbalance=(Q_bid-Q_ask)/(Q_bid+Q_ask); microprice=(bid*Q_ask+ask*Q_bid)/(Q_bid+Q_ask). Here symmetric depth carries little timing information; it can change or cancel.

Role is not permission

TIGHT means relative spread <=5%. Wider quotes suggest patient/passive research, not assumed fills. Queue position, adverse selection and multi-leg fill risk need explicit modeling.

Identity and freshness

Age=max(snapshot-bid_ts, snapshot-ask_ts). Historical future IDs, stale calendar spreads and interpolated forwards are research provenance flags; execution gates remain stricter.

Strategy framework: from observation to paper position

Surface opportunities

RR/BF mean reversion needs a prior-only same-expiry distribution and observed wings. Calendar/diagonal structures need month-by-month exposure accounting. Price alerts compare executable bid/ask packages after configured costs.

Entry cost and hedge

Entry debit=sum q_i*(ask for buys, bid for sells)+sum |q_i|*(fees+slippage). Matching-month futures hedge H=-sum q_i*delta_i; never net unrelated monthly exposures as if basis risk vanished.

Scenarios are not realized profit

Reprice Black-76 under forward, IV and elapsed-time shocks. This is model-mark P&L relative to starting mids, not realized P&L, and does not include futures hedge P&L.

Risk gates

Require causal decision/submission/fill times, displayed size, bounded-risk budget, current underlying identity, freshness and synchronized legs. No automatic live routing or naked short-volatility deployment.

Statistical methods: define the sampling unit

Prior-only z-score

z=(current-prior mean)/prior sample SD (ddof=1), same expiry and strictly earlier timestamps only. Insufficient history or zero variance gives missing z, not a neutral signal.

Time-of-day t-like ratios

Mean change / (sample SD/sqrt(session count)). The existing drift table uses session-row counts, potentially exceeding nonmissing changes. |t|>2 is exploratory, without multiplicity or dependence correction.

Robustness rather than one event

Use medians across sessions; report the largest session's share of slot RV; compare rank profiles across chronological halves and one-minute versus five-minute sampling.

Counts are not interchangeable

One quote, one snapshot, one expiry-gap and one session are different units. Multiple expiries across one weekend do not create multiple independent weekends. Missing slots can bias comparisons.

Time-of-day method and clock correction

Source: time_of_day/summary.json and time_of_day.py.

Panel construction

22 sessions; 353 priced option snapshots every 30 minutes from 08:30 to 18:00 Berlin; 87 failed snapshots. All observed expiry/strike instruments use mapped underlying routes.

Realized variance reference

RV=sum of squared log changes in two-sided June-future minute mids; gaps >5 minutes are excluded. Results are reported in Berlin and UTC slots. This is one June underlying proxy, not separate RV validation for every monthly future.

Time-zone issue is material

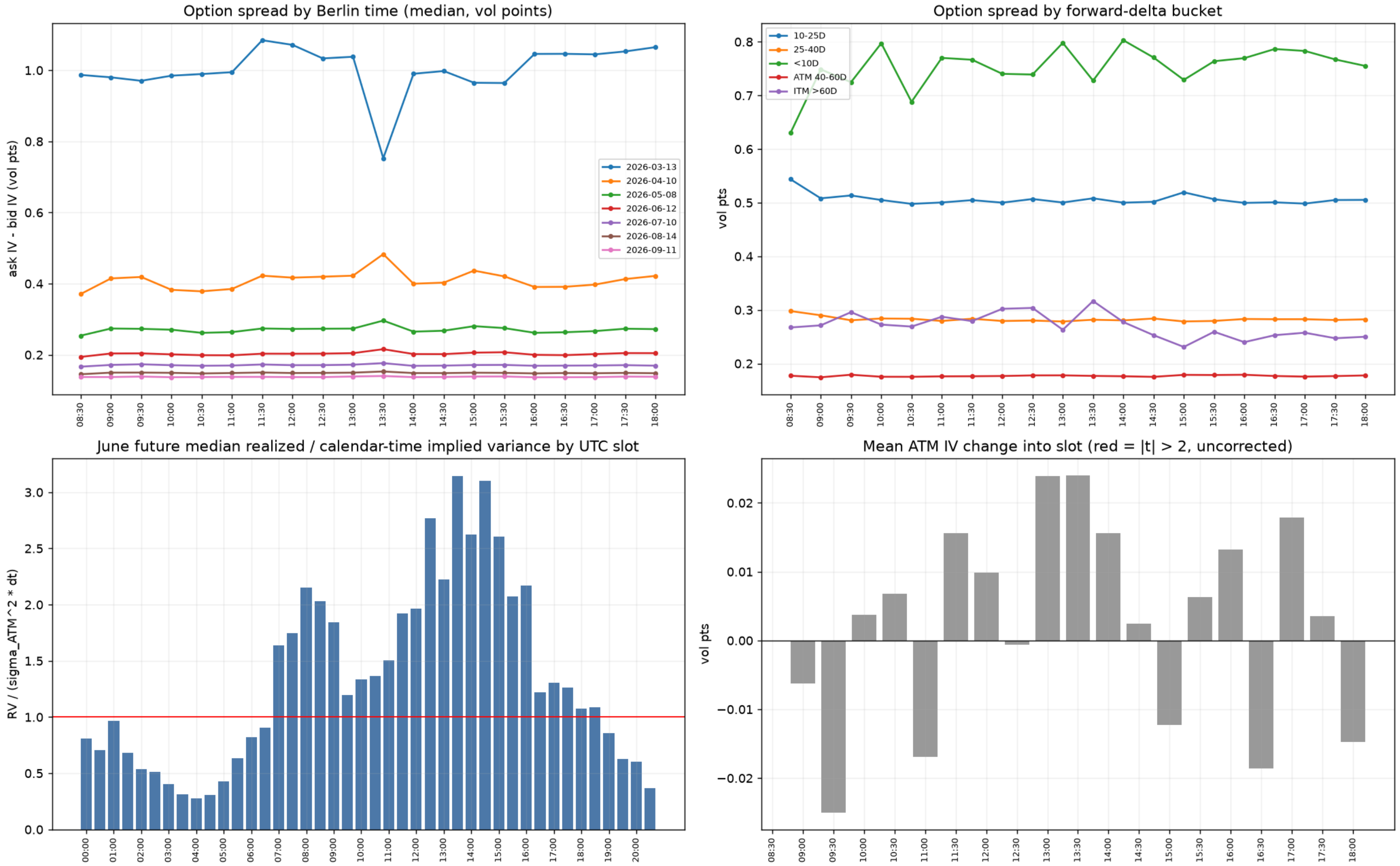
Phase04 timestamps are 19800 seconds early from naive UTC interpreted as IST. This study measures and corrects the offset; frozen phase04/phase05 outputs remain unchanged.

Calendar versus trading time

Calendar slot implied variance=sigma_ATM^2*slot_minutes/525600. Trading-day implied variance=sigma_ATM^2/365*7/5, assuming negligible weekend RV. A 252-day realized annualization is a different convention.

Dashboard 3: time-of-day liquidity, RV and IV drift

Source: outputs\time_of_day\time_of_day.png



Berlin option slots and UTC realized-variance slots. RV timing is stable in this sample, but option spread timing is not; gamma/theta edges exclude hedge costs.

Time-of-day findings: movement versus crossing cost

Source: time_of_day/summary.json. vp = volatility points of vega, not currency profit.

Measure	Observed value	Interpretation
June reference ATM IV	7.31%	Panel median; not the April-08 snapshot ATM level.
Daily RV / implied (1m / 5m)	1.08 / 0.99	Five-minute check removes much of the apparent excess; microstructure matters.
RV rank / liquidity rank stability	0.85 / -0.34	Movement ranking persists; spread ranking does not.
RV in option-quoting hours	68.1% in 39.6% of clock	Normalized median slot profile, not average daily share.
Option trade messages	6	Too sparse to support order-flow strategy inference.

Gamma/theta break-even: the spread wins

Source: time_of_day summary. Simplified June ATM model using five-minute RV; futures hedge costs, fees and fill uncertainty are excluded.

Holding schedule	Gross (vp)	Option spread (vp)	Net (vp)
Intraday only: buy 08:30, sell 18:00 Berlin, every day	+0.025	0.197	-0.172
Weekday only: buy Mon 08:30, sell Fri 18:00 (flat weekend)	+0.096	0.197	-0.101
Hold continuously for one week (turnover cost excluded)	-0.002	0.000	-0.002

theta/day in vega-normalized vol points = 100*sigma/(2*T*365). Gross edge = theta/day * [window RV/(sigma^2/365) - days charged]. Net here subtracts only the stated option spread. It is an approximation, not a delta-hedged trading backtest.

Opportunity verdicts and what to test next

Hedge overlay: candidate for validation

RV clusters around 12:30-15:30 UTC and the European morning. Test adaptive re-hedge bands with futures spread/fees, latency and turnover, rather than claiming the timing profile itself is alpha.

No established options entry-time edge

ATM spread is about 0.18 vol points across slots; rank ordering is unstable. No ATM drift slot has |t|>2 in the stored study. Day-only and weekday-only gamma estimates are negative after the option spread.

Weekend observation is small-sample

Mean overnight ATM change: +0.033 vp for weeknights versus +0.231 vp for weekend/holiday gaps. The latter is 23 expiry-gap rows but only 4 distinct observed date gaps, not independent weekend trades.

Promotion criteria

Obtain more independent sessions and trade/fill data; resolve clock/mapping provenance; pre-register target, fees and hedging rule; validate chronologically with overlap-aware splits and report all-in uncertainty.

SHAP: not computed does not mean zero

No fitted predictive model, prediction target or background/reference dataset is used by these dashboards. Consequently every metric here has SHAP status: not computed / not applicable, NOT zero. Black-76 pricing, historical z-scores, t-statistics, scenario shocks, rule-based alerts and slot rank correlations are not fitted strategy-outcome models. None establishes feature importance or profitability. SHAP can also explain a deterministic function such as Black-76 if its output, input features and reference distribution are specified, but that would attribute a modeled option price, not strategy profitability. No such attribution experiment is defined here.

SHAP (SHapley Additive exPlanations) attributes a specified model output for one observation to its input features relative to a reference population. In output units: f(x) = E_background[f(X)] + sum_j phi_j(x). For a feature set M: phi_j = sum over S subset M excluding j of [|S|! (|M|-|S|-1)! / |M|!] * [v_x(S union {j}) - v_x(S)], where v_x(S) is the model output averaged over the chosen missing-feature reference distribution. Conditional and interventional reference choices need not produce the same attribution.

Positive SHAP pushes that model prediction above its baseline; negative pushes it below. Global importance often means mean absolute SHAP over a specified evaluation set, not a coefficient, a significance test, a causal effect or a strategy return. Correlated inputs (ATM IV, total variance, RV ratios, RR/BF) make attribution dependent on the model and reference assumptions.

To compute defensible values, first choose a target and horizon (for example next-window realized variance or executable delta-hedged P&L after all costs), construct features available strictly at decision time, fit and validate a model with chronological splits and overlapping-label purging, freeze training-only reference data, and explain held-out predictions in explicit output units. Whole-session centered IV and full-sample summaries here are retrospective, not valid live features. Only 22 sessions and six option trade messages do not establish a reliable P&L prediction model. No model or SHAP numbers have been fabricated.

Limitations and project status

Data and model risk

Sparse option trades; cancellable top-of-book depth; historical future definitions; calendar-spread staleness and interpolated monthly forwards; risk-free-rate and expiry-time assumptions.

Statistical risk

Only 22 sessions; missing early/event-time snapshots; dependence across strikes/expiries; event-day outliers; no multiple-testing correction. Full-sample descriptive statistics are not decision-time features.

Legacy workflow is not repaired here

The original momentum decision remains rejected as recorded. Clock-time-dependent phase04/phase05 and related results need revalidation after the known time-zone bug; this PDF does not rerun or certify those frozen experiments.

Scope

This report includes every table and narrative section from 3 saved HTML dashboards; identical tables are cross-referenced and near-identical dashboard-2 tables are shown as differences. The companion collection shows each of the 39 unique saved figures once; reviewed duplicate PNGs are listed in the manifest.

Dashboard appendix 1: surface_framework

Source HTML: outputs\surface_framework\dashboard.html

Saved narrative and data tables follow. Identical tables are cross-referenced; tables sharing a section with an earlier dashboard are shown as differences only. Wide tables with few rows are transposed; constant columns are stated once in the subtitle.

Volatility surface research

As of 2026-04-08T14:00:00+00:00

68 / 111 quotes pass analytical quality. 0 price alerts. No live orders; all amounts are price units.

Liquidity-based strategies (research framework)

Not available: open-interest or dealer-position based liquidity (GEX/DEX, walls). Displayed size is resting liquidity, not inventory.

Interpretation and safeguards

RR25 = 25D call IV minus 25D put IV; negative means puts are richer. BF25 = average wing IV minus ATM IV. No extrapolation into absent wings.

ATM is interpolated at K=F from observed OTM quotes. Delta uses undiscounted Black forward convention.

Expensive puts or inverted IV term structure are not sufficient trading signals. Different option expiries reference different monthly futures.

PRICE_ALERT compares executable bid/ask packages after configured costs; it is not a fill, a verified arbitrage or a profit forecast. Freshness, depth, timing and risk gates apply at paper entry.

WATCH/RESEARCH_SIGNAL structures are not automatically tradable. Naked short volatility, dealer exposures and live routing are not enabled.

GEX, DEX and dealer-position volatility walls are unavailable: the source has no OI or signed positions. Displayed size is liquidity, not inventory.

Scenarios are model mark changes, not realized P&L. Hedge ratios are separate per underlying future month; no assumed contract multiplier or currency capital.

History uses strictly earlier same-expiry observations. A single snapshot has no mean-reversion evidence. Thresholds require development-only calibration and holdout validation.

Formulas used

Availability key: AVAILABLE: observed in the data, PARTIAL: observed with gaps or proxies, ASSUMED: user or convention parameter, not in the data, MISSING: not computable from the data

Reviewer addendum: notes on pages 48-92

Added 2026-11-01 after the report was generated. Original pages are unchanged; notes are inserted directly after the page they refer to and are labelled with that page number plus a letter.

Overall verdict. The report's conclusion stands: no strategy shows a cost-adjusted edge in this sample. The only actionable result is a hedge-timing overlay that lowers hedging cost for an existing delta-hedged book; it still needs a futures-cost backtest.

Where the notes are

Note	Original page(s)	Topic
47b	all	Full reviewer analysis (complete text)
48a, 49a, 51a	48-51 D1-T1..T3	Surface snapshot, forward volatility, prior-only signals never exercised
54a	52-54 D1-T4	PROFITABILITY VERDICT for each D1 strategy candidate
57a	55-57 D1-T5..T7	PROFITABILITY VERDICT for each liquidity strategy
61a	58-61 D1-T9	Assumed parameters
70a	62-70 D2	PROFITABILITY VERDICT for each D2 strategy candidate
71a, 73a	71-73 D3-T1/T2	Clock-bug scope; PROFITABILITY VERDICT for each time-of-day strategy
90a, 91a, 92a	82-92 D3-T5..T7	RV profile and New York time anchoring, weekday table, weekend sample
92b, 92c, 92d	after 92	Scorecard (every strategy as a position), what else can be done, conclusions

Corrections to the earlier summary of these pages

- "Median half-spread ~1.84 bp" does not appear in the report. Actual: 163 bps median relative spread on the development subset (page 22); 2.1-29.4% of premium per expiry on the snapshot (page 55).
- Surface strategies were not "obliterated by costs": every candidate has edge_price_units = None and there are 0 PRICE_ALERTs. No edge was ever established.
- Liquidity does not rule out hedged structures across the curve: July-September quotes are 16/16 TIGHT. Only the April front week is untradeable.
- Prior-only z-scores are the right design but never ran (all insufficient_history). "Truncation tests" are not in the report.
- The 19,800 s clock fix applies only to the time-of-day study; phase04/phase05 results remain uncertified (page 46).
- "Shift focus to passive market making" overstates the report: passive quoting is research-only and covers 9 April quotes.

Reviewer analysis: full report review

The complete written analysis, including the per-strategy verdicts, scorecard and further ideas. Tables in later notes are taken from this analysis.

Based on a full review of all 101 pages of report.pdf ("Options Research: Methods, Metrics and Dashboard Analysis", EUCO options on FCEU EUR/USD futures), here is an institutional-grade evaluation of the data, methods and options strategies presented.

(Note: The report explicitly labels itself as "Descriptive research; no live orders" and operates under the core principle that "measurement is not alpha." Dataset: 35,045,335 MBO records, 22 sessions from 2026-03-09 to 2026-04-08; surface snapshot 2026-04-08 14:00 UTC.)

Executive Summary: Strategy Verdicts

None of the evaluated strategies demonstrate a viable, cost-adjusted edge for live deployment. The evidence falls into three groups:

- **Tested and failed after costs:** the original EDA-led momentum signal (net -0.194 price units over 278 development trades) and the time-of-day long-gamma schedules, where crossing the option bid-ask spread (~0.197 vp round trip) exceeds the gross gamma/theta edge.
- **Never shown to have an edge:** the surface candidates (straddles, calendars, risk reversals, curvature trades). All 17 are status WATCH with edge_price_units = None, and the snapshot produced **0 price alerts**. Costs did not eliminate an edge here; no edge was established in the first place.
- **Only actionable finding:** a **hedge-timing overlay** that lowers path-dependent hedging cost for an *existing* delta-hedged book. It is a cost reduction, not standalone alpha, and still needs validation with futures spreads and fees.

The closest-to-viable idea is weekday-only long gamma, which would break even with roughly 16% higher realized volatility or a halved option spread, but it rests on only 3-4 weekends (see Section 6).

Per-strategy profitability (reviewer calculation, Section 4a and the scorecard in Section 9): every candidate was repriced leg by leg with Black-76 (the model reproduces the report's own Greeks), costed at the half-spread of each expiry, and run under the sample's trailing realized vol. Calendars, risk reversals, curvature trades and the April straddle are **not profitable** at any plausible realized vol. The May-September long straddles are **conditionally profitable**: delta-hedged to expiry they make roughly +7% to +11% of premium *if* realized vol repeats the sample's 7.27%, but lose about 0.0006-0.0008 price units if realized vol equals implied. That result exists only because implied vol on 8 April (6.4-6.6%) had fallen below trailing realized. It is a hypothesis to test, not a proven edge.

1. Data, Instrument Mapping and Pricing Method (pages 1-26)

- **Feed composition:** 17,515,881 adds, 17,516,289 cancels, 11,249 resets, 996 trades, 867 fills, 53 modifies. Almost all information is quote revision, not trading. 1,056 raw symbols: 985 options, 21 futures, 50 spreads/unparsed. Instrument IDs are reused across sessions, so joins must use session_date + instrument_id.
- **Trade sparsity:** 821 valid trade rows, of which 817 are futures and only 4 are options (the time-of-day study counts 6 option trade *messages* under a different definition). Option analysis therefore relies almost entirely on reconstructed quotes.
- **Contract mapping:** each option expiry is priced against its own monthly future. June and September are direct quotes; April and May are reconstructed via calendar spreads (quotes ~20-21 min old); July and August are time-interpolated between June and September. Put-call parity agrees within ~0.4-1.0 pips after remapping (the original single-quarterly-future mapping was 15-35 pips off).
- **Pricing:** Black-76 with assumed r = 3.5% and 13:00 UTC expiry (neither is in the data); IV solved by bisection separately for bid, mid and ask. Quotes outside no-arbitrage bounds are left missing rather than forced.

2. EDA and the Rejected Momentum Hypothesis (pages 5-24)

- **Statistical properties:** price levels are non-stationary (ADF p = 0.29-0.32); log returns are stationary (ADF ≈ -26 to -27). Returns have a large spike at zero and fat tails; options show far heavier tails than futures because a one-tick move on a tiny premium is a huge percentage (unclipped simple returns range -44 to +49).
- **Autocorrelation:** all |ACF| ≤ 0.087 (lag 3 on the 1-minute grid), consistent with bid/ask bounce. Statistically detectable but far too small to pay a 1-2 tick spread.

- **Spreads (development subset, 2 contracts):** relative spread p25/p50/p75/p90/p99 = 110/163/224/283/365 bps. Only 17.75% of decision rows passed the entry gate.
- **Hypothesis and outcome:** "a near-the-money option with a fresh quote and small spread gives a stronger signal when the futures return exceeds a calibrated threshold." **Rejected:** 278 development trades, net -0.194 price units, win rate 48.9%, bootstrap total-P&L 95% CI [-1.00, 0.57]; still negative on dev+validation. Phase 10 integrity checks (causality, future-quote guard, round-trip cost) passed, so the failure lies in the signal rather than in the code.

3. Volatility Surface Snapshot (2026-04-08 14:00 UTC)

Expiry	DTE	ATM IV	RR25 (vp)	BF25 (vp)	Rel. spread	IV spread (vp)	Tight / valid
2026-04-10	2.0	8.82%	-0.12	0.13	29.4%	1.10	0/14
2026-05-08	30	6.59%	-0.31	0.25	6.3%	0.32	6/21
2026-06-12	65	6.43%	-0.40	0.30	4.9%	0.23	14/28
2026-07-10	93	6.41%	n/a	n/a	2.8%	0.19	16/16
2026-08-14	128	6.48%	n/a	n/a	2.4%	0.16	16/16
2026-09-11	156	6.52%	n/a	n/a	2.1%	0.14	16/16

(RR25/BF25 and April ATM come from the 50% research gate; under the 5% conservative gate, April has no eligible quotes and no 25-delta coverage exists for any expiry.)

- **Skew:** puts are richer than calls, and the put premium and wing curvature both grow with maturity from April to June. July-September strikes do not reach 25-delta, so skew there is unobserved (NaN means unavailable, not zero).
- **Term structure:** the front week is elevated (8.82%, but only 2 days of variance), followed by a shallow dip and a gentle upslope. ATM forward vols are all positive and rising (6.29% May→Jun, 6.35% Jun→Jul, 6.66% Jul→Aug, 6.70% Aug→Sep), so there is no negative forward variance. Every pair references different monthly futures, so the term structure is a research diagnostic, not a calendar-arbitrage test.
- **Quality gates:** conservative 5% gate admits 68/111 quotes (9 candidates); research 50% gate admits 106/111 (17 candidates). Neither gate shows that a multi-leg package can fill.

4. Surface-Based Strategy Candidates (Straddles, Calendars, Risk Reversals)

The dashboards generate candidates across expiries April-September 2026. **All are WATCH status,** edge_price_units = None**, and there are 0 PRICE_ALERTs.**

- **Long Straddles** (entry debit 0.006-0.040 price units): require a horizon-matched realized-volatility forecast exceeding IV plus costs. The report explicitly states this forecast is "not supplied," so there is no measured edge to justify the debit.
- **Calendar Calls:** flagged for **basis/roll risk**: "front-rich IV alone is not calendar arbitrage." Front and back legs settle against different futures (calendar-spread-reconstructed, direct or interpolated), so hedges cannot be netted across months.
- **25-Delta Risk Reversals and Wing-vs-ATM Curvature:** need a prior-only same-expiry RR/BF history that does not yet exist; April candidates also carry near-expiry risk. "Expensive puts alone are not a sell signal."
- **Risk profile:** the April straddle carries gamma of ~106 against a 0.006 debit and theta of -0.0015/day, an extremely short-dated exposure in the widest market (29% spreads). Back-month straddles (July-September) are far more benign (gamma 16-21, vega 0.0047-0.0060 per vol point) and sit in tight markets, so for longer-dated structures liquidity is not the binding constraint. The missing realized-vol forecast is.

4a. Per-Strategy Profitability Verdicts (reviewer calculation)

Method: each candidate's legs were repriced with Black-76 at the snapshot (ATM IV for near-ATM legs, 25-delta IVs for wing legs); the recomputed Greeks match the report's tables. Cost = half the median IV spread of each expiry × leg vega, paid on entry and on exit (legs that expire are settled, no exit cost). Delta-hedged carry = theta × calendar days + ½ × gamma × $F^2 \times \sigma^2 / 252$ × trading days. "Break-even vol" is the realized vol (in the same units as IV) at which the position stops losing. Trailing realized vol = 7.27% (5-minute June-future mids, IV convention); implied = the snapshot ATM IV. Units are option price units per one option; no multiplier is verified. Futures hedge cost uses the 0.75-pip median futures spread with daily re-hedging.

ID (D1 / D2)	Strategy	Debit	Profitable?	Break-even realized vol	4-week hedged P&L at 7.27%	Held to expiry at 7.27%
D1-1	Long straddle May 1.1750	0.01807	**Conditional**	≥ 6.86% (4 wk) / 6.81% (expiry)	+0.00111	+0.00123 (+6.8%)
D2-5	Long straddle May 1.1725	0.01798	**Conditional**	≥ 6.89% / 6.81%	+0.00105	+0.00124 (+6.9%)
D1-3 / D2-9	Long straddle Jun 1.1725	0.02570	**Conditional**	≥ 6.91% / 6.60%	+0.00067	+0.00263 (+10.2%)
D1-5 / D2-13	Long straddle Jul 1.1750	0.03038	**Conditional**	≥ 6.92% / 6.56%	+0.00055	+0.00334 (+11.0%)
D1-7 / D2-15	Long straddle Aug 1.1750	0.03599	**Conditional**	≥ 7.08% / 6.61%	+0.00026	+0.00361 (+10.0%)
D1-9 / D2-17	Long straddle Sep 1.1775	0.03980	**Conditional**	≥ 7.16% / 6.65%	+0.00014	+0.00379 (+9.5%)
D2-1	Long straddle Apr 1.1700 (2 days)	0.00631	**No**	≥ 7.71% (trading-day basis, last 2 days) vs 7.15%	-0.00047 (2-day hold)	same: expires (-7%)
D1-2 / D2-8	Calendar call May/Jun	0.00507 / 0.00502	**No**	≤ 4.27%	-0.00101 / -0.00102	n/a
D1-4 / D2-12	Calendar call Jun/Jul	0.00351	**No**	none: loses at any vol	-0.00098	n/a
D1-6 / D2-14	Calendar call Jul/Aug	0.00403	**No**	none	-0.00100	n/a
D1-8 / D2-16	Calendar call Aug/Sep	0.00297	**No**	none	-0.00091	n/a
D2-4	Calendar call Apr/May	0.00719	**No**	≤ 5.72% (trading-day basis, 2 days)	-0.00034	n/a
D2-2	25D risk reversal Apr	0.00012	**No**	none	-0.00029	n/a
D2-6	25D risk reversal May	0.00086	**No**	≥ 8.45%	-0.00020	n/a
D2-10	25D risk reversal Jun	-0.00077 (credit)	**No**	none	-0.00058	n/a
D2-3	Wing-vs-ATM curvature Apr	-0.00302 (credit)	**No**	none	-0.00076	n/a
D2-7	Wing-vs-ATM curvature May	-0.00965 (credit)	**No**	none	-0.00250	n/a
D2-11	Wing-vs-ATM curvature Jun	-0.01347 (credit)	**No**	none	-0.00226	n/a

- **Straddles (conditional):** they win only because the 8 April implied vol (6.41-6.59%) sits 0.7-0.9 vp below the sample's trailing realized vol. If realized vol over the option's life equals implied, every straddle loses its costs (about -0.0006 to -0.0008). The report's own continuous-hold estimate (-0.0018 vp) is consistent with this: it uses the panel median IV of 7.31%, at which realized ≈ implied. The trailing figure includes the 23 March event, and the fall in IV suggests the market expects calmer conditions. Treat as a hypothesis for a pre-registered realized-vol forecast and paper-trading.
- **Calendars (no):** short gamma with tiny positive theta (0.000007-0.00005/day). Four weeks of carry cannot recover two spread crossings on two expiries (~0.0009). Any profit would have to come from back-month IV rising relative to the front, with cross-month basis risk.
- **Risk reversals (no):** essentially a long-delta position (delta 0.51-0.54) with almost no vega. The skew carry is negligible and the cost is 0.0003-0.0007. Profit depends on direction or a skew view that has no history yet.
- **Curvature trades (no):** short ATM gamma against long wings. In a sample where realized ≥ implied, short gamma loses, and four wide legs double the cost.

- **April expiry (no):** a 29% relative spread (1.10 vp) on 2-day options; the straddle needs about 7.7% realized vol (trading-day basis) over its last two days against 7.15% trailing, and loses about 7% of premium.

5. Liquidity-Based Execution Framework

Liquidity is strongly expiry-dependent: the front week is untradeable at 29% relative spreads, while July-September quotes are all TIGHT at 2-3% (0.14-0.19 vp). Displayed size is a symmetric market-maker grid (20 lots each side on every option), so depth imbalance = 0 and microprice = mid throughout the snapshot.

- **Liquidity-Filtered Universe:** take liquidity only on TIGHT quotes (relative spread ≤ 5%) where displayed size ≥ order size, choosing the lowest spread in vol points for the same exposure.
- **Passive Quoting / Spread Capture:** rest bids/offers inside the spread at surface fair value ± edge where the spread exceeds 1 vp, hedging in the matching future month. This applies only to 9 April-expiry quotes in the snapshot and needs MBO queue-position, fill and adverse-selection modelling; the report calls it "not a tradable signal."
- **Microprice / Imbalance Timing:** buy when microprice ≤ mid (offer side heavy). Execution overlay only. With symmetric 20-lot grids it is uninformative in this data, and displayed size can be cancelled instantly.
- **Patient Limit Orders on Wide Wings:** work wing legs at mid or better (rel. spread 5-20%) instead of crossing. Fill probability unknown.
- **Liquid Hedge Routing:** monthly futures (IDs 434/435/437/438) show only book resets and no trades, so hedges must route through June/September futures plus calendar spreads, with basis risk recorded separately.

Verdicts (these are execution tools, not profit sources):

Strategy	Profitable?	Value in this data
Liquidity-filtered universe	Not a P&L source; **use**	Removes April (0/14 tight) and the 37 non-tight quotes; caps round-trip cost at ≤0.37 vp in Jul-Sep
Passive quoting / spread capture	**No (insufficient flow)**	Up to half of 1.10 vp per fill on April quotes, but 4-6 option trades in 22 sessions means almost no fills; adverse selection unmeasured
Microprice / imbalance timing	**No effect**	Imbalance is 0 on every quote (20-lot grid); microprice = mid
Patient limit orders on wings	Cost saving if filled	Mid fills would save the half-spread: ~0.0004 per straddle entry (adds ~1.5-2.5 pp to the straddle returns above); fill rate unknown
Liquid hedge routing	Required cost	June/September futures plus calendar spread; basis risk must be booked separately
Liquidity-regime filter	Not computable yet	Needs multi-session liquidity history

6. Time-of-Day (ToD) and Long-Gamma Strategies

Panel: 22 sessions, 353 priced option snapshots every 30 minutes (08:30-18:00 Berlin), 87 failed snapshots; June ATM IV panel median 7.31%. Realized variance is measured on the June future only, as a proxy.

Strategy	Gross Edge	Round-Trip Cost	Net Edge	Break-even requires	Verdict
Intraday Long Gamma (Buy 08:30, Sell 18:00 Berlin)	+0.0249 vp/day	0.1967 vp	**-0.1718 vp/day**	~5.0x the realized variance (≈2.2x vol) or a spread ≤ 0.025 vp (8x tighter)	**Not Viable**
Weekday-Only Long Gamma (Flat weekend)	+0.0956 vp/week	0.1967 vp	**-0.1011 vp/week**	~1.34x the realized variance (≈1.16x vol) or a spread ≤ 0.096 vp (about half)	**Marginal / Unproven**
Continuous 1-Week Hold	-0.0018 vp	0.0000 vp (turnover excluded)	**-0.0018 vp**	n/a: realized ≈ implied	**No edge**

Break-even columns are derived here from the report's own formula: gross = theta/day (0.0451 vp) × (realized variance units − days charged); the spread equals 4.36 days of theta. Net figures exclude futures hedge costs and fees, so the true net is lower.

- **Realized vs implied:** daily RV/implied is 1.08 on 1-minute mids but 0.99 on 5-minute mids. Most of the apparent excess is bid/ask-bounce microstructure noise, so on a clean measure options are fairly priced over the sample.
- **Hedge-Timing Overlay (The Only Positive Finding):** median realized variance peaks at UTC 13:30 (5.6% of daily RV), 14:30 (5.5%), 12:30 (4.9%), 14:00 (4.7%) and 15:00 (4.6%), stable across sample halves (Spearman 0.85). Option quoting hours hold 68% of daily RV in 40% of the clock; Asian-session slots (02:30-04:30 UTC) realize only 0.3-0.5x calendar implied.
  - *Rule (per report):* if the desk is *already* running a delta-hedged book, use **narrower re-hedge bands** during 08:00-10:30 and 14:00-17:00 Berlin, and wider bands overnight; short-gamma books should reduce or pre-hedge before those windows. The benefit is strictly **lower hedge path cost**, not standalone alpha.
  - *Clock anchoring (added):* the UTC peaks line up with US events: 12:30 UTC = 08:30 ET data releases, and 13:30-14:30 UTC = the New York open / 10:00 ET releases. The sample spans the EU DST change (29 March 2026), while the US had already switched on 8 March, so Berlin clock time shifts by one hour relative to these events mid-sample. Hedge bands should be anchored to **New York (or UTC) time, not Berlin time**.
- **Event dominance:** mean-based statistics are dominated by 2026-03-23 (a ~120-pip EUR/USD rally, 10:15-13:00 UTC). That one session accounts for 91% of the 11:00 UTC slot's total RV (mean share 17.7% vs median 2.7%) and drives Monday's 11.5% annualized realized vol (vs ~7.5% on other weekdays). Only median-based figures should be used.
- **Intraday IV Seasonality: no time edge.** Median option spreads are flat (0.216-0.233 vp) across all slots, ATM spread ≈ 0.18 vp in every slot, and slot ranking is unstable (Spearman -0.34). No ATM drift slot reaches |t| > 2 across 20 tests; mean drifts are < 0.03 vp, far below the spread. Quotes resemble a fixed-vol market-maker grid.
- **Weekend effect:** mean overnight ATM change is +0.033 vp on weeknights (n = 96) versus +0.231 vp over weekend/holiday gaps (n = 23 expiry-gap rows, but only 4 distinct date gaps: 3 regular weekends plus the Easter closure of 3-6 April). This is not independent evidence.

7. Additional Observations (analysis not in the report)

1. **Implied vol cheap vs trailing realized on the snapshot date.** June ATM IV on 8 April (6.43%) sits ~0.9 vp below the panel median (7.31%), implying vol compression through the sample. Converted to the same trading-day convention (×0.983), it is ~6.32% against ~7.15% trailing 5-minute realized, a realized/implied variance ratio of ~1.28. This is a backward-looking comparison that includes the 23 March event, not a forecast. It is, however, exactly the horizon-matched realized-vol test the straddle candidates are waiting for, and it should be pre-registered before being examined on new data.
2. **Cost per unit of exposure favours the back months for vega, the front for gamma.** September has the lowest spread per vega (0.142 vp); April's gamma is cheap per unit of debit but its 1.10 vp spread makes it unusable. Strategy choice should be framed as cost per unit of the Greek being targeted.
3. **Skew is consistent and orderly.** The put premium (RR25 -0.12 → -0.40 vp) and curvature (BF25 0.13 → 0.30 vp) increase smoothly with maturity. There is no anomaly to fade, which matches the 0-alert result.
4. **Internal inconsistencies in the source report** (minor, but worth fixing before circulation):
   - 4 option trades (valid trade rows, page 23) vs 6 option trade messages (time-of-day study, page 42): different definitions, not reconciled in the text.
   - "Only 3 weekends" (D3-T1) vs "4 distinct observed date gaps" (page 44): the fourth gap is the Easter closure.
   - 22 sessions are claimed, but the weekday RV table sums to 21 (Mon 4, Tue 5, Wed 5, Thu 4, Fri 3).
   - The median spread is quoted on three different universes (163 bps on a 2-contract dev subset; 2-29% per expiry on the snapshot; ~3.2% pooled by time slot). These are not interchangeable.

8. Critical Methodological Red Flags (Bulge Bracket Stress Test)

1. **Cost Dominance where edge was measured:** for every tested schedule, the round-trip option spread exceeds the gross gamma/theta edge (e.g., 0.025 vp gross vs 0.197 vp cost). For the surface candidates, no edge was computed, so this cannot be framed as cost dominance.
2. **Data Sparsity & Fill Risk:** only 4-6 option trades in 22 sessions. The analysis relies on reconstructed MBO quote data, and fill probability, queue position and adverse selection are unquantified. Displayed depth is cancellable resting liquidity, not inventory or open interest (no OI, GEX/DEX or dealer-position data exists).
3. **Underlying Mapping / Basis Risk:** different expiries map to different underlying futures (direct, calendar spread or interpolated); monthly futures never trade. This basis risk invalidates simple calendar-arbitrage assumptions.
4. **Clock Bug Not Fully Resolved:** phase04 June-future BBO timestamps were 19,800 s (5.5 h) early because naive UTC was read as machine-local IST. The fix is applied **only** in the time-of-day study. The report states that clock-time-dependent phase04/phase05 results "need revalidation" and that the PDF "does not rerun or certify those frozen experiments."
5. **Look-Ahead Guardrails Designed but Not Exercised:** prior-only z-scores (strictly earlier same-expiry observations, ddof=1) are the correct design, but no history file was supplied, so every signal is insufficient_history / missing_current_coverage. Fractional differencing (d = 0.4) and the CUSUM threshold were chosen on the full sample and must be refit on training data before any predictive use. April data has already been viewed and is not a clean holdout.
6. **Statistical Power:** 22 sessions, dependence across strikes and expiries, no multiple-testing correction, and event-day outliers. SHAP is "not computed," not zero: no predictive model exists.
7. **Assumed Parameters:** r = 3.5% and the 13:00 UTC expiry time are assumptions; rate sensitivity of IV has not been validated. No contract multiplier is verified, so all results are in price units or vol points, not currency.

9. Strategy Scorecard: Every Strategy in the Report as a Position

Strategy (report location)	Positions	Profitable?	Key evidence	Action
Momentum option signal (p. 23)	278 dev trades	**No**	Net -0.194, win rate 48.9%, CI [-1.00, 0.57]	Drop
Long straddles May-Sep (D1-1/3/5/7/9, D2-5/9/13/15/17)	6 distinct	**Conditional**	Break-even realized 6.56-6.81% vs 7.27% trailing; +6.8% to +11.0% to expiry; loses ~0.0007 if realized = implied	Paper-trade with a pre-registered RV forecast
Long straddle Apr (D2-1)	1	**No**	Needs ~7.7% realized over 2 days vs 7.15%; 29% spread	Drop
Calendar calls (D1-2/4/6/8, D2-4/8/12/14/16)	6 distinct	**No**	Carry cannot cover ~0.0009 crossing cost; basis risk	Drop unless a term-structure view exists
25D risk reversals (D2-2/6/10)	3	**No**	Cost 0.0003-0.0007; delta 0.51-0.54, no skew history	Wait for RR history
Wing-vs-ATM curvature (D2-3/7/11)	3	**No**	-0.0008 to -0.0025 over 4 weeks; short gamma	Drop
Intraday long gamma (D3-T2)	schedule	**No**	Net -0.172 vp/day	Drop
Weekday-only long gamma (D3-T2)	schedule	**No (closest)**	Net -0.101 vp/week; needs half the spread	Retest with mid fills
Continuous hold (D3-T2)	schedule	**No edge**	-0.0018 vp at panel IV	Superseded by the straddle analysis
Hedge-timing overlay (D3-T1)	overlay	**Yes, as cost saving**	RV peak 12:30-15:00 UTC, rank stability 0.85	Validate with futures costs
Execution-window selection / IV seasonality / order-flow timing (D3-T1)	n/a	**No**	Flat spreads; no IV drift t-stat above 2; 6 option trades	Drop
Liquidity strategies (D1-T7)	overlays	Execution aids only	See Section 5	Use the filter; research the rest

10. What Else Can Be Done

Idea	Why the data supports it	How to test	Data needed	Priority
Conditional long-vol carry in Jul-Sep straddles	Only family with a positive break-even margin; Jul-Sep quotes all tight	Enter delta-hedged when forecast RV − ask IV > threshold set in advance; hold to expiry; daily NY-time hedging	Daily snapshots over new sessions	High
Realized-vol forecast (HAR-RV on futures)	4.1M futures BBO rows; futures are liquid even though options are not	Forecast next 30-90 day RV from daily/weekly/monthly RV; chronological validation vs IV	Longer futures history	High
Hedge-timing overlay backtest	Only robust pattern (Spearman 0.85)	Compare fixed vs time-varying hedge bands on futures P&L and turnover	Futures fees and latency	High
Mid / passive fill study	Halving the spread makes weekday-only long gamma break even (≤0.096 vp)	Replay MBO queues: how often a mid-priced order would have filled before the quote moved	Existing MBO (already available)	Medium
Event-day gamma	IV does not reprice intraday (fixed-vol grid) while RV concentrates at US releases	Test straddle carry only on scheduled US data days (CPI, NFP) vs other days	Economic calendar	Medium
Skew dynamics (sticky-strike vs sticky-delta)	22-session panel has RR25 changes and futures returns	Regress RR25 changes on futures returns; use for vanna-aware hedging and as a precondition for RR trades	Existing panel	Medium
Synthetic forwards for serial months	Monthly futures never trade; parity agrees within ~1 pip	Use put-call parity forwards instead of interpolated futures to hedge Apr/May/Jul/Aug	Existing quotes	Medium
Listed vs OTC vol relative value	Listed IV may deviate from OTC EUR/USD ATM vol	Compare EUCO ATM IV with OTC 1M/3M ATM by expiry	OTC vol quotes	Low
Snapshot history for z-scores	Every RR/BF signal is insufficient_history	Store daily snapshots at a fixed time for 3+ months	Ongoing capture	Prerequisite

Final Recommendation for the Portfolio Manager

Reject for live deployment as standalone alpha strategies.

The research is rigorous and correctly concludes that, on this sample, no strategy shows a cost-adjusted edge. The market is quote-driven with almost no trading, front-week options are very wide, and futures roll mapping adds basis risk. Back-month liquidity is adequate, so the binding constraint there is the absence of a validated realized-vol forecast rather than execution cost.

Next Steps / Actionable Refinements:

1. **Validate the Hedge-Timing Overlay:** backtest adaptive re-hedge bands on the futures leg with real futures spreads, fees, latency and turnover, anchoring the high-RV window (12:30-15:30 UTC) to New York time and using median-based profiles that exclude 23 March-type outliers.
2. **Build the Missing Realized-Vol Forecast:** pre-register a horizon-matched RV forecast (target, fees and hedge rule fixed in advance) and test it chronologically on new sessions. This is the gate for every straddle candidate and the weekday-only schedule.
3. **Collect More Independent Data:** more sessions (especially more weekends), trade/fill data and a multi-session snapshot history so prior-only RR/BF z-scores can actually be computed.
4. **Revalidate Legacy Results:** rerun the clock-dependent phase04/phase05 experiments with the 19,800 s correction before citing them.

5. **Research Only, No Deployment:** passive quoting and calendar structures remain research items until queue-position/adverse-selection simulation and a tick-level model of calendar-spread liquidity exist.

Revision Notes (changes from the previous version)

- **Scope:** extended from pages 48-92 to the full 101-page report (added Sections 1-3 and the momentum rejection).
- **Corrected:** "median half-spread ~1.84 bp" is not in the report. The actual median is 163 bps (dev subset) or 2-29% of premium per expiry (snapshot).
- **Corrected:** surface strategies were never assigned an edge (0 price alerts), so "costs obliterate surface-arbitrage edges" was overstated.
- **Corrected:** liquidity is inadequate only at the front end; July-September quotes are all tight.
- **Corrected:** the look-ahead protections were designed but never exercised (insufficient history). "Truncation tests" do not appear in the report and were removed.
- **Corrected:** the clock bug is fixed only in the time-of-day study; legacy results remain uncertified.
- **Corrected:** "shift focus to passive market making" went beyond the report's conclusions; it is now framed as research-only.
- **Clarified:** option trade count is 4 (valid trade rows) or 6 (trade messages), depending on definition.
- **Added:** break-even analysis, DST/time-zone anchoring, event dominance (23 March), the snapshot implied-vs-realized comparison, cost per Greek, and internal inconsistencies in the source report.
- **Added (this revision):** per-strategy profitability verdicts for all 17 surface candidates (Section 4a), verdicts for the liquidity strategies (Section 5), a full strategy scorecard (Section 9) and a list of further ideas (Section 10).

Reviewer note follows: 48a →

D1-T1: Expiry metrics (transposed: one column per row)

Source: outputs\surface_framework\dashboard.html; 6 saved data rows. NaN is unavailable, not zero. Same value in every row shown: asof = 2026-04-08T14:00:00+00:00; call_10d_iv = NaN; put_10d_iv = NaN; rr10 = NaN; bf10 = NaN; call_25d_iv = NaN; put_25d_iv = NaN; rr25 = NaN; bf25 = NaN; coverage_25d = False.

expiry	2026-04-10	2026-05-08	2026-06-12	2026-07-10	2026-08-14	2026-09-11
n_quotes	14	21	28	16	16	16
n_eligible	0	6	14	16	16	16
atm_iv	NaN	0.0659427	0.0643192	0.064088	0.0647876	0.0651918
dte	1.95833	29.9583	64.9583	92.9583	127.958	155.958
underlying_routes	spread 2026-04/2026-06 vs 2026-06 (quote age 20m)	spread 2026-05/2026-06 vs 2026-06 (quote age 21m)	direct	interp 2026-06..2026-09	interp 2026-06..2026-09	direct
atm_total_variance	NaN	0.00035691	0.000736247	0.00104604	0.0014715	0.00181594

Note on D1-T1: the snapshot under both quote gates

Combines D1-T1, D2-T1 (page 63) and D1-T5 (page 55). RR25/BF25 in volatility points; ATM in percent.

Expiry	DTE	ATM IV	RR25 (vp)	BF25 (vp)	Rel. spread	IV spread (vp)	Tight / valid	Route
2026-04-10	2.0	8.82% *	-0.12 *	0.13 *	29.4%	1.10	0/14	calendar spread (20m old)
2026-05-08	30	6.59%	-0.31 *	0.25 *	6.3%	0.32	6/21	calendar spread (21m old)
2026-06-12	65	6.43%	-0.40 *	0.30 *	4.9%	0.23	14/28	direct
2026-07-10	93	6.41%	n/a	n/a	2.8%	0.19	16/16	interpolated
2026-08-14	128	6.48%	n/a	n/a	2.4%	0.16	16/16	interpolated
2026-09-11	156	6.52%	n/a	n/a	2.1%	0.14	16/16	direct

* Available only under the 50% research gate (dashboard 2). Under the conservative 5% gate shown on page 48, April has no eligible quotes and no expiry has 25-delta coverage.

- **Skew is orderly.** Puts are richer than calls and both the put premium and wing curvature grow smoothly with maturity. There is no anomaly to fade, consistent with 0 price alerts.
- **The front week is a different market.** April's 8.82% ATM carries only 2 days of variance (total variance 4.2e-5) and trades at a 29% relative spread; it should not anchor term-structure conclusions.
- **Cost depends on the Greek.** September has the lowest spread per unit of vega (0.142 vp); short-dated options give more gamma per unit of debit but the April spread (1.10 vp) makes them unusable. Compare structures by cost per unit of the Greek being targeted.
- **Spread figures are not interchangeable.** The report quotes medians on three universes: 163 bps (2-contract dev subset, page 22), 2.1-29.4% (snapshot by expiry), about 3.2% (pooled by time slot, page 74).

Reviewer note follows: 49a →

D1-T2: Term structure (research only)

Source: outputs\surface_framework\dashboard.html; 5 saved data rows. NaN is unavailable, not zero.

front_expiry	back_expiry	atm_slope_vol_points	forward_variance	forward_vol	interpretation
2026-04-10	2026-05-08	NaN	NaN	NaN	research_only_different_underlying_months
2026-05-08	2026-06-12	-0.162346	0.003956	0.062896	research_only_different_underlying_months
2026-06-12	2026-07-10	-0.023121	0.004038	0.063548	research_only_different_underlying_months
2026-07-10	2026-08-14	0.069955	0.004437	0.066610	research_only_different_underlying_months
2026-08-14	2026-09-11	0.040427	0.004490	0.067008	research_only_different_underlying_months

REVIEWER NOTE

Note on D1-T2: forward volatility

Derived from atm_total_variance in D1-T1 and D2-T1; forward_vol = sqrt[(w2 - w1) / (T2 - T1)].

Interval	Forward vol	Comment
Apr → May (research gate)	6.41%	Front week elevated in IV but tiny in variance
May → Jun	6.29%	Lowest point of the curve
Jun → Jul	6.35%	Direct June vs interpolated July
Jul → Aug	6.66%	Both legs interpolated between Jun and Sep
Aug → Sep	6.70%	Interpolated vs direct

- All forward variances are positive and gently rising: there is no negative forward variance and therefore no inversion flag on this snapshot.
- Because July and August are interpolated between June and September, the forward vols across Jun → Sep are partly a construction of the interpolation, not independent market quotes.
- As the report states, every pair references different monthly futures; this is a term-structure diagnostic, not a calendar-arbitrage test.

D1-T3: Prior-only statistical signals

Source: outputs\surface_framework\dashboard.html; 18 saved data rows. NaN is unavailable, not zero. Same value in every row shown: asof = 2026-04-08T14:00:00+00:00; n_prior = 0; zscore = NaN.

expiry	metric	value	status
2026-04-10	atm_iv	NaN	missing_current_coverage
2026-04-10	rr25	NaN	missing_current_coverage
2026-04-10	bf25	NaN	missing_current_coverage
2026-05-08	atm_iv	0.065943	insufficient_history
2026-05-08	rr25	NaN	missing_current_coverage
2026-05-08	bf25	NaN	missing_current_coverage
2026-06-12	atm_iv	0.064319	insufficient_history
2026-06-12	rr25	NaN	missing_current_coverage
2026-06-12	bf25	NaN	missing_current_coverage
2026-07-10	atm_iv	0.064088	insufficient_history
2026-07-10	rr25	NaN	missing_current_coverage
2026-07-10	bf25	NaN	missing_current_coverage
2026-08-14	atm_iv	0.064788	insufficient_history
2026-08-14	rr25	NaN	missing_current_coverage
2026-08-14	bf25	NaN	missing_current_coverage
2026-09-11	atm_iv	0.065192	insufficient_history
2026-09-11	rr25	NaN	missing_current_coverage

Reviewer note follows: 51a →

expiry	metric	value	status
2026-09-11	bf25	NaN	missing_current_coverage

Note on D1-T3: the look-ahead guardrail was never exercised

Applies to D1-T3 (pages 50-51) and D2-T3 (page 65).

> **Correction.** The prior-only z-score design (strictly earlier, same-expiry observations, ddof = 1) is correct, but no history file was supplied. Every row is n_prior = 0 with status insufficient_history or missing_current_coverage. These tables are evidence of a sound design, not of a tested signal.

- Risk-reversal and butterfly mean-reversion candidates (D2-T4) depend entirely on this history; until it exists they cannot be evaluated.
- Other full-sample choices also need training-only refitting before predictive use: fractional differencing d = 0.4 and the CUSUM threshold 0.00066 (page 15).
- April 2026 has already been viewed by the exploratory work and is not a clean holdout (page 4). New sessions are needed for validation.
- Minimum requirement: repeated snapshots across sessions, stored at the same clock time, so each expiry accumulates its own prior distribution.

D1-T4: Strategy candidates | columns 1/2

Source: outputs\surface_framework\dashboard.html; 9 saved data rows. NaN is unavailable, not zero. Same value in every row shown: status = WATCH; edge_price_units = None.

id	strategy	expiry	reason	entry_debit_price_units	legs
1	long_straddle	2026-05-08	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.01807	+1 EUCO SI 20260508 PS EU C 1.1750 0; +1 EUCO SI 20260508 PS EU P 1.1750 0
2	calendar_call	2026-05-08/2026-06-12	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	0.00507	-1 EUCO SI 20260508 PS EU C 1.1700 0; +1 EUCO SI 20260612 PS EU C 1.1700 0
3	long_straddle	2026-06-12	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.02570	+1 EUCO SI 20260612 PS EU C 1.1725 0; +1 EUCO SI 20260612 PS EU P 1.1725 0
4	calendar_call	2026-06-12/2026-07-10	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	0.00351	-1 EUCO SI 20260612 PS EU C 1.1725 0; +1 EUCO SI 20260710 PS EU C 1.1725 0
5	long_straddle	2026-07-10	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.03038	+1 EUCO SI 20260710 PS EU C 1.1750 0; +1 EUCO SI 20260710 PS EU P 1.1750 0
6	calendar_call	2026-07-10/2026-08-14	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	0.00403	-1 EUCO SI 20260710 PS EU C 1.1750 0; +1 EUCO SI 20260814 PS EU C 1.1750 0
7	long_straddle	2026-08-14	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.03599	+1 EUCO SI 20260814 PS EU C 1.1750 0; +1 EUCO SI 20260814 PS EU P 1.1750 0

id	strategy	expiry	reason	entry_debit_price_units	legs
8	calendar_call	2026-08-14/2026-09-11	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	0.00297	-1 EUCO SI 20260814 PS EU C 1.1750 0; +1 EUCO SI 20260911 PS EU C 1.1750 0
9	long_straddle	2026-09-11	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.03980	+1 EUCO SI 20260911 PS EU C 1.1775 0; +1 EUCO SI 20260911 PS EU P 1.1775 0

Profitability verdicts follow: 54a →

D1-T4: Strategy candidates | columns 2/2

Source: outputs\surface_framework\dashboard.html; 9 saved data rows. NaN is unavailable, not zero. Same value in every row shown: status = WATCH; edge_price_units = None.

id	delta	gamma	vega_per_vol_point	theta_per_day
1	-0.114116	35.869463	0.002643	-0.000287
2	0.009756	-5.525638	0.000619	0.000050
3	0.026940	24.859107	0.003922	-0.000192
4	0.011841	-2.074117	0.000377	0.000017
5	0.001302	20.879907	0.004687	-0.000158
6	0.014619	-1.705260	0.000398	0.000012
7	0.033954	17.446091	0.005483	-0.000135
8	0.008709	-0.890866	0.000276	0.000007
9	0.014840	15.686603	0.006048	-0.000122

Profitability verdicts: D1-T4 strategy candidates

One row per candidate on pages 52-54. Reviewer calculation; not in the source report. Units: option price units per option.

How to read this. The report gives no edge for these candidates (edge_price_units = None, 0 price alerts). Each one is repriced here leg by leg with Black-76 (the recomputed Greeks match the table on page 54), costed at the half-spread of each expiry on entry and exit, and run delta-hedged under the sample's trailing realized vol of 7.27% (IV units). **Conditional** means profitable only if realized vol over the holding period stays above the break-even.

ID (D1 / D2)	Strategy	Debit	Profitable?	Break-even realized vol	4-week hedged P&L at 7.27%	Held to expiry at 7.27%
D1-1	Long straddle May 1.1750	0.01807	**Conditional**	≥ 6.86% (4 wk) / 6.81% (expiry)	+0.00111	+0.00123 (+6.8%)
D1-3 / D2-9	Long straddle Jun 1.1725	0.02570	**Conditional**	≥ 6.91% / 6.60%	+0.00067	+0.00263 (+10.2%)
D1-5 / D2-13	Long straddle Jul 1.1750	0.03038	**Conditional**	≥ 6.92% / 6.56%	+0.00055	+0.00334 (+11.0%)
D1-7 / D2-15	Long straddle Aug 1.1750	0.03599	**Conditional**	≥ 7.08% / 6.61%	+0.00026	+0.00361 (+10.0%)
D1-9 / D2-17	Long straddle Sep 1.1775	0.03980	**Conditional**	≥ 7.16% / 6.65%	+0.00014	+0.00379 (+9.5%)
D1-2 / D2-8	Calendar call May/Jun	0.00507 / 0.00502	**No**	≤ 4.27%	-0.00101 / -0.00102	n/a
D1-4 / D2-12	Calendar call Jun/Jul	0.00351	**No**	none: loses at any vol	-0.00098	n/a
D1-6 / D2-14	Calendar call Jul/Aug	0.00403	**No**	none	-0.00100	n/a
D1-8 / D2-16	Calendar call Aug/Sep	0.00297	**No**	none	-0.00091	n/a

Method: each candidate's legs were repriced with Black-76 at the snapshot (ATM IV for near-ATM legs, 25-delta IVs for wing legs); the recomputed Greeks match the report's tables. Cost = half the median IV spread of each expiry × leg vega, paid on entry and on exit (legs that expire are settled, no exit cost). Delta-hedged carry = theta × calendar days + $\frac{1}{2} \times \text{gamma} \times F^2 \times \sigma^2 / 252 \times$ trading days. "Break-even vol" is the realized vol (in the same units as IV) at which the position stops losing. Trailing realized vol = 7.27% (5-minute June-future mids, IV convention); implied = the snapshot ATM IV. Units are option price units per one option; no multiplier is verified. Futures hedge cost uses the 0.75-pip median futures spread with daily re-hedging.

- **Straddles (conditional):** they win only because the 8 April implied vol (6.41-6.59%) sits 0.7-0.9 vp below the sample's trailing realized vol. If realized vol over the option's life equals implied, every straddle loses its costs (about -0.0006 to -0.0008). The report's own continuous-hold estimate (-0.0018 vp) is consistent with this: it uses the panel median IV of 7.31%, at which realized ≈ implied. The trailing figure includes the 23 March event, and the fall in IV suggests the market expects calmer conditions. Treat as a hypothesis for a pre-registered realized-vol forecast and paper-trading.
- **Calendars (no):** short gamma with tiny positive theta (0.000007-0.00005/day). Four weeks of carry cannot recover two spread crossings on two expiries (~0.0009). Any profit would have to come from back-month IV rising relative to the front, with cross-month basis risk.

D1-T5: Liquidity by expiry (transposed: one column per row)

Source: outputs\surface_framework\dashboard.html; 6 saved data rows. NaN is unavailable, not zero. Same value in every row shown: median_depth_imbalance = 0.

expiry	2026-04-10	2026-05-08	2026-06-12	2026-07-10	2026-08-14	2026-09-11
n_valid_books	14	21	28	16	16	16
n_tight	0	6	14	16	16	16
median_rel_spread	0.2941	0.06316	0.04878	0.02804	0.02392	0.02088
median_spread_vol_points	1.104	0.3198	0.2271	0.1851	0.1566	0.1422
displayed_bid_size	280	420	560	320	320	320
displayed_ask_size	280	420	560	320	320	320
median_quote_age_seconds	3.532	3.841	3.532	3.099	2.776	2.776

D1-T6: Liquidity roles by quote

Source: outputs\surface_framework\dashboard.html; 6 saved data rows. NaN is unavailable, not zero.

liquidity_role	expiry	passive_quote_research	patient_limit_order	taker_eligible
	2026-04-10	9	5	0
	2026-05-08	0	15	6
	2026-06-12	0	14	14
	2026-07-10	0	0	16
	2026-08-14	0	0	16
	2026-09-11	0	0	16

Profitability verdicts follow: 57a →

D1-T7: Liquidity-based strategies (research framework)

Source: outputs\surface_framework\dashboard.html; 6 saved data rows. NaN is unavailable, not zero.

Strategy	Liquidity input	Rule	Status / limitation
Liquidity-filtered universe	rel_spread, spread_vol_points, depth	Only take liquidity on TIGHT quotes (rel spread ≤ 5%) whose displayed size ≥ order size; choose strikes/expiries by lowest spread in vol points for the same exposure.	Implemented as liquidity_role = taker_eligible.
Passive quoting / spread capture around fitted surface	spread_vol_points, quote age	Where the spread exceeds 1 vol point, rest bids/offers inside the spread at surface fair value ± edge; hedge fills delta-neutral in the matching future month.	passive_quote_research: needs MBO queue-position and fill simulation, adverse-selection measurement; not a tradable signal.
Microprice / imbalance timed execution	bid_size, ask_size	Buy when microprice ≤ mid (offer side heavy), defer when microprice is near the ask; mirror for sells. Reduces crossing cost on legs already selected by the surface study.	Execution overlay only; displayed size can be cancelled instantly.
Patient limit orders on wide wings	rel_spread 5–20%	Work orders at mid or better for wing legs of risk reversals/butterflies instead of crossing.	patient_limit_order; fill probability unknown.
Liquid hedge routing	future route and quote age	Hedge illiquid monthly-future exposure via the direct June/September futures plus calendar spread; record basis risk separately.	Monthly futures show no trades in the sample.
Liquidity-regime filter	median spread and depth per expiry over time	Suspend taking when spreads widen beyond their prior-only distribution; stand aside or quote wider.	Requires multi-session liquidity history (not yet computed).

Profitability verdicts: liquidity-based strategies (D1-T7)

One row per strategy on page 57 (identical in dashboard 2). These are execution tools; the question is how much cost they save, not what they earn.

Strategy	Profitable?	Value in this data
Liquidity-filtered universe	Not a P&L source; **use**	Removes April (0/14 tight) and the 37 non-tight quotes; caps round-trip cost at ≤0.37 vp in Jul-Sep
Passive quoting / spread capture	**No (insufficient flow)**	Up to half of 1.10 vp per fill on April quotes, but 4-6 option trades in 22 sessions means almost no fills; adverse selection unmeasured
Microprice / imbalance timing	**No effect**	Imbalance is 0 on every quote (20-lot grid); microprice = mid
Patient limit orders on wings	Cost saving if filled	Mid fills would save the half-spread: ~0.0004 per straddle entry (adds ~1.5-2.5 pp to the straddle returns above); fill rate unknown
Liquid hedge routing	Required cost	June/September futures plus calendar spread; basis risk must be booked separately
Liquidity-regime filter	Not computable yet	Needs multi-session liquidity history

- **Liquidity is a front-end problem.** Tight quotes: April 0/14, May 6/21, June 14/28, July-September 16/16; median quote age 2.8-3.8 s.
- **Depth is a fixed grid.** 20 lots each side on every option, so imbalance is 0 and microprice = mid.
- **Option trades:** 4 valid option trade rows (page 23) vs 6 option trade messages (page 42); too few for any fill or order-flow model.
- No open interest, GEX/DEX or dealer-position data exists; displayed size is cancellable resting liquidity.

D1-T9: Formulas used

Source: outputs\surface_framework\dashboard.html; 27 saved data rows. NaN is unavailable, not zero.

Quantity	Formula	Notes	Reason
Black-76 d1, d2	$d1 = [\ln(F/K) + \tfrac{1}{2}\sigma^2 T] / (\sigma\sqrt{T})$; $d2 = d1 - \sigma\sqrt{T}$	PARTIAL: F comes from the same-month FCEU future: June and September are direct quotes; April and May use calendar spreads (quotes about 20 minutes old); July and August are time-interpolated. ASSUMED: expiry time 13:00 UTC; T = ACT/365 (exact cut-off time not in data)	EUCO options are written on FCEU futures. A future has no carry, so Black-76 (forward F, no dividend or foreign-rate term) is used instead of Black-Scholes or Garman-Kohlhagen. That avoids needing EUR and USD deposit rates, which are not in the data. European exercise is assumed (the feed symbol carries 'EU').
Call / put price	$C = DF\cdot[F\cdot N(d1) - K\cdot N(d2)]$; $P = DF\cdot[K\cdot N(-d2) - F\cdot N(-d1)]$	ASSUMED: r = 3.5% continuous; the dataset has no rate curve. Both prices and inferred IV depend on this assumption; rate sensitivity has not been validated	Standard Black-76 premium, discounted from expiry because the premium is paid upfront.
Implied volatility	σ: Black76(σ) = price, bisection on [0.0001, 5]	AVAILABLE: bid, ask and mid rebuilt from MBO events at the snapshot; all 111 books have a bid IV and an ask IV	Bisection always converges and is vectorised. Newton's method divides by vega, which is near zero for deep ITM/OTM and short-dated strikes. Prices outside the no-arbitrage bounds return NaN rather than a forced value.
Put-call parity forward	$F = K + (Cmid - Pmid) / DF$	PARTIAL: needs a two-sided call and put at the same strike; used as a validator and as an alternative underlying mode, not as the primary F	Model-free check that needs no volatility. It caught the original quarterly-future mapping error (15–35 pips off); the remapped futures agree within about 1 pip.
Calendar-spread forward	$Ffront = Fback + Sfront{-}back$	PARTIAL: spread books are often pulled; the last two-sided quote is used if it is at most 120 minutes old, otherwise linear time interpolation between direct futures	Monthly FCEU futures (IDs 434/435/437/438) show only book resets and no trades. The exchange calendar spread is the only traded link to a liquid quarterly future. Spread price = front − back.

Quantity	Formula	Notes	Reason
Log-moneyness	$k = \ln(K/F)$	AVAILABLE: strike from symbology; F as above	Scale-free and centred at the forward, so smiles of different expiries line up. A call is ITM if $K < F$; a put is ITM if $K > F$.
Delta	$\Delta C = DF \cdot N(d1)$; $\Delta P = -DF \cdot N(-d1)$	AVAILABLE: model output from mid IV, F and T	Black-76 spot-premium delta for hedging. Wing selection uses forward delta $\lvert\Delta\rvert/DF$, so that the 25Δ and 10Δ points do not move with the assumed rate (forward-delta FX convention, not premium-adjusted).
Gamma	$\Gamma = DF \cdot \varphi(d1) / (F\sigma\sqrt{T})$	AVAILABLE: model output	Closed-form sensitivity to the future price; used for scenario and risk inspection.
Vega (per vol point)	$V = DF \cdot F \cdot \varphi(d1) \cdot \sqrt{T} \times 0.01$	AVAILABLE: model output	Scaled to one vol point so it can be compared directly with bid/ask spreads quoted in vol points.
Theta (per day)	$\Theta = [r \cdot \text{Price} - DF \cdot F \cdot \varphi(d1) \cdot \sigma/(2\sqrt{T})] / 365$	ASSUMED: depends on the assumed r and the calendar-day convention	Black-76 theta: the first term is discount accrual and the second is volatility decay. Expressed per calendar day.
ATM IV	$\sigma ATM = \text{interp}(k \to 0)$ on OTM quotes	AVAILABLE: all six expiries have OTM strikes on both sides of F	Uses ATM-forward ($K = F$) rather than the FX delta-neutral straddle convention, because it needs no iteration. OTM quotes are used because ITM premiums are mostly intrinsic value and their IV is sensitive to forward errors. Values are never extrapolated.
Risk reversal	$RRx = \sigma x\Delta\ \text{call} - \sigma x\Delta\ \text{put}$	PARTIAL: 25Δ: April, May and June only (in the 50% spread research view). 10Δ: no expiry, because the wings are not quoted. July to September strikes do not reach 25Δ	Standard FX skew measure, quoted at fixed delta rather than fixed strike so expiries are comparable. Negative means puts are richer.
Butterfly	$BFx = (\sigma x\Delta\ \text{call} + \sigma x\Delta\ \text{put})/2 - \sigma ATM$	PARTIAL: same coverage as the risk reversal	Smile-strangle (simple) butterfly. It measures curvature without the broker-strangle matching that the market-quoted butterfly needs.

Quantity	Formula	Notes	Reason
Total variance	$w = \sigma\mathrm{ATM}^2 \cdot T$	AVAILABLE: ATM IV for all six expiries	Total variance puts maturity comparisons on a variance-time scale rather than comparing raw IV. Here the expiries reference different futures, so monotonicity is not a standalone no-arbitrage test.
Forward variance / vol	$\sigma\mathrm{fwd}^2 = (w2 - w1) / (T2 - T1)$	PARTIAL: ATM-only approximation. Each expiry references a different future, so basis and roll differences are not removed	Differences total variance over the maturity interval rather than raw IV. With different underlying futures this is a term-structure diagnostic, not an executable forward-volatility quote or proof of arbitrage.
Prior-only z-score	$z = (xt - \mathrm{mean}(x<t)) / \mathrm{sd}(x<t)$	MISSING: no history file was supplied, so every signal is insufficient_history. Repeated snapshots across sessions are needed	Uses only strictly earlier observations, which avoids look-ahead. Sample sd (ddof=1) because the histories are short.
Mid / relative spread	$\mathrm{mid} = (\mathrm{bid}+\mathrm{ask})/2$; $\mathrm{rel} = (\mathrm{ask}-\mathrm{bid})/\mathrm{mid}$	AVAILABLE: top of book from MBO reconstruction	Relative spread is scale-free across cheap wings and expensive ITM options; it is used as the quality gate (rel ≤ max_rel_spread).
Spread in vol points	$(\sigma\mathrm{ask} - \sigma\mathrm{bid}) \times 100$, fallback $(\mathrm{ask}-\mathrm{bid})/V$	AVAILABLE: fallback not needed in this snapshot	Relative spread overstates cost on cheap wings. Spread in vol points is the cost per unit of the exposure actually traded, so strikes and expiries can be compared. The fallback applies when one side has no IV.
Depth imbalance	$I = (\mathrm{Qbid} - \mathrm{Qask}) / (\mathrm{Qbid} + \mathrm{Qask})$	PARTIAL: top-of-book size only. Every option shows 20 lots on each side (a market maker grid), so I = 0 and the measure is uninformative here	Simplest bounded measure of order-book pressure, on a scale from −1 to 1.
Microprice	$\mu = (\mathrm{bid}\cdot\mathrm{Qask} + \mathrm{ask}\cdot\mathrm{Qbid}) / (\mathrm{Qbid} + \mathrm{Qask})$	PARTIAL: same size limitation: equals mid in this snapshot	Size-weighted fair price that leans toward the thinner side. Reported as $(\mu - \mathrm{mid})/\mathrm{spread}$ for execution timing.

Reviewer note follows: 61a →

Quantity	Formula	Notes	Reason
Quote age	max(tsnap − tbid, tsnap − task)	PARTIAL: MBO has no explicit quote-refresh event. The timestamp is that of the latest surviving order at the best price	Uses the older side (conservative), because a package is only as fresh as its stalest leg.
Executable entry cost	Σ qi·(ask if qi>0 else bid) + |qi|·(fee+slippage)	MISSING: fees and the contract multiplier are unknown, so fee and slippage are user parameters (default 0) and results are in option price units	Crosses the spread on every leg (no passive fill assumed), so any edge must survive the worst displayed price.
Vertical edge	edge = −cost (long payoff) or −cost − DF·(K2−K1) (reverse)	AVAILABLE: executable bid/ask; there are currently no violations (0 PRICE_ALERT)	No-arbitrage bounds: a vertical's value must lie between 0 and DF·(K2−K1). These are model-free, so they do not depend on the volatility surface.
Butterfly edge	edge = −cost(+1 K1, −2 K2, +1 K3)	AVAILABLE: adjacent listed strikes	Price convexity in strike (non-negative butterfly) is a model-free condition. Equal spacing keeps the 1:−2:1 weights exact.
Hedge units	Hfuture = −Σ qi·Δi per underlying month	PARTIAL: monthly futures do not trade, so a hedge may need a quarterly future plus a calendar spread. Units are per option unit (no multiplier)	Each expiry settles against its own monthly future. Netting across months would hide basis and roll risk.
Scenario P&L	Σ qi·[Black76(F(1+m), σ+s, T−d/365) − midi]	ASSUMED: no smile dynamics (sticky-strike vs sticky-delta) are estimated. This is a model mark, not realized P&L	Full revaluation, rather than a Greek Taylor expansion, captures convexity for large moves. The parallel vol shift is a deliberate simplification.
Paper realized P&L	Σ qi·exiti − |qi|·fee − entry debit	MISSING: no currency P&L because the multiplier and capital are unverified; price units only	Exits also cross the spread (sell at bid minus slippage, buy back at ask plus slippage), symmetric with entry, so P&L is not flattered.

Note on D1-T9: assumptions behind every IV and Greek

Applies to the formula table on pages 58-61.

Input	Status	Effect if wrong
Risk-free rate r = 3.5% continuous	ASSUMED	Shifts discounting, theta's r·Price term and every inverted IV; rate sensitivity not validated
Expiry time 13:00 UTC, ACT/365	ASSUMED	Material for the 2-day April expiry, where T is small
Forward for Apr/May	PARTIAL	Calendar-spread quotes ~20 minutes old
Forward for Jul/Aug	PARTIAL	Linear time interpolation; no traded monthly future
Fees, slippage, multiplier	MISSING	Defaults of 0; all results in price units or vol points only

- Recommended: re-solve the surface at r = 2.5% and 4.5% and at 09:00/15:00 UTC expiry to bound the sensitivity of the April figures before relying on them.

Dashboard appendix 2: surface_framework_research

Source HTML: outputs\surface_framework_research\dashboard.html

Saved narrative and data tables follow. Identical tables are cross-referenced; tables sharing a section with an earlier dashboard are shown as differences only. Wide tables with few rows are transposed; constant columns are stated once in the subtitle.

Volatility surface research

106 / 111 quotes pass analytical quality. 0 price alerts. No live orders; all amounts are price units.

11 narrative paragraphs identical to an earlier dashboard appendix are not repeated.

D2-T1: Expiry metrics - changed rows, differing columns only (transposed: one column per row)

Source: outputs\surface_framework_research\dashboard.html; 6 saved data rows. NaN is unavailable, not zero. Same value in every row shown: coverage_25d = True.

Compared with D1-T1: 3 identical rows omitted; 3 rows changed (8 columns identical to D1-T1 omitted); 0 rows only here; rows only in D1-T1: none.

expiry	2026-04-10	2026-05-08	2026-06-12
n_eligible	9	21	28
atm_iv	0.0882128	0.0659218	0.0643192
call_25d_iv	0.0889048	0.0668828	0.0653667
put_25d_iv	0.0901057	0.0700096	0.0693505
rr25	-0.00120091	-0.00312688	-0.00398383
bf25	0.00129252	0.00252437	0.00303938
atm_total_variance	4.175e-05	0.000356684	0.000736247

D2-T2: Term structure (research only) - changed rows, differing columns only

Source: outputs\surface_framework_research\dashboard.html; 5 saved data rows. NaN is unavailable, not zero.

Compared with D1-T2: 3 identical rows omitted; 2 rows changed (2 columns identical to D1-T2 omitted); 0 rows only here; rows only in D1-T2: none.

front_expiry	atm_slope_vol_points	forward_variance	forward_vol
2026-04-10	-2.229093	0.004105	0.064073
2026-05-08	-0.160260	0.003958	0.062915

D2-T3: Prior-only statistical signals - changed rows, differing columns only

Source: outputs\surface_framework_research\dashboard.html; 18 saved data rows. NaN is unavailable, not zero.

Compared with D1-T3: 10 identical rows omitted; 8 rows changed (3 columns identical to D1-T3 omitted); 0 rows only here; rows only in D1-T3: none.

expiry	metric	value	status
2026-04-10	atm_iv	0.088213	insufficient_history
2026-04-10	rr25	-0.001201	insufficient_history
2026-04-10	bf25	0.001293	insufficient_history
2026-05-08	atm_iv	0.065922	insufficient_history
2026-05-08	rr25	-0.003127	insufficient_history
2026-05-08	bf25	0.002524	insufficient_history
2026-06-12	rr25	-0.003984	insufficient_history
2026-06-12	bf25	0.003039	insufficient_history

D2-T4: Strategy candidates - changed rows, differing columns only | columns 1/2

Source: outputs\surface_framework_research\dashboard.html; 17 saved data rows. NaN is unavailable, not zero.

Compared with D1-T4: 0 identical rows omitted; 9 rows changed (2 columns identical to D1-T4 omitted); 8 rows only here; rows only in D1-T4: none.

id	strategy	expiry	reason	entry_debit_price_units	legs
1	long_straddle	2026-04-10	near_expiry_risk;Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.00631	+1 EUCO SI 20260410 PS EU C 1.1700 0; +1 EUCO SI 20260410 PS EU P 1.1700 0
2	risk_reversal_25d	2026-04-10	near_expiry_risk;Needs prior-only same-expiry RR history; expensive puts alone are not a sell signal	0.00012	+1 EUCO SI 20260410 PS EU C 1.1750 0; -1 EUCO SI 20260410 PS EU P 1.1650 0
3	wing_vs_atm_curvature	2026-04-10	near_expiry_risk;25D wing strangle versus ATM straddle; requires prior BF extreme, cost and scenario review	-0.00302	-1 EUCO SI 20260410 PS EU C 1.1700 0; -1 EUCO SI 20260410 PS EU P 1.1700 0; +1 EUCO SI 20260410 PS EU C 1.1750 0; +1 EUCO SI 20260410 PS EU P 1.1650 0
4	calendar_call	2026-04-10/2026-05-08	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	0.00719	-1 EUCO SI 20260410 PS EU C 1.1700 0; +1 EUCO SI 20260508 PS EU C 1.1700 0
5	long_straddle	2026-05-08	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.01798	+1 EUCO SI 20260508 PS EU C 1.1725 0; +1 EUCO SI 20260508 PS EU P 1.1725 0

id	strategy	expiry	reason	entry_debit_price_units	legs
6	risk_reversal_25d	2026-05-08	Needs prior-only same-expiry RR history; expensive puts alone are not a sell signal	0.00086	+1 EUCO SI 20260508 PS EU C 1.1850 0; -1 EUCO SI 20260508 PS EU P 1.1550 0
7	wing_vs_atm_curvature	2026-05-08	25D wing strangle versus ATM straddle; requires prior BF extreme, cost and scenario review	-0.00965	-1 EUCO SI 20260508 PS EU C 1.1725 0; -1 EUCO SI 20260508 PS EU P 1.1725 0; +1 EUCO SI 20260508 PS EU C 1.1850 0; +1 EUCO SI 20260508 PS EU P 1.1550 0
8	calendar_call	2026-05-08/2026-06-12	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	0.00502	-1 EUCO SI 20260508 PS EU C 1.1725 0; +1 EUCO SI 20260612 PS EU C 1.1725 0
9	long_straddle	2026-06-12	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	0.02570	+1 EUCO SI 20260612 PS EU C 1.1725 0; +1 EUCO SI 20260612 PS EU P 1.1725 0

D2-T4: Strategy candidates - changed rows, differing columns only | columns 2/2

Source: outputs\surface_framework_research\dashboard.html; 17 saved data rows. NaN is unavailable, not zero.

id	delta	gamma	vega_per_vol_point	theta_per_day
1	-0.028328	106.454495	0.000683	-0.001524
2	0.512993	-1.720144	-0.000013	0.000032
3	0.005882	-22.194772	-0.000132	0.000268
4	0.045546	-34.884076	0.000989	0.000623
5	-0.023918	36.092353	0.002669	-0.000291
6	0.513149	2.054934	0.000091	-0.000003
7	0.065500	-8.057062	-0.000514	0.000046
8	0.023658	-5.545836	0.000626	0.000050
9	0.026940	24.859107	0.003922	-0.000192

D2-T4: Strategy candidates - rows not in D1-T4 (transposed: one column per row)

Source: outputs\surface_framework_research\dashboard.html; 17 saved data rows. NaN is unavailable, not zero. Same value in every row shown: status = WATCH; edge_price_units = None.

id	10	11	12	13	14	15	16	17
strategy	risk_reversal_25d	wing_vs_atm_curvature	calendar_call	long_straddle	calendar_call	long_straddle	calendar_call	long_straddle
expiry	2026-06-12	2026-06-12	2026-06-12/2026-07-10	2026-07-10	2026-07-10/2026-08-14	2026-08-14	2026-08-14/2026-09-11	2026-09-11
reason	Needs prior-only same-expiry RR history; expensive puts alone are not a sell signal	25D wing strangle versus ATM straddle; requires prior BF extreme, cost and scenario review	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied	Different future months: basis/roll risk; front-rich IV alone is not calendar arbitrage	Requires a horizon-matched realized-vol/event forecast exceeding IV plus costs; not supplied
entry_debit_price_units	-0.00077	-0.01347	0.00351	0.03038	0.00403	0.03599	0.00297	0.03980
legs	+1 EUCO SI 20260612 PS EU C 1.19500; -1 EUCO SI 20260612 PS EU P 1.15500	-1 EUCO SI 20260612 PS EU C 1.17250; -1 EUCO SI 20260612 PS EU P 1.17250; +1 EUCO SI 20260612 PS EU C 1.19500; +1 EUCO SI 20260612 PS EU P 1.15500	-1 EUCO SI 20260612 PS EU C 1.17250; +1 EUCO SI 20260710 PS EU C 1.17250	+1 EUCO SI 20260710 PS EU C 1.17500; +1 EUCO SI 20260710 PS EU P 1.17500	-1 EUCO SI 20260710 PS EU C 1.17500; +1 EUCO SI 20260814 PS EU C 1.17500	+1 EUCO SI 20260814 PS EU C 1.17500; +1 EUCO SI 20260814 PS EU P 1.17500	-1 EUCO SI 20260814 PS EU C 1.17500; +1 EUCO SI 20260911 PS EU C 1.17500	+1 EUCO SI 20260911 PS EU C 1.17750; +1 EUCO SI 20260911 PS EU P 1.17750
delta	0.539752	-0.058835	0.011841	0.001302	0.014619	0.033954	0.008709	0.014840

Profitability verdicts follow: 70a →

id	10	11	12	13	14	15	16	17
gamma	-0.215544	-4.841452	-2.074117	20.879907	-1.705260	17.446091	-0.890866	15.686603
vega_per_vol_point	-0.000096	-0.000664	0.000377	0.004687	0.000398	0.005483	0.000276	0.006048
theta_per_day	0.000008	0.000026	0.000017	-0.000158	0.000012	-0.000135	0.000007	-0.000122

D2-T5: Liquidity by expiry

Identical saved table reproduced at D1-T5: Liquidity by expiry (surface_framework). All values and rows are shared.

D2-T6: Liquidity roles by quote

Identical saved table reproduced at D1-T6: Liquidity roles by quote (surface_framework). All values and rows are shared.

D2-T7: Liquidity-based strategies (research framework)

Identical saved table reproduced at D1-T7: Liquidity-based strategies (research framework) (surface_framework). All values and rows are shared.

D2-T9: Formulas used

Identical saved table reproduced at D1-T9: Formulas used (surface_framework). All values and rows are shared.

Profitability verdicts: D2-T4 research-gate candidates

All 17 candidates on pages 66-70. IDs 9 and 12-17 repeat D1 candidates 3-9 (note 54a). Reviewer calculation; not in the source report.

ID (D1 / D2)	Strategy	Debit	Profitable?	Break-even realized vol	4-week hedged P&L at 7.27%	Held to expiry at 7.27%
D2-5	Long straddle May 1.1725	0.01798	**Conditional**	≥ 6.89% / 6.81%	+0.00105	+0.00124 (+6.9%)
D1-3 / D2-9	Long straddle Jun 1.1725	0.02570	**Conditional**	≥ 6.91% / 6.60%	+0.00067	+0.00263 (+10.2%)
D1-5 / D2-13	Long straddle Jul 1.1750	0.03038	**Conditional**	≥ 6.92% / 6.56%	+0.00055	+0.00334 (+11.0%)
D1-7 / D2-15	Long straddle Aug 1.1750	0.03599	**Conditional**	≥ 7.08% / 6.61%	+0.00026	+0.00361 (+10.0%)
D1-9 / D2-17	Long straddle Sep 1.1775	0.03980	**Conditional**	≥ 7.16% / 6.65%	+0.00014	+0.00379 (+9.5%)
D2-1	Long straddle Apr 1.1700 (2 days)	0.00631	**No**	≥ 7.71% (trading-day basis, last 2 days) vs 7.15%	-0.00047 (2-day hold)	same: expires (-7%)
D1-2 / D2-8	Calendar call May/Jun	0.00507 / 0.00502	**No**	≤ 4.27%	-0.00101 / -0.00102	n/a
D1-4 / D2-12	Calendar call Jun/Jul	0.00351	**No**	none: loses at any vol	-0.00098	n/a
D1-6 / D2-14	Calendar call Jul/Aug	0.00403	**No**	none	-0.00100	n/a
D1-8 / D2-16	Calendar call Aug/Sep	0.00297	**No**	none	-0.00091	n/a
D2-4	Calendar call Apr/May	0.00719	**No**	≤ 5.72% (trading-day basis, 2 days)	-0.00034	n/a
D2-2	25D risk reversal Apr	0.00012	**No**	none	-0.00029	n/a
D2-6	25D risk reversal May	0.00086	**No**	≥ 8.45%	-0.00020	n/a
D2-10	25D risk reversal Jun	-0.00077 (credit)	**No**	none	-0.00058	n/a
D2-3	Wing-vs-ATM curvature Apr	-0.00302 (credit)	**No**	none	-0.00076	n/a
D2-7	Wing-vs-ATM curvature May	-0.00965 (credit)	**No**	none	-0.00250	n/a
D2-11	Wing-vs-ATM curvature Jun	-0.01347 (credit)	**No**	none	-0.00226	n/a

Method: each candidate's legs were repriced with Black-76 at the snapshot (ATM IV for near-ATM legs, 25-delta IVs for wing legs); the recomputed Greeks match the report's tables. Cost = half the median IV spread of each expiry × leg vega, paid on entry and on exit (legs that expire are settled, no exit cost). Delta-hedged carry = theta × calendar days + ½ × gamma × F^2 × σ^2 / 252 × trading days. "Break-even vol" is the realized vol (in the same units as IV) at which the position stops losing. Trailing realized vol = 7.27% (5-minute June-future mids, IV convention); implied = the snapshot ATM IV. Units are option price units per one option; no multiplier is verified. Futures hedge cost uses the 0.75-pip median futures spread with daily re-hedging.

- **Straddles (conditional):** they win only because the 8 April implied vol (6.41-6.59%) sits 0.7-0.9 vp below the sample's trailing realized vol. If realized vol over the option's life equals implied, every straddle loses its costs (about -0.0006 to -0.0008). The report's own continuous-hold estimate (-0.0018 vp) is consistent with this: it uses the panel median IV of 7.31%, at which realized ≈ implied. The trailing figure includes the 23 March event, and the fall in IV suggests the market expects calmer conditions. Treat as a hypothesis for a pre-registered realized-vol forecast and paper-trading.
- **Calendars (no):** short gamma with tiny positive theta (0.000007-0.00005/day). Four weeks of carry cannot recover two spread crossings on two expiries (~0.0009). Any profit would have to come from back-month IV rising relative to the front, with cross-month basis risk.

- **Risk reversals (no):** essentially a long-delta position (delta 0.51-0.54) with almost no vega. The skew carry is negligible and the cost is 0.0003-0.0007. Profit depends on direction or a skew view that has no history yet.
- **Curvature trades (no):** short ATM gamma against long wings. In a sample where realized ≥ implied, short gamma loses, and four wide legs double the cost.
- **April expiry (no):** a 29% relative spread (1.10 vp) on 2-day options; the straddle needs about 7.7% realized vol (trading-day basis) over its last two days against 7.15% trailing, and loses about 7% of premium.

Reviewer note follows: 71a →

Dashboard appendix 3: time_of_day

Source HTML: outputs\time_of_day\dashboard.html

Saved narrative and data tables follow. Identical tables are cross-referenced; tables sharing a section with an earlier dashboard are shown as differences only. Wide tables with few rows are transposed; constant columns are stated once in the subtitle.

Time-of-day study: all option expiries/strikes and the June FCEU future

22 sessions, 353 option surface snapshots every 30 min (Berlin 08:30-18:00), 87 slots without a priceable book. June ATM IV median 7.31%. Descriptive only; no multiple-testing correction; April data already viewed.

Data fix: phase04 June-future BBO timestamps were 19800 s early (naive UTC read as machine-local IST); corrected here by a verified constant offset. Slot statistics use medians across sessions; means are shown for reference and are dominated by 2026-03-23 (~120-pip EUR/USD rally, 10:15-13:00 UTC). Median trading-day realized vol 7.47% (252-day) vs June ATM IV 7.31%; realized/implied per trading day = 1.08 (1-minute mids include bid/ask bounce, which biases realized upward).

Long-gamma break-even (June ATM, vol points of vega)

gross = theta/day × (realized variance in units of $\sigma^2/365$ − days charged). theta/day in vp = $\sigma/(2\cdot T\cdot 365)\times 100$. Daily RV from 5-minute mids (1-minute RV/implied 1.08, 5-minute 0.99). Futures hedge costs, the 2026-03-23 tail and fill uncertainty are excluded, so the true net is lower.

Formulas

Spread in vol points = (IVask − IVbid) × 100, per quote; medians across quotes/sessions.

Slot realized variance RVs = $\Sigma\, ri^2$, r = 1-minute log change of two-sided June mid (gaps > 5 min dropped); summarised by the median across sessions.

Share of daily RV = median RVs / Σs median RVs; concentration = share / (slot minutes / 1440).

RV / calendar implied = median RVs / ($\sigma\mathrm{ATM,June}^2$ × slot minutes / 525600): above 1 means the slot realizes more than a uniform calendar-time IV charges for it.

Trading-day implied variance = $\sigma\mathrm{ATM}^2/365 \times 7/5$ (weekend variance assumed ~0).

ATM drift = mean over sessions of (ATM IVslot − ATM IVprevious slot), same expiry; t = mean / (sd / √n).

Split-half consistency = Spearman rank correlation of slot statistics between first and second half of sessions.

Note on dashboard 3: clock fix scope and RV measurement

Applies to the time-of-day appendix (pages 71-92).

Correction. The 19,800 s offset (naive UTC read as IST) is corrected only inside this time-of-day study. Clock-time-dependent phase04/phase05 results elsewhere are not rerun or certified (page 46) and should not be cited until revalidated.

- **Most apparent excess realized vol is noise.** Daily RV / implied is 1.08 on 1-minute mids but 0.99 on 5-minute mids. On a clean measure, options were fairly priced over the sample.
- **One underlying proxy.** All RV figures use the June future; no separate RV validation exists for other monthly futures.
- **87 failed snapshots** out of 440 requested, concentrated early in the day, so early slots (08:30: 2 sessions) carry little weight.

D3-T1: Time-of-day strategies: evidence and verdict (research only)

Source: outputs\time_of_day\dashboard.html; 6 saved data rows. NaN is unavailable, not zero.

Strategy	Evidence in this sample	Rule	Verdict
Hedge-timing for delta-hedged options (gamma scalping windows)	Realized variance peaks at UTC 13:30 (5.6%), 14:30 (5.5%), 12:30 (4.9%), 14:00 (4.7%), 15:00 (4.6%); stable across halves (Spearman 0.85). Overnight slots realize 0.3–0.9× calendar implied.	Long gamma: narrower re-hedge bands 08:00–10:30 and 14:00–17:00 Berlin, wider overnight. Short gamma: reduce or pre-hedge before those windows.	Usable as an overlay. Robust pattern; benefit is hedge-cost/path, not standalone alpha.
Day-only long gamma (flat overnight)	Option hours hold 68% of daily RV in 40% of the clock. Gross edge 0.025 vp/day vs ATM round-trip spread 0.20 vp.	Buy delta-hedged ATM 08:30, sell 18:00 Berlin.	Not viable: net -0.172 vp per day.
Weekday-only long gamma (flat weekend)	Gross 0.096 vp/week vs 0.20 vp spread. Overnight ATM IV change: weeknight +0.033 vp (n=96), weekend/holiday +0.231 vp (n=23).	Buy Monday open, sell Friday close; the converse is short options over the weekend.	Marginal / unproven: net -0.101 vp; only 3 weekends; uses 5-minute RV.
Execution-window selection	Median option spread 0.216–0.233 vp across all slots; ATM 0.18 vp in every slot; slot ranking not stable (Spearman -0.34).	Trade when quotes are present; avoid the open, before 09:30 Berlin, when some underlyings are not priced.	No time edge: quotes look like a fixed-vol market-maker grid.
Intraday IV seasonality	No slot with |t| > 2 (20 tests); mean drifts < 0.03 vp.	None.	Not supported. Moves are far below the 0.18 vp spread.
Option order-flow timing	6 option trades in 22 sessions.	None.	Not implementable.

Profitability verdicts follow: 73a →

D3-T2: Long-gamma break-even (June ATM, vol points of vega)

Source: outputs\time_of_day\dashboard.html; 3 saved data rows. NaN is unavailable, not zero. Same value in every row shown: june_T_years = 0.222; theta_per_day_vp = 0.04509.

strategy	realized_var_units	days_charged	gross_edge_vp	round_trip_cost_vp	net_edge_vp
Intraday only: buy 08:30, sell 18:00 Berlin, every day	0.9478	0.3958	0.02489	0.1967	-0.1718
Weekday only: buy Mon 08:30, sell Fri 18:00 (flat weekend)	6.515	4.396	0.09556	0.1967	-0.1011
Hold continuously for one week (turnover cost excluded)	6.959	7	-0.00183	0	-0.00183

Profitability verdicts: time-of-day strategies (D3-T1, D3-T2)

Every strategy on pages 72-73, with the break-even requirement derived from the report's own formula (theta/day 0.0451 vp; round-trip spread 0.1967 vp = 4.36 days of theta).

Strategy	Gross Edge	Round-Trip Cost	Net Edge	Break-even requires	Verdict
Intraday Long Gamma (Buy 08:30, Sell 18:00 Berlin)	+0.0249 vp/day	0.1967 vp	**-0.1718 vp/day**	~5.0x the realized variance (≈2.2x vol) or a spread ≤ 0.025 vp (8x tighter)	**Not Viable**
Weekday-Only Long Gamma (Flat weekend)	+0.0956 vp/week	0.1967 vp	**-0.1011 vp/week**	~1.34x the realized variance (≈1.16x vol) or a spread ≤ 0.096 vp (about half)	**Marginal / Unproven**
Continuous 1-Week Hold	-0.0018 vp	0.0000 vp (turnover excluded)	**-0.0018 vp**	n/a: realized ≈ implied	**No edge**

Strategy (D3-T1)	Profitable?	Evidence	What would change the verdict
Hedge-timing for delta-hedged options	**Yes, as cost saving**	RV peaks 12:30-15:00 UTC; split-half rank 0.85	Validated futures-cost backtest; anchor to New York time (note 90a)
Day-only long gamma	**No**	Net -0.172 vp/day	Realized vol ≈2.2× higher or a spread ≤0.025 vp
Weekday-only long gamma	**No (closest)**	Net -0.101 vp/week; 3 weekends + Easter	Spread ≤0.096 vp (mid fills) or realized vol ≈16% higher
Execution-window selection	**No**	Spreads flat 0.216-0.233 vp; rank -0.34	None in this data
Intraday IV seasonality	**No**	No |t| > 2 in 20 tests; drifts < 0.03 vp	None in this data
Option order-flow timing	**No**	6 option trades in 22 sessions	Trade data at scale

- The continuous one-week hold (-0.0018 vp) uses the panel median IV of 7.31%, where realized ≈ implied. At the lower 8 April IV the back-month straddles become conditionally positive (notes 54a and 92b), which is why the straddle verdict is 'conditional' while this row shows no edge.

D3-T3: Option liquidity by Berlin slot (all expiries and strikes) | columns 1/2

Source: outputs\time_of_day\dashboard.html; 20 saved data rows. NaN is unavailable, not zero.

slot	sessions	median_spread_vol_points	median_rel_spread	valid_books_per_session	tight_share
08:30	2	0.2326	0.03156	117.5	0.6766
09:00	11	0.2288	0.03207	110	0.6603
09:30	15	0.2229	0.03242	116.5	0.663
10:00	17	0.227	0.0322	114.4	0.6658
10:30	18	0.2246	0.03186	112.5	0.6711
11:00	19	0.2195	0.03187	117.8	0.6731
11:30	21	0.2197	0.03163	113.5	0.6706
12:00	21	0.2165	0.03159	118	0.6788
12:30	21	0.2203	0.03196	112.9	0.6698
13:00	20	0.2204	0.03228	112.5	0.6646
13:30	13	0.229	0.03518	97.31	0.6206
14:00	20	0.2219	0.03181	112.4	0.6713
14:30	20	0.2301	0.03221	103.5	0.6588
15:00	17	0.2318	0.0339	101.2	0.6349
15:30	20	0.2264	0.03295	107.8	0.6521
16:00	19	0.231	0.03279	102.1	0.6521

slot	sessions	median_spread_vol_points	median_rel_spread	valid_books_per_session	tight_share
16:30	19	0.2233	0.03219	112.5	0.6623
17:00	20	0.2249	0.03172	107.5	0.6676
17:30	21	0.2212	0.03201	111.2	0.666
18:00	19	0.2313	0.03304	102.7	0.6474

D3-T3: Option liquidity by Berlin slot (all expiries and strikes) | columns 2/2

Source: outputs\time_of_day\dashboard.html; 20 saved data rows. NaN is unavailable, not zero.

slot	median_quote_age_s
08:30	23.31
09:00	24.21
09:30	18.13
10:00	13.18
10:30	24.27
11:00	27.78
11:30	36.26
12:00	48.21
12:30	28.97
13:00	34.05
13:30	17.92
14:00	21.81
14:30	18.45
15:00	19
15:30	22.17
16:00	17.61
16:30	16.95

slot	median_quote_age_s
17:00	32.71
17:30	21.69
18:00	29.8

D3-T4: ATM IV drift by Berlin slot | columns 1/2

Source: outputs\time_of_day\dashboard.html; 20 saved data rows. NaN is unavailable, not zero.

slot	sessions	mean_d_atm_vp	sd_d_atm_vp	mean_atm_vs_session_mean_vp	mean_d_rr25_vp
08:30	2	NaN	NaN	-0.09911	NaN
09:00	11	-0.006247	0.03335	-0.02345	-0.05772
09:30	15	-0.02502	0.2145	-0.01259	0.008461
10:00	17	0.00376	0.05741	0.0007304	-0.07738
10:30	18	0.006849	0.1718	-0.009488	0.04152
11:00	19	-0.01692	0.07372	-0.02366	-0.007059
11:30	21	0.01559	0.08101	-0.003167	0.008082
12:00	21	0.009873	0.05436	0.003913	0.001875
12:30	21	-0.0005876	0.1012	0.009892	0.0002913
13:00	20	0.02391	0.1385	0.03464	0.01113
13:30	13	0.02397	0.05591	0.01763	-0.04904
14:00	20	0.01558	0.1088	0.0006983	0.01173
14:30	20	0.002485	0.05986	0.003673	-0.08553
15:00	17	-0.01224	0.07363	0.007242	0.07713
15:30	20	0.006307	0.09069	0.002246	0.01069
16:00	19	0.01324	0.06865	0.01181	-0.04589
16:30	19	-0.01865	0.07182	-0.01811	0.04545

slot	sessions	mean_d_atm_vp	sd_d_atm_vp	mean_atm_vs_session_mean_vp	mean_d_rr25_vp
17:00	20	0.01791	0.06749	0.01214	-0.00658
17:30	21	0.003548	0.1046	-0.0007429	0.01835
18:00	19	-0.01474	0.1077	-0.002935	-0.00998

D3-T4: ATM IV drift by Berlin slot | columns 2/2

Source: outputs\time_of_day\dashboard.html; 20 saved data rows. NaN is unavailable, not zero.

slot	t_stat_d_atm
08:30	NaN
09:00	-0.6213
09:30	-0.4516
10:00	0.2701
10:30	0.1691
11:00	-1
11:30	0.8817
12:00	0.8323
12:30	-0.02661
13:00	0.772
13:30	1.546
14:00	0.6402
14:30	0.1857
15:00	-0.6852
15:30	0.311
16:00	0.8406
16:30	-1.132
17:00	1.187

slot	t_stat_d_atm
17:30	0.1554
18:00	-0.5961

D3-T5: June future realized variance by UTC slot | columns 1/3

Source: outputs\time_of_day\dashboard.html; 42 saved data rows. NaN is unavailable, not zero. Same value in every row shown: share_of_clock = 0.02083.

slot_utc	sessions	mean_rv	median_rv	sd_rv	median_spread_pips
00:00	18	3.564e-07	2.472e-07	3.232e-07	4.5
00:30	18	3.661e-07	2.151e-07	3.619e-07	4.75
01:00	19	5.661e-07	2.938e-07	1.078e-06	4
01:30	19	3.221e-07	2.085e-07	3.921e-07	4
02:00	19	3.093e-07	1.644e-07	3.261e-07	1
02:30	19	1.564e-07	1.561e-07	6.192e-08	1
03:00	19	1.281e-07	1.223e-07	5.305e-08	1
03:30	19	1.169e-07	9.541e-08	6.674e-08	0.9
04:00	20	9.757e-08	8.516e-08	5.493e-08	0.9
04:30	21	1.001e-07	9.333e-08	4.605e-08	0.9
05:00	21	3.223e-07	1.297e-07	6.816e-07	0.8
05:30	21	3.337e-07	1.931e-07	4.7e-07	0.8
06:00	21	3.606e-07	2.502e-07	2.672e-07	0.8
06:30	20	3.93e-07	2.756e-07	3.044e-07	0.8
07:00	21	6.638e-07	4.989e-07	5.35e-07	0.75
07:30	21	6.318e-07	5.318e-07	3.838e-07	0.7
08:00	21	7.532e-07	6.556e-07	3.904e-07	0.7

slot_utc	sessions	mean_rv	median_rv	sd_rv	median_spread_pips
08:30	21	6.958e-07	6.182e-07	3.958e-07	0.75
09:00	21	6.2e-07	5.609e-07	4.208e-07	0.75
09:30	20	4.206e-07	3.641e-07	2.468e-07	0.7
10:00	21	4.35e-07	4.076e-07	2.386e-07	0.7
10:30	21	5.613e-07	4.154e-07	4.164e-07	0.75
11:00	21	5.183e-06	4.587e-07	2.145e-05	0.75
11:30	21	9.929e-07	5.859e-07	1.855e-06	0.8
12:00	21	8.08e-07	5.985e-07	7.329e-07	0.8
12:30	21	1.02e-06	8.431e-07	6.905e-07	0.7
13:00	21	9.775e-07	6.78e-07	7.537e-07	0.7
13:30	21	1.299e-06	9.572e-07	1.148e-06	0.8
14:00	21	1.318e-06	7.994e-07	1.316e-06	0.8
14:30	21	1.403e-06	9.457e-07	1.374e-06	0.7
15:00	21	1.09e-06	7.949e-07	8.055e-07	0.7
15:30	21	6.777e-07	6.312e-07	3.526e-07	0.7
16:00	21	6.794e-07	6.608e-07	3.919e-07	0.7
16:30	21	6.098e-07	3.725e-07	7.38e-07	0.75
17:00	21	7.272e-07	3.97e-07	5.964e-07	0.9
17:30	21	4.818e-07	3.842e-07	3.614e-07	1
18:00	20	6.923e-07	3.282e-07	7.47e-07	1

slot_utc	sessions	mean_rv	median_rv	sd_rv	median_spread_pips
18:30	20	5.816e-07	3.306e-07	5.534e-07	1.45
19:00	20	5.985e-07	2.616e-07	1.025e-06	1
19:30	20	3.199e-07	1.904e-07	2.969e-07	1
20:00	19	9.436e-07	1.832e-07	2.544e-06	1
20:30	19	2.537e-07	1.122e-07	4.244e-07	4

D3-T5: June future realized variance by UTC slot | columns 2/3

Source: outputs\time_of_day\dashboard.html; 42 saved data rows. NaN is unavailable, not zero. Same value in every row shown: share_of_clock = 0.02083.

slot_utc	minutes_obs	share_of_daily_rv	median_share_of_daily_rv	max_session	max_session_share_of_slot
00:00	26.39	0.01214	0.01446	2026-04-08	0.2179
00:30	28.61	0.01246	0.01258	2026-03-31	0.2045
01:00	28.37	0.01928	0.01719	2026-04-02	0.4614
01:30	29.42	0.01097	0.0122	2026-04-02	0.2986
02:00	29.47	0.01053	0.009614	2026-03-19	0.2214
02:30	29.11	0.005325	0.009131	2026-03-12	0.0896
03:00	29.05	0.004361	0.007153	2026-03-12	0.1041
03:30	28.79	0.00398	0.005581	2026-03-24	0.1188
04:00	28.65	0.003322	0.004981	2026-03-23	0.1169
04:30	28.95	0.003407	0.005459	2026-04-08	0.1061
05:00	29.48	0.01097	0.007589	2026-03-09	0.4768
05:30	29.52	0.01136	0.01129	2026-03-09	0.3216
06:00	28.57	0.01228	0.01464	2026-03-09	0.1345
06:30	29.95	0.01338	0.01612	2026-03-09	0.1546
07:00	27.81	0.0226	0.02918	2026-03-19	0.1465
07:30	29.19	0.02151	0.03111	2026-04-01	0.1455

slot_utc	minutes_obs	share_of_daily_rv	median_share_of_daily_rv	max_session	max_session_share_of_slot
08:00	29.62	0.02565	0.03835	2026-04-07	0.1121
08:30	29.67	0.02369	0.03616	2026-03-24	0.1228
09:00	29.48	0.02111	0.03281	2026-03-11	0.1391
09:30	30	0.01432	0.0213	2026-03-19	0.1478
10:00	29.52	0.01481	0.02385	2026-04-07	0.1073
10:30	30	0.01911	0.0243	2026-03-13	0.1464
11:00	29.9	0.1765	0.02683	2026-03-23	0.9076
11:30	30	0.03381	0.03427	2026-03-23	0.4317
12:00	29.95	0.02751	0.03501	2026-03-23	0.2189
12:30	29.95	0.03473	0.04932	2026-03-23	0.14
13:00	29.14	0.03329	0.03966	2026-03-19	0.1431
13:30	29.95	0.04424	0.05599	2026-03-19	0.1837
14:00	30	0.04489	0.04676	2026-03-11	0.1907
14:30	30	0.04778	0.05532	2026-04-02	0.2206
15:00	29.95	0.03711	0.0465	2026-03-20	0.1309
15:30	30	0.02308	0.03692	2026-03-12	0.08988
16:00	30	0.02314	0.03865	2026-03-23	0.1233
16:30	30	0.02077	0.02179	2026-03-31	0.2859
17:00	30	0.02476	0.02322	2026-03-31	0.1469
17:30	28.76	0.01641	0.02247	2026-03-24	0.1631

slot_utc	minutes_obs	share_of_daily_rv	median_share_of_daily_rv	max_session	max_session_share_of_slot
18:00	29.9	0.02358	0.0192	2026-03-20	0.1877
18:30	29.7	0.01981	0.01934	2026-03-10	0.1716
19:00	29.65	0.02038	0.0153	2026-03-19	0.4034
19:30	29.25	0.01089	0.01114	2026-03-19	0.1707
20:00	28.26	0.03213	0.01071	2026-03-26	0.6035
20:30	21.95	0.008639	0.006561	2026-03-24	0.3835

D3-T5: June future realized variance by UTC slot | columns 3/3

Source: outputs\time_of_day\dashboard.html; 42 saved data rows. NaN is unavailable, not zero. Same value in every row shown: share_of_clock = 0.02083.

slot_utc	rv_concentration	realized_vol_ann_if_slot_all_day	rv_over_calendar_implied	median_rv_over_calendar_implied	t_stat_vs_uniform
00:00	0.694	0.06581	1.17	0.8111	-4.5
00:30	0.604	0.06139	1.201	0.7059	-3.906
01:00	0.8249	0.07174	1.858	0.9641	-0.5379
01:30	0.5855	0.06045	1.057	0.6844	-4.193
02:00	0.4615	0.05366	1.015	0.5394	-5.213
02:30	0.4383	0.0523	0.5132	0.5123	-38.22
03:00	0.3433	0.04629	0.4203	0.4013	-46.93
03:30	0.2679	0.04089	0.3836	0.3131	-38.03
04:00	0.2391	0.03863	0.3202	0.2795	-48.99
04:30	0.262	0.04044	0.3284	0.3063	-59.62
05:00	0.3643	0.04768	1.058	0.4258	-2.535
05:30	0.5422	0.05816	1.095	0.6336	-3.564
06:00	0.7025	0.06621	1.183	0.821	-5.809
06:30	0.7737	0.06948	1.29	0.9043	-4.498
07:00	1.401	0.09349	2.178	1.637	-0.3031
07:30	1.493	0.09652	2.073	1.745	-0.8051

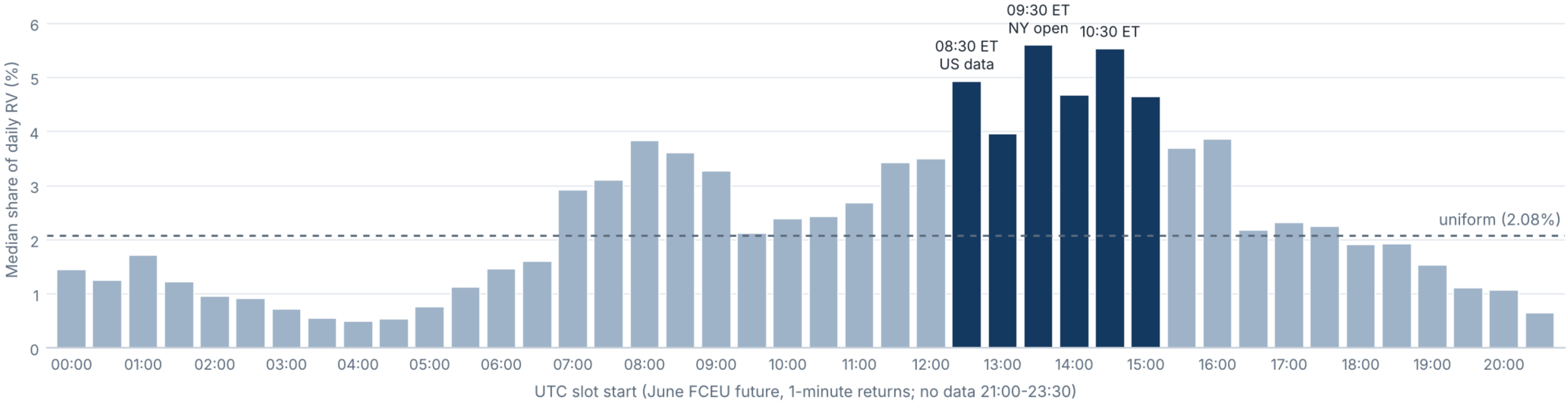
slot_utc	rv_concentration	realized_vol_ann_if_slot_all_day	rv_over_calendar_implied	median_rv_over_calendar_implied	t_stat_vs_uniform
08:00	1.841	0.1072	2.472	2.151	0.6338
08:30	1.736	0.1041	2.283	2.029	-0.03905
09:00	1.575	0.09913	2.035	1.841	-0.8623
09:30	1.022	0.07987	1.38	1.195	-5.05
10:00	1.145	0.08451	1.427	1.338	-5.075
10:30	1.166	0.08531	1.842	1.363	-1.518
11:00	1.288	0.08965	17.01	1.505	0.958
11:30	1.645	0.1013	3.258	1.923	0.7256
12:00	1.681	0.1024	2.652	1.964	0.6802
12:30	2.367	0.1215	3.347	2.767	2.128
13:00	1.904	0.109	3.208	2.225	1.692
13:30	2.688	0.1295	4.264	3.141	2.396
14:00	2.244	0.1183	4.326	2.623	2.155
14:30	2.655	0.1287	4.604	3.103	2.347
15:00	2.232	0.118	3.576	2.608	2.222
15:30	1.772	0.1052	2.224	2.071	-0.2799
16:00	1.855	0.1076	2.23	2.169	-0.2313
16:30	1.046	0.08079	2.001	1.222	-0.5549
17:00	1.115	0.0834	2.386	1.303	0.2148
17:30	1.079	0.08204	1.581	1.261	-2.757

Reviewer note follows: 90a →

slot_utc	rv_concentration	realized_vol_ann_if_slot_all_day	rv_over_calendar_implied	median_rv_over_calendar_implied	t_stat_vs_uniform
18:00	0.9216	0.07583	2.272	1.077	-0.04121
18:30	0.9282	0.0761	1.909	1.085	-0.9501
19:00	0.7346	0.0677	1.964	0.8586	-0.4394
19:30	0.5346	0.05776	1.05	0.6248	-5.714
20:00	0.5143	0.05665	3.096	0.6011	0.4187
20:30	0.3149	0.04433	0.8326	0.368	-4.575

Note on D3-T5: anchor the hedge overlay to New York time

Median share of daily realized variance by UTC slot, from D3-T5 columns 2/3. Dark bars: the 12:30-15:00 UTC high-variance window.



- **The peaks are US events.** In this sample the US was already on daylight time (from 8 March 2026), so 12:30 UTC = 08:30 ET data releases and 13:30-14:30 UTC = the New York open and 10:00 ET releases.
- **Berlin clock time drifts mid-sample.** Berlin moved to summer time on 29 March 2026, so the report's rule (narrow bands 14:00-17:00 Berlin) covers different US events before and after that date. Define the window in New York time (about 08:30-11:00 ET) or UTC.
- **Use medians only.** At 11:00 UTC, one session (23 March, a ~120-pip rally) supplies 91% of the slot's total RV: mean share 17.7% vs median 2.7%. Mean-based columns in D3-T5 are dominated by that day.
- Asian-session slots (02:30-04:30 UTC) realize only 0.3-0.5× calendar implied: wider bands there are well supported.

Reviewer note follows: 91a →

D3-T6: Realized variance by weekday

Source: outputs\time_of_day\dashboard.html; 5 saved data rows. NaN is unavailable, not zero.

weekday	mean_daily_rv	sessions	realized_vol_ann_252
Friday	1.973e-05	3	0.07052
Monday	5.244e-05	4	0.115
Thursday	2.845e-05	4	0.08468
Tuesday	2.245e-05	5	0.07521
Wednesday	2.217e-05	5	0.07474

Note on D3-T6: realized variance by weekday

Applies to page 91.

- **Monday's 11.5% is one event.** 23 March 2026 was a Monday; with only 4 Monday sessions, it lifts Monday's annualized realized vol from the ~7.5% seen on other weekdays.
- **21 vs 22 sessions.** The weekday counts (Mon 4, Tue 5, Wed 5, Thu 4, Fri 3) sum to 21, not the 22 sessions stated elsewhere. Good Friday (3 April) and Easter Monday (6 April) are absent, as expected; the 22nd session label should be reconciled.
- No weekday conclusion is supportable with 3-5 sessions per day.

Reviewer note follows: 92a →

D3-T7: Overnight vs intraday ATM IV change (vol points, mean by expiry)

Source: outputs\time_of_day\dashboard.html; 7 saved data rows. NaN is unavailable, not zero.

expiry	intraday_vp / mean	intraday_vp / count	overnight_vp / mean	overnight_vp / count
2026-03-13	-0.6228	4	2.658	4
2026-04-10	-0.005535	20	0.04678	20
2026-05-08	-0.02324	20	-0.03855	20
2026-06-12	-0.01148	20	-0.04522	20
2026-07-10	-0.01123	20	-0.04658	20
2026-08-14	-0.03796	20	-0.01847	20
2026-09-11	-0.05107	15	-0.009372	15

Note on D3-T7: weekend sample and implied vs realized

Applies to page 92 and the weekend observation on page 44.

- **Weekend evidence is four gaps.** Weekend/holiday ATM change +0.231 vp (23 expiry-gap rows) comes from 3 regular weekends plus the Easter closure (2-7 April). This reconciles "only 3 weekends" (page 72) with "4 distinct date gaps" (page 44). The rows are not independent.
- **The March expiry row is an outlier.** 2026-03-13 shows intraday -0.62 vp and overnight +2.66 vp from 4 observations each, near expiry; exclude it from any pooled weekend figure.

Implied vs trailing realized on the snapshot date (reviewer calculation)

Measure	Value	Source
June ATM IV, panel median	7.31%	Page 71
June ATM IV, 8 April 14:00 UTC snapshot	6.43% (≈6.32% trading-day basis, ×0.983)	Page 48
Trailing realized vol, 5-minute mids	≈7.15% (trading-day basis)	Derived: √0.99 × 7.31% × 0.983
Realized / implied variance on snapshot date	≈1.28	Derived

- Implied vol compressed by about 0.9 vp over the sample, so on 8 April long volatility looked cheap against trailing realized. This is backward-looking and includes the 23 March event; it is a hypothesis for the missing horizon-matched forecast, to be pre-registered and tested on new sessions, not a signal.

Strategy scorecard: every strategy as a position

All strategies in the report in one view, with the profitability verdict and the action. Reviewer calculation where marked in notes 54a, 57a, 70a and 73a.

Strategy (report location)	Positions	Profitable?	Key evidence	Action
Momentum option signal (p. 23)	278 dev trades	**No**	Net -0.194, win rate 48.9%, CI [-1.00, 0.57]	Drop
Long straddles May-Sep (D1-1/3/5/7/9, D2-5/9/13/15/17)	6 distinct	**Conditional**	Break-even realized 6.56-6.81% vs 7.27% trailing; +6.8% to +11.0% to expiry; loses ~0.0007 if realized = implied	Paper-trade with a pre-registered RV forecast
Long straddle Apr (D2-1)	1	**No**	Needs ~7.7% realized over 2 days vs 7.15%; 29% spread	Drop
Calendar calls (D1-2/4/6/8, D2-4/8/12/14/16)	6 distinct	**No**	Carry cannot cover ~0.0009 crossing cost; basis risk	Drop unless a term-structure view exists
25D risk reversals (D2-2/6/10)	3	**No**	Cost 0.0003-0.0007; delta 0.51-0.54, no skew history	Wait for RR history
Wing-vs-ATM curvature (D2-3/7/11)	3	**No**	-0.0008 to -0.0025 over 4 weeks; short gamma	Drop
Intraday long gamma (D3-T2)	schedule	**No**	Net -0.172 vp/day	Drop
Weekday-only long gamma (D3-T2)	schedule	**No (closest)**	Net -0.101 vp/week; needs half the spread	Retest with mid fills
Continuous hold (D3-T2)	schedule	**No edge**	-0.0018 vp at panel IV	Superseded by the straddle analysis
Hedge-timing overlay (D3-T1)	overlay	**Yes, as cost saving**	RV peak 12:30-15:00 UTC, rank stability 0.85	Validate with futures costs
Execution-window selection / IV seasonality / order-flow timing (D3-T1)	n/a	**No**	Flat spreads; no IV drift t-stat above 2; 6 option trades	Drop
Liquidity strategies (D1-T7)	overlays	Execution aids only	See Section 5	Use the filter; research the rest

Bottom line. Nothing is profitable as tested. The only family with a positive margin is delta-hedged long straddles in May-September, and only if realized vol stays near the sample's 7.27% while implied sits at 6.4-6.6%. That is a forecastable, testable hypothesis, not an established edge.

What else can be done

Further work ranked by how directly the data in this report supports it.

Idea	Why the data supports it	How to test	Data needed	Priority
Conditional long-vol carry in Jul-Sep straddles	Only family with a positive break-even margin; Jul-Sep quotes all tight	Enter delta-hedged when forecast RV − ask IV > threshold set in advance; hold to expiry; daily NY-time hedging	Daily snapshots over new sessions	High
Realized-vol forecast (HAR-RV on futures)	4.1M futures BBO rows; futures are liquid even though options are not	Forecast next 30-90 day RV from daily/weekly/monthly RV; chronological validation vs IV	Longer futures history	High
Hedge-timing overlay backtest	Only robust pattern (Spearman 0.85)	Compare fixed vs time-varying hedge bands on futures P&L and turnover	Futures fees and latency	High
Mid / passive fill study	Halving the spread makes weekday-only long gamma break even (≤0.096 vp)	Replay MBO queues: how often a mid-priced order would have filled before the quote moved	Existing MBO (already available)	Medium
Event-day gamma	IV does not reprice intraday (fixed-vol grid) while RV concentrates at US releases	Test straddle carry only on scheduled US data days (CPI, NFP) vs other days	Economic calendar	Medium
Skew dynamics (sticky-strike vs sticky-delta)	22-session panel has RR25 changes and futures returns	Regress RR25 changes on futures returns; use for vanna-aware hedging and as a precondition for RR trades	Existing panel	Medium
Synthetic forwards for serial months	Monthly futures never trade; parity agrees within ~1 pip	Use put-call parity forwards instead of interpolated futures to hedge Apr/May/Jul/Aug	Existing quotes	Medium
Listed vs OTC vol relative value	Listed IV may deviate from OTC EUR/USD ATM vol	Compare EUCO ATM IV with OTC 1M/3M ATM by expiry	OTC vol quotes	Low
Snapshot history for z-scores	Every RR/BF signal is insufficient_history	Store daily snapshots at a fixed time for 3+ months	Ongoing capture	Prerequisite

Reviewer conclusions and next steps

Final recommendation (from the reviewer analysis, note 47b).

Reject for live deployment as standalone alpha strategies.

The research is rigorous and correctly concludes that, on this sample, no strategy shows a cost-adjusted edge. The market is quote-driven with almost no trading, front-week options are very wide, and futures roll mapping adds basis risk. Back-month liquidity is adequate, so the binding constraint there is the absence of a validated realized-vol forecast rather than execution cost.

Next Steps / Actionable Refinements:

1. **Validate the Hedge-Timing Overlay:** backtest adaptive re-hedge bands on the futures leg with real futures spreads, fees, latency and turnover, anchoring the high-RV window (12:30-15:30 UTC) to New York time and using median-based profiles that exclude 23 March-type outliers.
2. **Build the Missing Realized-Vol Forecast:** pre-register a horizon-matched RV forecast (target, fees and hedge rule fixed in advance) and test it chronologically on new sessions. This is the gate for every straddle candidate and the weekday-only schedule.
3. **Collect More Independent Data:** more sessions (especially more weekends), trade/fill data and a multi-session snapshot history so prior-only RR/BF z-scores can actually be computed.
4. **Revalidate Legacy Results:** rerun the clock-dependent phase04/phase05 experiments with the 19,800 s correction before citing them.
5. **Research Only, No Deployment:** passive quoting and calendar structures remain research items until queue-position/adverse-selection simulation and a tick-level model of calendar-spread liquidity exist.

Shared metric dictionary: every defined field

Units, exact calculation, interpretation and limitations. All 103 currently registered fields are included, even if a particular snapshot does not display them.

Metric / field	Units	Definition / method	Interpretation / limitations
asof	UTC timestamp	Snapshot decision time.	Only information available at this time belongs in the snapshot.
atm_iv	annualized decimal volatility	Interpolate OTM mid IV at log(K/F) = 0.	0.0731 = 7.31% annualized IV. Measures the volatility level; no extrapolation beyond observed strikes.
atm_slope_vol_points	volatility points	100 * (back ATM IV - front ATM IV).	Positive = back expiry higher IV. This is a difference, not a per-year slope.
atm_total_variance	squared log return	atm_iv^2 * dte / 365.	Cumulative model variance to expiry. Different underlying months prevent a standalone calendar-arbitrage conclusion.
avoid	quote count	Count of invalid displayed books.	Excluded from execution research.
back_expiry	date	Later member of an adjacent expiry pair.	Not necessarily a constant-underlying calendar spread.
bf10	decimal volatility difference	(call_10d_iv + put_10d_iv)/2 - atm_iv.	10-delta butterfly: positive = wings richer than ATM (smile curvature). Not the executable price of a butterfly package.
bf25	decimal volatility difference	(call_25d_iv + put_25d_iv)/2 - atm_iv.	25-delta butterfly: positive = wings richer than ATM (smile curvature). Not the executable price of a butterfly package.
call_10d_iv	annualized decimal volatility	Interpolate OTM call mid IV at absolute forward delta 10/100, where forward delta = abs(delta)/DF.	No wing extrapolation. Missing means insufficient eligible strikes, not zero volatility.
call_25d_iv	annualized decimal volatility	Interpolate OTM call mid IV at absolute forward delta 25/100, where forward delta = abs(delta)/DF.	No wing extrapolation. Missing means insufficient eligible strikes, not zero volatility.
count	observation count	Number of nonmissing observations under the parent metric.	Expiry-gap observations can share a session/weekend and are not independent.
coverage_25d	boolean	True when both rr25 and bf25 are finite.	False means missing 25-delta/ATM coverage, not zero skew.
days_charged	calendar days	Time decay charged: 9.5/24 intraday, 4+9.5/24 weekday-only, 7 continuously.	Simplified fixed-IV holding schedule, not exact path-dependent theta.

Metric / field	Units	Definition / method	Interpretation / limitations
delta	price sensitivity to future	Black-76 DF*N(d1) for calls, -DF*N(-d1) for puts; candidates sum quantity-weighted legs.	Forward-price sensitivity, not spot delta. Hedge separately by underlying month.
delta_bucket	absolute forward-delta band	abs(Black-76 discounted delta)/DF: <10%, 10-25%, 25-40%, 40-60%, >60%.	D denotes delta, not days. ATM 40-60D is a bucket, not exact K=F.
displayed_ask_size	displayed contracts	Sum of best-ask size over valid books in expiry.	Sizes may cancel before a fill.
displayed_bid_size	displayed contracts	Sum of best-bid size over valid books in expiry.	Resting displayed size, not trades, inventory or open interest.
dte	calendar days	365 * T, with T = seconds to expiration / (365 * 86400).	Fractional remaining life, not trading days.
edge_price_units	option price units	Executable package edge for a bounded payoff: -entry cost, less discounted payoff width for a reverse vertical.	Missing for candidates without a defined price-consistency edge; not expected return.
entry_debit_price_units	option price units	Sum q_i * (ask_i for buys, bid_i for sells) + sum |q_i| * (fee + slippage).	Positive = debit; negative = credit. No currency P&L without a verified multiplier.
expiry	date	Option expiration date; model expiration is 13:00 UTC.	Each expiry is mapped to its own monthly future.
failed_snapshots	snapshot count	Requested slots without a priceable underlying or usable IV points.	Missingness is not random; early slots/holidays and event-time gaps can bias comparisons.
forward_variance	annualized variance	(back total variance - front total variance) / (T_back - T_front).	Negative is a research inconsistency flag, not proof of arbitrage across different futures.
forward_vol	annualized decimal volatility	sqrt(forward_variance) if nonnegative; otherwise missing.	Model diagnostic; not an executable forward-volatility quote.
front_expiry	date	Earlier member of an adjacent expiry pair.	Term comparisons reference different monthly futures.
future_bbo_clock_offset_seconds_corrected	seconds	Measured raw-MBO first timestamp minus phase04-BBO first timestamp, verified approximately constant across sessions.	Added to BBO timestamps in this study only; does not repair earlier frozen experiments.
gamma	price sensitivity per future-price squared	Black-76 DF*phi(d1)/(F*sigma*sqrt(T)); candidates sum quantity-weighted legs.	Positive = convex exposure; Greek sums do not remove cross-month basis risk.

Metric / field	Units	Definition / method	Interpretation / limitations
gross_edge_vp	volatility points of vega	theta_per_day_vp * (realized_var_units - days_charged).	Approximate delta-hedged gamma/theta edge, before option spread and hedge costs; not a backtested return.
id	integer	Candidate identifier within this generated study.	Use with the matching frozen opportunities manifest only.
interpretation	text	Qualifier attached to the term-structure comparison.	research_only_different_underlying_months forbids treating it as a same-underlying arbitrage test.
intraday_vp	volatility points	100*(last available ATM IV - first available ATM IV) for same expiry/session.	Not necessarily exchange open-to-close. Paired with an observed next session in the overnight table.
june_T_years	ACT/365 years	Median June-option dte / 365.	Single representative maturity used in the approximate break-even calculation.
june_atm_iv_median	annualized decimal volatility	Median ATM IV over observed June expiry snapshots with a direct underlying route.	Reference sigma in the RV/implied and gamma break-even comparisons; 0.0731 = 7.31%.
legs	signed quantities and symbols	Each option instrument and its quantity.	Positive quantities buy; negative sell. Multi-leg execution is not assumed atomic.
liquidity_role	category	avoid for invalid books; taker_eligible if tight; passive_quote_research if vol spread >= 1 point; otherwise patient_limit_order.	Role is a research suggestion, not entry approval.
max_session	session date	Session with the largest RV in this slot.	Identifies the event dominating the mean.
max_session_share_of_slot	fraction	Largest session RV / sum of RV across sessions for this slot.	Near 1 means the slot statistic is dominated by one session.
mean	units of parent metric	Arithmetic mean of nonmissing observations under the parent column (intraday_vp or overnight_vp).	In the table it is by expiry; weekend summary pools expiry-gaps, not independent weekends.
mean_atm_vs_session_mean_vp	volatility points	100*(ATM IV - same-session same-expiry average IV), averaged across expiries then sessions.	Descriptive centering uses the whole session, including later snapshots; not a live signal.
mean_d_atm_vp	volatility points per observed interval	Within session/expiry: 100*(ATM IV - preceding available snapshot IV); average expiries per session/slot, then sessions.	A positive mean is rising IV, not a proven trade edge. Gaps can exceed the nominal interval.

Metric / field	Units	Definition / method	Interpretation / limitations
mean_d_rr25_vp	volatility points per observed interval	100*(RR25 - preceding available same-session/expiry RR25), averaged across expiries then sessions.	Positive = call-minus-put skew increasing. Missing wings are excluded.
mean_daily_rv	squared log return	Mean across sessions of sum of retained one-minute return squares, grouped by weekday.	Uses all retained sessions here; unlike the summary daily RV calculation, does not exclude sessions with <=600 observations.
mean_rv	squared log return	Mean across sessions of slot RV, where RV = sum of squared one-minute June-future mid log returns.	Sensitive to outliers and microstructure noise. Returns spanning >5 minutes are excluded.
median_daily_rv5_over_implied_trading_day	ratio	Median daily RV from five-minute sampled mids / (June reference ATM IV^2/365 * 7/5), on the retained daily session set.	Robustness check, not an independent experiment; sampled returns can bridge missing observations.
median_daily_rv_over_implied_trading_day	ratio	Median daily one-minute RV / (June reference ATM IV^2/365 * 7/5), retaining sessions with >600 returns.	Weekend variance is assumed zero; one-minute mids can inflate RV through microstructure noise.
median_depth_imbalance	ratio [-1, 1]	Median of (bid_size - ask_size)/(bid_size + ask_size).	Positive = more displayed bid size; not a proven directional signal.
median_quote_age_s	seconds	Median quote_age_seconds across valid quotes in slot.	Same age convention as median_quote_age_seconds; unknown ages are excluded.
median_quote_age_seconds	seconds	Median of max(asof - bid_ts, asof - ask_ts) over valid books.	Age of latest surviving best-price order; not necessarily time since the price last changed. Missing timestamps reduce provenance.
median_rel_spread	fraction of option mid	Median of (ask - bid) / ((ask + bid)/2).	0.05 = 5% of premium. Surface: books within expiry; time table: pooled valid quote observations within slot.
median_rv	squared log return	Median across observed sessions of slot RV.	Robust typical-slot movement, not an option price or return forecast.
median_rv_over_calendar_implied	ratio	median_rv / (June reference ATM IV^2 * slot_minutes / 525600).	Above 1 = typical slot realizes more than uniform calendar IV allocates. Not evidence of tradable profit after costs.
median_share_of_daily_rv	fraction	median_rv in slot / sum of median_rv over observed slots.	Median-based normalized profile, not the median of each day's variance shares.
median_spread_pips	EUR/USD pips	Median across sessions of median futures ask-bid spread in slot, multiplied by 10000.	1 pip = 0.0001 in underlying price; not an option volatility point.

Metric / field	Units	Definition / method	Interpretation / limitations
median_spread_vol_points	volatility points	Median of 100*(iv_ask - iv_bid); fallback (ask - bid)/vega_per_vol_point.	Cost in comparable IV units; time slot table pools valid quotes across sessions/expiries/strikes, so composition matters.
metric	metric name	Statistic being compared with its strictly earlier same-expiry observations: atm_iv, rr25 or bf25.	Click the metric value for its definition. This is not a feature-importance score.
minutes_obs	observations per session	Mean number of retained one-minute return observations in slot.	May include gaps up to 5 minutes; not a guarantee of complete clock coverage.
n_eligible	quote count	Number of rows with study_eligible = True.	Passes model/book/spread checks; not proof of executable liquidity.
n_prior	observation count	Count of finite, strictly earlier same-expiry history values.	Not necessarily independent sessions. No current/future rows; at least min_history observations required.
n_quotes	quote count	Number of input option rows in this expiry.	Includes quotes that fail the analytical quality gate.
n_tight	book count	Valid books with relative spread <= 0.05.	5% is a research classification threshold, not guaranteed cheap execution.
n_valid_books	book count	Books with bid > 0, ask > bid and both displayed sizes >= 1.	Book validity is weaker than analytical or paper-entry eligibility.
net_edge_vp	volatility points of vega	gross_edge_vp - round_trip_cost_vp.	Net of the stated option spread only, not all-in realized profit.
option_quoting_hours_share_of_clock	fraction	Number of observed option-hour RV buckets * slot_minutes / 1440.	Full 9.5-hour window is 39.6% of a 24-hour day.
option_quoting_hours_share_of_daily_rv	fraction	Sum of median_share_of_daily_rv for Berlin slots from 08:30 inclusive to 18:00 exclusive.	68% means that fraction of the normalized median RV profile, not the average of daily shares.
option_snapshots	snapshot count	Distinct successfully priced (session, slot) pairs.	An option snapshot contains multiple instruments and expiries; it is not an independent trade.
option_trades_in_sample	trade-message count	Option-kind MBO records with action T, joined on session_date and instrument_id.	Too few here to estimate reliable order-flow or fill-timing effects; not quote count.
overnight_vp	volatility points	100*(next observed session's first ATM IV - preceding session's last ATM IV), same expiry.	Includes weekend/holiday gaps and quote changes; not option P&L.

Metric / field	Units	Definition / method	Interpretation / limitations
passive_quote_research	quote count	Count of valid non-tight books with spread >= 1 volatility point.	Requires queue/fill and adverse-selection research.
patient_limit_order	quote count	Count of valid non-tight books not meeting passive spread threshold.	Working a limit order needs a fill model.
put_10d_iv	annualized decimal volatility	Interpolate OTM put mid IV at absolute forward delta 10/100, where forward delta = abs(delta)/DF.	No wing extrapolation. Missing means insufficient eligible strikes, not zero volatility.
put_25d_iv	annualized decimal volatility	Interpolate OTM put mid IV at absolute forward delta 25/100, where forward delta = abs(delta)/DF.	No wing extrapolation. Missing means insufficient eligible strikes, not zero volatility.
realized_var_units	calendar implied-day units	Window RV / (June reference ATM IV^2/365); intraday uses option-hours share * median 5-minute daily RV; weekday uses (4 + option-hours share) * that daily RV.	Continuous week uses 5 trading days. Assumes representative days and negligible weekend RV.
realized_vol_ann_252	annualized decimal volatility	sqrt(mean_daily_rv * 252).	Trading-day annualization. Not directly comparable to calendar-time IV without weekend assumptions.
realized_vol_ann_from_median_trading_day	annualized decimal volatility	sqrt(median daily one-minute RV * 252), using sessions with >600 returns.	Typical trading-day annualization; differs from the 365-day IV convention.
realized_vol_ann_if_slot_all_day	annualized decimal volatility	sqrt(median_rv * 525600 / slot_minutes).	Hypothetical calendar annualization if this slot's variance rate lasted all day; not realized daily vol.
reason	text	Rule/coverage evidence that generated the candidate.	Rules and thresholds are not fitted predictive explanations.
round_trip_cost_vp	volatility points of vega	Median June ATM bid-ask IV spread for opening and closing at unchanged quotes; zero in the continuous-hold row.	A full spread = two half-spread crossings. Continuous hold intentionally excludes turnover; all rows omit futures hedge costs/fees.
rr10	decimal volatility difference	call_10d_iv - put_10d_iv.	10-delta risk reversal: negative = puts richer than calls. Multiply by 100 for volatility points; -0.002 = -0.20 points.
rr25	decimal volatility difference	call_25d_iv - put_25d_iv.	25-delta risk reversal: negative = puts richer than calls. Multiply by 100 for volatility points; -0.002 = -0.20 points.
rv_concentration	ratio	median_share_of_daily_rv / share_of_clock.	Above 1 = more daily variance per clock minute than a uniform profile; no cost-adjusted alpha implication.

Metric / field	Units	Definition / method	Interpretation / limitations
rv_over_calendar_implied	ratio	mean_rv / (June reference ATM IV^2 * slot_minutes / 525600).	Mean-based comparison to uniform calendar variance; reference IV is the panel median, not a contemporaneous option hedge.
sd_d_atm_vp	volatility points	Sample standard deviation (ddof=1) of session-level mean ATM changes into slot.	Dispersion, not the standard error of the mean.
sd_rv	squared log return	Sample standard deviation (ddof=1) across session slot RV values.	High values indicate event/outlier sensitivity.
sessions	session count	Number of distinct observed sessions in the group.	For IV drift, includes session/slot rows with missing changes; actual nonmissing drift count can be smaller. Small samples and dependencies weaken inference.
share_of_clock	fraction of 24 hours	Slot length in minutes / 1440.	Calendar-time benchmark, not exchange trading-hours share.
share_of_daily_rv	fraction	mean_rv in slot / sum of mean_rv over observed slots.	Mean-based normalized share; not the mean of daily shares. Different slot coverage can bias it.
slot	HH:MM Europe/Berlin	Snapshot time for options; start of time bucket for futures returns.	Berlin observes DST. Missing snapshots mean adjacent IV changes can span more than one nominal slot.
slot_utc	HH:MM UTC	UTC snapshot time or start of futures-return bucket.	Use UTC for fixed-UTC event comparisons; Berlin offset changes with DST.
split_half_spearman	rank correlation [-1, 1]	Pearson correlation of slot ranks between chronological first and second session halves, using common finite slots.	Near +1 = stable ordering; near 0 = unstable. Not out-of-sample validation or SHAP importance.
status	category	History: insufficient_history, missing_current_coverage, zero_history_variance, normal or extreme (|z| >= configured threshold). Candidates: PRICE_ALERT, WATCH or RESEARCH_SIGNAL.	PRICE_ALERT is a cost-adjusted price-consistency alert, not a fill or a verified arbitrage; other candidates are research only.
strategy	text	Candidate structure or hypothetical holding schedule.	A label is not a profitability claim.
t_stat_d_atm	t-like ratio	mean_d_atm_vp / (sd_d_atm_vp/sqrt(sessions)).	Unadjusted exploratory statistic; uses session row count, which may exceed nonmissing changes. |t| > 2 is not validated significance or a multiple-testing-corrected signal.

Metric / field	Units	Definition / method	Interpretation / limitations
t_stat_vs_uniform	t-like ratio	(mean_rv - sum of slot mean_rv / number of observed slots) / (sd_rv/sqrt(max(sessions, 2))).	Exploratory uniform-slot benchmark estimated from the same data; unadjusted for dependence, skewed RV, missing coverage and multiple testing.
taker_eligible	quote count	Count assigned liquidity_role = taker_eligible (relative spread <= 5%).	Still requires size, freshness and risk checks.
theta_per_day	option price units per calendar day	(r*model_price - DF*F*phi(d1)*sigma/(2*sqrt(T))) / 365; candidates sum quantity-weighted legs.	Holding F and IV fixed; excludes hedge/funding execution costs.
theta_per_day_vp	volatility points of vega per calendar day	100 * June reference ATM IV / (2 * june_T_years * 365).	Positive magnitude of approximate ATM decay normalized by vega; unlike signed theta_per_day, omits the r*price term.
tight_share	fraction [0, 1]	Count of valid books with relative spread <= 5% / count of valid books.	Quote-weighted across sessions, not an equal-session average.
underlying_routes	route labels	Unique future-mapping routes used within the expiry.	Direct future, calendar-spread reconstruction or interpolation; inspect provenance/basis risk.
valid_books_per_session	books per observed session	Pooled valid book observations in slot / distinct observed sessions.	All quoted expiries and strikes; not the number of tradable structures.
value	units of named metric	Current value of the statistic named in metric.	atm_iv, rr25 and bf25 here are decimal volatilities, not volatility points.
vega_per_vol_point	option price units per volatility point	Black-76 DF*F*phi(d1)*sqrt(T)*0.01; candidates sum quantity-weighted legs.	Sensitivity to a +1 percentage-point IV move, e.g. 7% to 8%.
weekday	weekday name	Berlin-local weekday of the session's retained futures returns.	Only a few sessions per weekday; events can dominate.
zscore	standard deviations	(current - prior mean) / prior sample standard deviation (ddof=1).	Positive = above own history. Missing for insufficient history, missing coverage or zero prior variance; not a probability or SHAP value.

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